

Call: HORIZON-CL6-2024-GOVERNANCE-01

(Innovative governance, environmental observations and digital solutions in support of the Green Deal)

Topic: HORIZON-CL6-2024-GOVERNANCE-01-7

Type of Action: HORIZON-RIA

(HORIZON Research and Innovation Actions)

Proposal number: 101182954

Proposal acronym: AGROBOOST

Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

Table of contents

Section	Title	Action
1	General information	
2	Participants	
3	Budget	
4	Ethics and security	

Administrative forms

Proposal ID 101182954
Acronym AGROBOOST

1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-CL6-2024-GOVERNANCE-01-7	Type of Action	HORIZON-RIA
Call	HORIZON-CL6-2024-GOVERNANCE-01	Type of Model Grant Agreement	HORIZON-AG
Acronym	AGROBOOST		
Proposal title	Agricultural Ground-breaking sOlutions based on roBotics and augmented reality for boOsting sOcial sustainability and competitiveness while increasing the safeTy of workers		
	Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &		
Duration in months	60		
Free keywords	Robotics, augmented reality, artificial intelligence, automation, digitalisation, sustainable agriculture, enhanced working conditions, training activities, safety, multi-actor approach		

Abstract *

AGROBOOST is a pilot driven, 5-year multi-disciplinary project that aims to transform the agricultural sector into a more attractive, sustainable and efficient industry by harnessing the power of advanced agro-robotics, augmented reality (AR), artificial intelligence (AI), automation and digitalisation revealing the societal and economic benefits for all the people working in agriculture, especially women, young professionals and farmers with moving disabilities. Through the development of cutting-edge, ground-breaking solutions, the project will improve the safety and appeal of agricultural work, reduce environmental impacts, enhance productivity and address labour shortages. AGROBOOST will also emphasize on the development of flexible business models, concrete system analysis tools and training curricula to ensure the widespread adoption and the rapid uptake of these technology innovations. Six distinct and diverse pilot validation campaigns will be demonstrated in 6 countries across Europe (Portugal, Ireland, Greece, England, Belgium and Switzerland), covering multiple agricultural practices and sectors including (1) tree pruning and flower thinning, (2) mushroom selective harvesting, (3) automated weeding, (4) strawberries gentle picking, (5) animal husbandry and livestock farm management and (6) crop monitoring in steep slope viticulture. Financial support to third parties for testing and validation of technology offerings as part of the project's funding and implementation strategy will be awarded through an Open Call, enabling a dynamic response to a changing policy and technology landscape. Through the engagement of a diverse range of stakeholders e.g., farmers (young and women), farm workers, advisors, technology providers and scientists in decision-making processes and the incorporation of social sciences perspectives, AGROBOOST will create a more inclusive and forward-looking agricultural landscape, securing a promising and sustainable future for farming.

Remaining characters 0

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ☐ Yes ☒ No

Please give the proposal reference or contract number.

Previously submitted proposals should be with either 6 or 9 digits.

Administrative forms

Proposal ID **101182954**

Acronym **AGROBOOST**

Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * ☒

2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). * ☒

3) We declare:

- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the [EU Financial Regulation 2018/1046](#)
- to have the financial and operational capacity to carry out the proposed project. *

☒

4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the [Funding & Tenders Portal Terms and Conditions](#). * ☒

5) We have read, understood and accepted the [Funding & Tenders Portal Terms & Conditions](#) and [Privacy Statement](#) that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). * ☒

6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the [ALLEA European Code of Conduct for Research Integrity](#), as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. [Appropriate procedures, policies and structures](#) are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. * ☒

7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of [Regulation 2021/821](#), or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). * ☒

8) We confirm that the activities proposed do not

- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.
- lead to the destruction of human embryos (for example, for obtaining stem cells)

These activities are excluded from funding. * ☒

9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. * ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

Administrative forms

Proposal ID 101182954
Acronym AGROBOOST

2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Role	Action
1	UNIVERSITY COLLEGE DUBLIN, NATIONAL UNIVERSITY OF	Ireland	Coordinator	
2	ETHNICON METSOVION POLYTECHNION	Greece	Partner	
3	NETCOMPANY-INTRASOFT SA	LU	Partner	
4	WISEFOUR INNOVATION AND DEVELOPMENTORGANISAT	Cyprus	Partner	
5	KATHOLIEKE UNIVERSITEIT LEUVEN	BE	Partner	
6	HERIOT-WATT UNIVERSITY	United Kingdom	Partner	
7	GEOPONIKO PANEPISTIMION ATHINON	EL	Partner	
8	INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E CC	Portugal	Partner	
9	UNIVERSITY OF LINCOLN	United Kingdom	Partner	
10	DANMARKS TEKNISKE UNIVERSITET	Denmark	Partner	
11	INSTITUTO NACIONAL DE INVESTIGAÇÃO AGRARIA E VETEPT		Partner	
12	SINOPSIS SOCIALLY INTEGRATED INNOVATION S.R.L.	Romania	Partner	
13	GREEN & DIGITAL IDIOTIKI KEFALAIOUCHIKI ETAIREIA	Greece	Partner	
14	ASSOCIACAO DOS JOVENS AGRICULTORES DE PORTUGAL	Portugal	Partner	
15	Federación de Asociaciones de Mujeres Rurales- FADEMU	ES	Partner	
16	MARTEL GMBH	CH	Associated	
17	FORSCHUNGSINSTITUT FUR BIOLOGISCHEN LANDBAU STI	Switzerland	Associated	
18	ECOLE POLYTECHNIQUE FEDERALE DE LAUSANNE	Switzerland	Associated	

Organisation data

PIC	Legal name
999974359	UNIVERSITY COLLEGE DUBLIN, NATIONAL UNIVERSITY OF IRELAND, DUBLIN

Short name: NUID UCD

Address

Street	BELFIELD
Town	DUBLIN
Postcode	4
Country	Ireland
Webpage	www.ucd.ie

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	07/08/2014 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	School of Biosystems and Food Engineering	☐ not applicable
	☒ Same as proposing organisation's address	
Street	BELFIELD	
Town	DUBLIN	
Postcode	4	
Country	Ireland	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Dimitrios

Last name*

Argyropoulos

E-Mail*

dimitrios.argyropoulos@ucd.ie

Position in org.

Assistant Professor

Department

School of Biosystems and Food Engineering

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

BELFIELD

Town

DUBLIN

Post code

4

Country

Ireland

Website

https://www.ucd.ie/biosystems/

Phone

+xxx xxxxxxxxx

Phone 2

+xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Sarah	Nolan	sarah.nolan1@ucd.ie	+xxx xxxxxxxxx
Navab	Karimi	navab.karimi@ucd.ie	+xxx xxxxxxxxx
Junli	Xu	junli.xu@ucd.ie	+xxx xxxxxxxxx
Chantal	Keijzer	euresearchoffice@ucd.ie	+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Dimitrios	Argyropoulos	Man	Greece	dimitrios.argyropoulos@ucd.ie	Category B Senior resea	Leading	0000-0002-0632-4634	Orcid ID
Ms	Sarah	Nolan	Woman	Ireland	sarah.nolan1@ucd.ie	Category D First stage r	Team member		
Mr	Navab	Karimi	Man	Iran	navab.karimi@ucd.ie	Category D First stage r	Team member		
Dr	Junli	Xu	Woman	China	junli.xu@ucd.ie	Category B Senior resea	Team member	0000-0002-4442-7538	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Tsoulias N, Argyropoulos D, Paraforos D, 2023. <i>Digital Farming and Field Robots</i> . Encyclopedia of Smart Agriculture Technologies, 1-13.
Publication	Charisis C, Gyalai-Korpos M, Nagy A S, Karantzas K, Argyropoulos D, 2023. <i>Detecting and locating mushroom clusters by a Mask R-CNN model in farm environment</i> . Precision agriculture '23, 393-400.
Publication	Wieme J, Mollazade K, Malounas I, Zude-Sasse M, Zhao M, Gowen A, Argyropoulos D, Fountas S, Van Beek J, 2022. <i>Hyperspectral Imaging Systems and Artificial Intelligence for Quality Assessment of Fruit, Vegetables and Mushrooms: A Review</i> . Biosystems Engineering, 222, 156-176.
Publication	Chai J J K, Xu J L, O'Sullivan C, 2023. <i>Real-Time Detection of Strawberry Ripeness Using Augmented Reality and Deep Learning</i> . Sensors (Basel), 23(17), 7639.
Publication	Argyropoulos D, Paraforos D S, 2022. <i>Predicting equilibrium moisture content of mushrooms by NARX neural network and first order kinetic modelling approaches</i> , IFAC-PapersOnLine 55 (32), 241-246.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
WATSON (Project Coordinator) (Horizon Europe)	A holistic frameWork with Anticounterfeit and inTelligence-based technologieS that will assist food chain stakeHolders in rapidly identifying and preventing the spread of fraudulent practices (WATSON) aims to develop a set of tools and systems that detect and prevent fraudulent practices across the entire food supply chain. Watson will rely upon digital and intelligence-based technologies, including AI, IoT, Blockchain, Distributed Ledger Technology and Spectroscopy-based Technology.
FARMTOPIA (Horizon Europe)	Democratizing digital farming through smart solutions for small farms (FARMTOPIA) aims to democratize digital farming by creating a paradigm shift in the way Agricultural Digital Technologies (ADTs) for small farms are created, deployed and paid for. UCD is the leader of Mushroom pilot on Digital Agricultural Technologies.
MUSHNOMICS (Horizon 2020)	Unlocking data-driven innovation for improving productivity and data sharing in mushroom value chain (MUSHNOMICS) aims to demonstrate the feasibility of dynamic data-driven analytics for multi-domain mushroom production environments, through developing best-performing AI-driven algorithms using IoT devices, ICT platforms to exchange data and co-develop innovative business models for growers. UCD is the WP Leader of AI Algorithms development for mushroom growth monitoring.
SPECTROFOOD	Information agrifood quality estimation using hyperspectral techniques (SPECTROFOOD) aspires to develop digital technology solutions for use in the agrifood value chain, combining innovative Hyperspectral Imaging Systems, AI techniques, analytic tools, and data platform-related solutions in order to reduce food waste while optimizing the use of production inputs. UCD is the leader of the Mushroom pilot on Hyperspectral Imaging Technologies.
FOSTERXR	The FoOd induStry of the digiTal Era: new practices in tRaining and skills development through Extended Reality (FosterxR) aims to introduce new technologies in the food industry and food science and improve the capacity of students, researchers, academics and industry professionals, in these technologies. UCD is responsible for development of Computer Vision Systems for detection and Segmentation of Mushroom Instances in natural environments.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)
-------------------------------------	--

Administrative forms

DIGITAL AGTECH LAB	<i>Led by Dr Dimitrios Argyropoulos, the Digital AgTech Lab within the UCD School of Biosystems Engineering is focused on the application of "smart systems" to the agri-food and bio-resource sectors, with applications in food chain integrity, precision farming and innovation.</i>
CONSUS DIGITAL AGRICULTURE	<i>CONSUS Digital Agriculture programme at UCD, Europe's largest university-based research and innovation programme in this domain. Jointly funded by Science Foundation Ireland (SFI) and Origin Enterprises Ltd.</i>
UCD LYONS RESEARCH FARM	<i>The university research farm (at Lyons Estate) provides state-of-the-art research facilities on agricultural production systems.</i>
AGTECHUCD	<i>AgTechUCD, based at UCD Lyons Farm and part of NovaUCD, is focused on accelerating the launch and scaling of AgTech and FoodTech start-ups. AgTechUCD supports such start-ups as they build their innovative businesses into leading enterprises, by providing a 12-week pre acceleration programme.</i>

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999978142	ETHNICON METSOVION POLYTECHNION

Short name: NATIONAL TECHNICAL UNIVERSITY OF ATHENS - NTUA

Address

Street	HEROON POLYTECHNIOU 9 ZOGRAPHOU CAMP
Town	ATHINA
Postcode	157 80
Country	Greece
Webpage	www.ntua.gr

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	25/11/2022 - no
SME self-assessment	unknown
SME validation	08/01/2009 - no

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Industrial Engineering Lab of School of Mechanical Engineering	☐ not applicable
	☒ Same as proposing organisation's address	
Street	HEROON POLYTECHNIOU 9 ZOGRAPHOU CAMPUS	
Town	ATHINA	
Postcode	157 80	
Country	Greece	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Prof.

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Nikolaos

Last name*

Panayiotou

E-Mail*

panayiot@central.ntua.gr

Position in org.

Professor/ Head of the Laboratory

Department

Industrial Engineering Lab of School of Mechanical Engineering

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

HEROON POLYTECHNIOU 9 ZOGRAPHOU CAMPUS

Town

ATHINA

Post code

157 80

Country

Greece

Website

Please enter website

Phone

+30 210 7721195

Phone 2

+30 210 7724141

Other contact persons

First Name	Last Name	E-mail	Phone
Sotiris	Gayialis	sotga@mail.ntua.gr	+XXX XXXXXXXXX

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Nikolaos	Panayiotou	Man	Greece	panayiot@central.ntua.gr	Category A Top grade re	Leading	0000-0002-0603-0302	Orcid ID
Dr	Sotiris	Gayialis	Man	Greece	sotga@mail.ntua.gr	Category B Senior resea	Team member	0000-0002-7524-0215	Orcid ID
Dr	Georgios	Papadopoulos	Man	Greece	gpapado@mail.ntua.gr	Category B Senior resea	Team member	0000-0002-8305-7971	Orcid ID
Mr	Evipidis	Kechagias	Man	Greece	eurikechagias@mail.ntua.gr	Category D First stage r	Team member	0000-0001-9466-6185	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	<i>Kechagias E P, Gayialis S P, Papadopoulos G A, Papoutsis G, 2023. An Ethereum-Based Distributed Application for Enhancing Food Supply Chain Traceability. Foods, 12(6), 1220.</i>
Publication	<i>Gayialis S P, Kechagias E P, Papadopoulos G A, Kanakis E, 2022, September. A smart-contract enabled blockchain traceability system against wine supply chain counterfeiting. In IFIP International Conference on Advances in Production Management Systems, 477-484. Cham: Springer Nature Switzerland.</i>
Publication	<i>Kechagias E P, Chatzistelios G, Papadopoulos G A, Apostolou P, 2022. Digital transformation of the maritime industry: A cybersecurity systemic approach. International Journal of Critical Infrastructure Protection, 37, 100526.</i>
Publication	<i>Kechagias E P, Gayialis S P, Konstantakopoulos G D, Papadopoulos G A, 2020. An application of an urban freight transportation system for reduced environmental emissions. Systems, 8(4), 49.</i>
Publication	<i>Gayialis S P, Konstantakopoulos G D, Kechagias E P, Papadopoulos G A, 2020, March. An advanced transportation system based on internet of things. In Proceedings of the 10th Annual International Conference on Industrial Engineering and Operations Management (IEOM 2020), Dubai, United Arab Emirates, 10-12.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
SOPRANO (Horizon Europe)	<i>Socially-Acceptable and Trustworthy Human-Robot Teaming for Agile Industries (SOPRANO) coalesces multidisciplinary research and innovation in human-robot collaboration and intelligent multi-agent systems, aspiring to design the next generation of manufacturing floors, construction sites, and agri-food production, where humans and intelligent machines will seamlessly work together.</i>
Co4Robots (Horizon Europe)	<i>Achieving Complex Collaborative Missions via Decentralized Control and Coordination of Interacting Robots (Co4Robots) aims to build a systematic methodology to accomplish complex specifications given to a team of potentially heterogeneous robots; control schemes appropriate for the mobility and manipulation capabilities of the robots; perceptual capabilities that enable robots to localize themselves, estimate the state of the dynamic environment; and their systematic integration approach.</i>
RECONFIG (Horizon Europe)	<i>Cognitive, Decentralized Coordination of Heterogeneous Multi-Robot Systems via Reconfigurable Task Planning (RECONFIG) proposes a reconfigurable and adaptive decentralized coordination framework for a team of multi-DoF (degrees-of-freedom) robots. This would allow, for example, a team of robots to individually distinguish objects in the environment, then agree that the same symbol represents the same object for each robot, then develop joint actions and change their plans accordingly.</i>
FRIETS (Horizon Europe)	<i>Sustainable optimization of the value chain of added-value fresh and dried berries through the integration of Precision Agriculture management strategies and innovative dehydration and edible coating (FRIETS) will create a solid framework for the development and marketisation of customised, nutritious and innovative, soft fruit smart snacks that contain no added sugar, sodium or chemical preservatives, produced via innovative precision agriculture management strategies.</i>
WATSON (Horizon Europe)	<i>A holistic frameWork with Anticounterfeit and inTelligence-based technologieS that will assist food chain stakehOlders in rapidly identifying and prevenTing the spread of fraudulent practices (WATSON) will develop a set of tools and systems that detect and prevent fraudulent practices across the entire food supply chain. It will increase transparency by improving track-and-trace mechanisms containing accurate, time-relevant and untampered information on the food product throughout its journey.</i>

Administrative forms

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Lab Infrastructure	<i>The Industrial Engineering Laboratory (IELab NTUA) is located at the Sector of Industrial Engineering and Operational Research. It has laboratory areas that sum up to a total of 250 m2. Additionally, the IELab NTUA uses 5 Classrooms, 7 Offices and a Conference room.</i>
Lab Equipment	<i>The laboratory is equipped with desktops and laptops, a server, mini PCs with IoT sensors and programming kits, scanners, printers, video cameras, teleconference equipment. The Lab also has enterprise software (erp, wms), business process management software, process mining and simulation tools.</i>

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999702371	NETCOMPANY-INTRASOFT SA

Short name: NETCOMPANY-INTRASOFT SA

Address

Street	RUE NICOLAS BOVE 2B
Town	LUXEMBOURG
Postcode	1253
Country	Luxembourg
Webpage	www.netcompany-intrasoft.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2021 - no
SME self-assessment	31/12/2021 - no
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Research & Innovation Development	☐ not applicable
	☒ Same as proposing organisation's address	
Street	RUE NICOLAS BOVE 2B	
Town	LUXEMBOURG	
Postcode	1253	
Country	Luxembourg	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Mr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Konstantinos

Last name*

Thivaios

E-Mail*

ridlux@intrasoft-intl.com

Position in org.

Director

Department

Research & Innovation Development

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

RUE NICOLAS BOVE 2B

Town

LUXEMBOURG

Post code

1253

Country

Luxembourg

Website

Please enter website

Phone

+xxx xxxxxxxxxx

Phone 2

+xxx xxxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Despina	Anastasopoulos	despina.anastasopoulos@netcompany-intrasoft	+xxx xxxxxxxxxx
Amalia	Ntemou	amalia.ntemou@netcompany-intrasoft.com	+xxx xxxxxxxxxx
Ioanna	Drigkopoulou	ioanna.drigkopoulou@netcompany-intrasoft.co	+xxx xxxxxxxxxx
Marios	Logothetis	marios.logothetis@netcompany-intrasoft.com	+xxx xxxxxxxxxx
Spyros	Evangelatos	spyros.evangelatos@netcompany-intrasoft.com	+xxx xxxxxxxxxx
Dorine	Karvouniari-Matzakou	dorine.karvouniari-matzakou@netcompany-intr	+xxx xxxxxxxxxx
Ioannis	Chrysakis	ioannis.chrysakis@netcompany.com	+xxx xxxxxxxxxx
Paraskevas	Bourgos	paraskevas.bourgos@netcompany.com	+xxx xxxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	<i>INTRA has an extensive experience in designing, developing, integrating and testing complex IT systems and services. For such activities, INTRA capitalizes on DevOps frameworks and a fully-fledged Continuous Integration/Continuous Deployment environment and relevant tools, that, among others, consist of GitLab for source control, acting as code repository and allowing code versioning, Jenkins for automated build and testing, Docker for containerization of services and components.</i>
Service	<i>The Big Data StreamHandler platform is a fully featured industrial grade solution which is capable to scale out and accommodate various and from different domains big data, interoperating with all modern data storage technologies as well as other persistence approaches and can support all important Big Data languages including Python, Java, R and Scala as well as other traditional programming approaches.</i>
Publication	<i>Christou I T, Kefalakis N, Zalonis A, Soldatos J, 2020, June. Predictive and Explainable Machine Learning for Industrial Internet of Things Applications, International Workshop on IoT Applications and Industry 4.0 at the 16th IEEE International Conference on Distributed Computing in Sensor Systems (DCOSS 2020), Los Angeles, CA (virtual).</i>
Publication	<i>Soldatos J, Kefalakis N, Serrano M, Hauswirth M, Zaslavsky A, Jayaraman P, Dimitropoulos P, 2014. Practical IoT deployment on Smart Manufacturing and Smart Agriculture based on an Open-Source Platform, in Philippe Cousin (editor) Internet of Things Success Stories, Series 1.</i>
Publication	<i>Soldatos J, Kefalakis N, Despotopoulou A M, Bodin U, Musumeci A, Scandura A, Aliprandi C, Arabsolgar D, Colledani M, 2021. A Digital Platform for Cross-Sector Collaborative Value Networks in The Circular Economy. Proc. Manufacturing, 54, 64-69.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
<i>DATABIO (Horizon 2020)</i>	<i>DataBio proposed the deployment of a state-of-the-art, big data platform on top of the existing partners' infrastructure and solutions - the Big DATABIO Platform. Based on the preparation and requirement specifications work the pilots were implemented utilizing and selecting the best suitable market ready or almost market ready Big Data and Earth Observation methods, technologies, tools and services integrated to the common Big DATABIO Platform.</i>
<i>WATSON (Horizon Europe)</i>	<i>WATSON will provide a methodological framework combined with a set of tools and systems that can detect and prevent fraudulent activities throughout the whole food chain thus accelerating the deployment of transparency solutions in the EU food systems.</i>
<i>CYBELE (Horizon Europe)</i>	<i>The CYBELE project will generate innovation and create value in the field of agri-food in precision agriculture and precision livestock farming to increase capacity building within the industrial and research communities. The project envisages demonstrating that high-performance computing (HPC), Big Data, cloud computing, and the Internet of Things can transform farming, reduce scarcity, increase food supply, and bring social, economic and environmental benefits.</i>
<i>DATA4FOOD2030 (Horizon Europe)</i>	<i>The Data4Food2030 project aims to improve the data economy for food systems (DE4FS) by expanding its definition, mapping its development, performance and impact to create new insights and opportunities.</i>
<i>CiROCCO (Horizon Europe)</i>	<i>CiROCCO's vision is to establish an end-to-end sensing system, composed of a distributed network of cost-effective sensing nodes coupled with state-of-the-art data fusion remote sensing and assimilation modelling techniques. Four services will be developed i) Renewable energy systems planning ii) Air quality early warning system for human health iii) Land use and ecosystem management iv) Modelling of GHGs and particles emissions, to validate the quality of the collected data.</i>

Administrative forms

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☐ Yes ☒ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
944556513	WISEFOUR INNOVATION AND DEVELOPMENTORGANISATION LTD

Short name: Wisefour Innovation and Development Organisation Ltd

Address

Street	THERMOPYLON 70
Town	NICOSIA-STROVOLOS
Postcode	2007
Country	Cyprus
Webpage	www.wisefour.eu

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	03/09/2019 - yes
SME self-assessment	31/12/2013 - yes
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

No department involved

Department name

Name of the department/institute carrying out the work.

☒ not applicable

☐ Same as proposing organisation's address

Street

Please enter street name and number.

Town

Please enter the name of the town.

Postcode

Area code.

Country

Please select a country

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

TitleMrs

Gender

Woman

Man

Non Binary

First name*Katerina

Last name*Kadena

E-Mail*katerina@wisefour.eu

Position in org.General Manager

DepartmentWISEFOUR INNOVATION AND DEVELOPMENTORGANISATION LTD

Same as organisation name

Same as proposing organisation's address

StreetTHERMOPYLON 70

TownNICOSIA-STROVOLOS

Post code2007

CountryCyprus

Websitewww.wisefour.eu

Phone+35722316576

Phone 2+xxx xxxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Maria	Kadena	maria@wisefour.eu	+xxx xxxxxxxxxx
Jenny	Lazarou	jenny@wisefour.eu	+xxx xxxxxxxxxx
Nikos	Arslanoglou	nikos@wisefour.eu	+xxx xxxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Lucia	Fais	Woman	Israel	lucia@wisefour.eu	Category B Senior researcher	Leading		
Mrs	Katerina	Kadena	Woman	Greece	katerina@wisefour.eu	Category C Recognised	Team member		
Mrs	Konstantina	Skolariki	Woman	Greece	konstantina@wisefour.eu	Category C Recognised	Team member		
Mrs	Jenny	Lazarou	Woman	Greece	jenny@wisefour.eu	Category D First stage researcher	Team member		
Mrs	Maria	Kadena	Woman	Greece	maria@wisefour.eu	Category D First stage researcher	Team member		
Mr	Andreas	Gregoriou	Man	Cyprus	legal@wisefour.eu	Category C Recognised	Team member		

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☒

Legal and ethics

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Fais L, 2012. Energy and environmental performance of tall buildings: State of the art. Advances in Building Energy Research, 6(1), 36-60.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ALLIANCE	ALLIANCE represents a paradigm shift in the Food Supply Chain Systems' management for the combat against Food Fraud, distinguishing from the traditionally approaches that leverage monolithic digitalized logistic solutions and standalone FSC interoperability protocols.
DELOS	DELOS project aims to promote the process of anaerobic digestion of agricultural waste (eg from olive processing, cheese-making, animal husbandry) as a promising waste treatment process with a high content of organic matter.
NARA	The main goal of NARA Project is to improve the professionalization of young farmers in Europe through the analysis of training needs in six key areas for their sector: Food security; Technologies postharvest; Commercialization and logistic; Soil, water and byproducts; Packaging; and Entrepreneurship.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999991334	KATHOLIEKE UNIVERSITEIT LEUVEN

Short name: KU Leuven

Address

Street	OUDE MARKT 13
Town	LEUVEN
Postcode	3000
Country	Belgium
Webpage	www.kuleuven.be

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2022 - no
SME self-assessment	31/12/2022 - no
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Department of Biosystems	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Biosystems 1, Kasteelpark Arenberg 30	
Town	Leuven/Herverlee	
Postcode	3001	
Country	Belgium	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title	Prof.	Gender	☐ Woman	☒ Man	☐ Non Binary
First name*	Tomas	Last name*	Norton		
E-Mail*	tomas.norton@kuleuven.be				
Position in org.	Associate Professor				
Department	Department of Biosystems			☐	Same as organisation name
	☐ Same as proposing organisation's address				
Street	Biosystems 1, Kasteelpark Arenberg 30				
Town	Leuven/Herverlee	Post code	3001		
Country	Belgium				
Website	Please enter website				
Phone	+xxx xxxxxxxxx	Phone 2	+xxx xxxxxxxxx		

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier	
Mr	Sam	Willems	Man	Belgium	Sam.willems@kuleuven.be	Category A Top grade re	Team member	https://www.kuleuven.be/wieiswie/nl/person/00109578	Other ID	
Dr	Liu	Dong	Man	China	Dong.liu@kuleuven.be	Category B Senior resea	Team member	https://www.kuleuven.be/wieiswie/nl/person/00138031	Other ID	
Prof	Tomas	Norton	Man	Ireland	Tomas.norton@kuleuven.be	Category C Recognised	Leading			

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Parmiggiani A, Liu D, Psota E, Fitzgerald R, Norton T, 2023. Don't get lost in the crowd: Graph convolutional network for online animal tracking in dense groups. Computers And Electronics In Agriculture, 212, Art.No. 108038, 1-13. doi: 10.1016/j.compag.2023.108038
Publication	Siegford J M, Steibel J P, Han J, Benjamin M, Brown-Brandl T, Dorea J R R, Morris D, Norton T, Psota E, Rosa G J M, 2023. The quest to develop automated systems for monitoring animal behavior. APPLIED ANIMAL BEHAVIOUR SCIENCE, 265, Art.No. 106000. doi: 10.1016/j.applanim.2023.106000
Publication	Liu D, Parmiggiani A, Psota E, Fitzgerald R, Norton T, 2023. Where's your head at? Detecting the orientation and position of pigs with rotated bounding boxes. Computers And Electronics In Agriculture, 212, Art.No. 108099, 1-13. doi: 10.1016/j.compag.2023.108099
Publication	Larsen M, Wang M, Willems S, Liu D, Norton T, 2023. Automatic detection of locomotor play in young pigs: A proof of concept. Biosystems Engineering, 229, 154-166. doi: 10.1016/j.biosystemseng.2023.03.006
Publication	Verlinden S, Larsen M, Debontridder P, Youssef A, Everaert N, Norton T, 2022. Effect of lower temperature stimuli during incubation on fear and social-related behaviours in broilers. Applied Animal Behaviour Science, 248, Art.No. 105572. doi: 10.1016/j.applanim.2022.105572

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ATHLETIC-CHICKS	Health and welfare effects of early nutrition for broilers feeding - ATHLETIC-CHICKS (Agentschap Innoveren en Ondernemen 3E230111). This LA project aims to provide the broiler sector with scientifically validated protocols for optimal implementation of early feeding for both barn hatchery and hatchery systems, both for chicks from conventionally fast-growing broilers and slower-growing broilers.
A Coordinated Innovation Network for Advancing...	A Coordinated Innovation Network for Advancing Computer Vision in Precision Livestock Farming (USDA: AFRI COMPETITIVE GRANT). Two main objectives: 1) Creating and releasing reference datasets and benchmarking data to facilitate the development of CV applications addressing key challenges in PLF. 2) Building a coordinated innovation network of stakeholders, researchers, and students to speed the development of CV applications in PLF.
FARMTOPIA (Horizon Europe)	Democratizing digital farming through smart solutions for small farms (FARMTOPIA). The overall objective of the Farmtopia project is to democratize digital farming by creating a paradigm shift in the way Agricultural Digital Technologies (ADTs) for small farms are created, deployed and paid for.
QuantiFarm (Horizon Europe)	Testing a high-tech generation of digital agriculture solutions (QUANTIFARM). The project will introduce a comprehensive assessment framework for independent qualitative and quantitative assessment of the multiple costs and benefits of digital technologies in agriculture, as well as of their sustainability impact.
AutoPlayPig (Horizon 2020)	Automatic detection of play behaviour in young pigs as a measure of positive affective states (AutoPlayPig). The project aims to investigate the relation between play behaviour in pigs and sensor data from image and sound analysis combining ethology and computer science in Computational Ethology (CE) field.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Algorithms library	M3-BIORES have a library of algorithms available for the computer vision-based monitoring of animals, which will be of course made available to the student during this project.

Administrative forms

Computer facilities	M3-BIORES have high power computer facilities with state-of-the-art GPU processing for deep learning based model development.
Hardware and software	M3-BIORES have hardware and software for data acquisition with 2/3D activity cameras and sound monitoring systems.

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999853400	HERIOT-WATT UNIVERSITY

Short name: HWU

Address

Street	Riccarton
Town	EDINBURGH
Postcode	EH14 4AS
Country	United Kingdom
Webpage	www.hw.ac.uk

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is unknown (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	School of Engineering and Physical Sciences	☐ not applicable
	☒ Same as proposing organisation's address	
Street	Riccarton	
Town	EDINBURGH	
Postcode	EH14 4AS	
Country	United Kingdom	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Fernando Alfredo

Last name*

Auat Cheein

E-Mail*

f.auat@hw.ac.uk

Position in org.

Associate Professor

Department

School of Engineering and Physical Sciences

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

Riccarton

Town

EDINBURGH

Post code

EH14 4AS

Country

United Kingdom

Website

https://researchportal.hw.ac.uk/en/persons/fernando-auat-cheein

Phone

+xxx xxxxxxxxx

Phone 2

+xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
RES	Europe	res.eu@hw.ac.uk	+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Marta	Vallejo	Woman	Spain	Marta.vallejo@hw.ac.uk	Category A Top grade re	Team member	0000-0001-9957-954X	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Estrada J S, Vasconez J P, Fu L, Cheein F A, 2024. Deep Learning based flower detection and counting in highly populated images: A peach grove case study. <i>Journal of Agriculture and Food Research</i> , 15, art. no. 100930
Publication	Deniz N, Jorquera F, Torres-Torriti M, Cheein F A, 2024. Model predictive path-following controller for Generalised N-Trailer vehicles with noisy sensors and disturbances. <i>Control Engineering Practice</i> , 142, art. no. 105747
Publication	Jiang H, Sun X, Fang W, Fu L, Li R, Cheein F A, Majeed Y, 2023. Thin wire segmentation and reconstruction based on a novel image overlap-partitioning and stitching algorithm in apple fruiting wall architecture for robotic picking. <i>Computers and Electronics in Agriculture</i> , 209, art. no. 107840
Publication	Bustos J A T, Orchard M E, Torres-Torriti M, Cheein F A, 2023. GNSS-Based Estimation of Average Instantaneous Power Consumption in Electric Vehicles. <i>IEEE Transactions on Industrial Electronics</i> , 70 (9), 9281 – 9290. DOI: 10.1109/TIE.2022.3206748
Publication	Vásconez J P, Auat Cheein F A, 2022. Workload and production assessment in the avocado harvesting process using human-robot collaborative strategies. <i>Biosystems Engineering</i> , 223, 56-77. DOI: 10.1016/j.biosystemseng.2022.08.010

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
CDT D2AIR	UKRI AI Centre For Doctoral Training In Dependable And Deployable Artificial Intelligence For Robotics. Co-Lead: Fernando Auat Cheein. Corresponds to a joint program with the University of Edinburgh.
Solving the Big Data problem in Agriculture	UKRI-Innovate UK: Solving the Big Data problem in Agriculture: Extracting Minimum Viable Datasets for Plant Detection and Crop Treatment. Lead PI: Fernando Auat Cheein. Project focused on phenotyping and characterization of specific crops, in association with the industry.
Adopting Agri - Robotics	UKRI-Innovate UK: Adopting Agri - Robotics To Optimise Sustainable Weed Management Practises For The Reduction Of Chemical Herbicide On Farm. Lead PI: Fernando Auat Cheein. Project with the industry, focused on the use of new mobile platforms in agriculture.
Energy efficient navigation of N-trailers	Energy efficient navigation strategies for autonomous N-trailer vehicles in agricultural applications. Principal Investigator: Fernando Auat Cheein. Research project focused on developing new strategies for the manoeuvrability of N trailer systems.
Enhancing motion performance of machinery	Enhancing Motion Performance of Automated Machinery in Agricultural Environments based on Bayesian Estimation Approaches. Principal Investigator: Fernando Auat Cheein. Research project focused on the development of new motion controllers for agricultural machinery.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spot legged robot	Quadruped legged robot manufactured by Boston Dynamics
LiDAR Ouster 32	Light detection and ranging sensors for 3D monitoring of the scene
UR5 robot manipulator	Robotic manipulator to be placed on the top of the legged robot
Three finger grippers	Own design of grippers to be placed on the robotic arm for egg manipulation

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999872218	GEOPONIKO PANEPISTIMION ATHINON

Short name: AGRICULTURAL UNIVERSITY OF ATHENS

Address

Street	IERA ODOS 75
Town	ATHINA
Postcode	11855
Country	Greece
Webpage	www.aua.gr

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	23/01/2023 - no
SME self-assessment	07/05/2021 - no
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Natural Resource Management & Agricultural Engineering	☐ not applicable
	☒ Same as proposing organisation's address	
Street	IERA ODOS 75	
Town	ATHINA	
Postcode	11855	
Country	Greece	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Prof.

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Spyros

Last name*

Fountas

E-Mail*

sfountas@aua.gr

Position in org.

Head of Agricultural Engineering Lab

Department

Natural Resource Management & Agricultural Engineering

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

IERA ODOS 75

Town

ATHINA

Post code

11855

Country

Greece

Website

www.aua.gr

Phone

+2105294035

Phone 2

+xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Giannis	Malounas	gmalounas@aua.gr	+xxx xxxxxxxxx
Kalliopi	Kounani	kalliopi.kounani@aua.gr	+xxx xxxxxxxxx
Antonis	Prekas	a.prekas@aua.gr	+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Spyros	Fountas	Man	Greece	sfountas@aua.gr	Category A Top grade re	Leading	0000-0001-9787-6268	Orcid ID
Dr	Ioannis	Zafeiriou	Man	Greece	ioannis.zafeiriou@aua.gr	Category C Recognised	Team member	0009-0000-4214-2734	Orcid ID
Mrs	Georgia	Nikolakopoulou	Woman	Greece	gnikolakopoulou@aua.gr	Category D First stage r	Team member	0000-0003-2309-1163	Orcid ID
Mrs	Kallia	Kasimati	Woman	Greece	k.kasimati@aua.gr	Category D First stage r	Team member	0000-0002-2747-7138	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	<i>Espejo-Garcia B, Panoutsopoulos H, Anastasiou E, Rodríguez-Rigueiro F J, Fountas S, 2023. Top-tuning on transformers and data augmentation transferring for boosting the performance of weed identification. Computers and Electronics in Agriculture, 211, 108055.</i>
Publication	<i>Kasimati A, Psiroukis V, Panoutsopoulos H, Mouseti S, Mylonas N, Fountas S, 2023. Management Information Systems and Emerging Technologies. Advanced Automation for Tree Fruit Orchards and Vineyards, 195-218.</i>
Publication	<i>Darra N, Anastasiou E, Kriezi O, Lazarou E, Kalivas D, Fountas S, 2023. Can Yield Prediction Be Fully Digitized? A Systematic Review. Agronomy, 13(9), 2441.</i>
Dataset	<i>EDEN library (https://edenlibrary.ai/) offering high quality agri-food image data and using them to provide tailor- made AI tools and solutions to boost productivity.</i>
Software	<i>Software Architecture for Agricultural Robots - SAFAR (robot control, route planning, robot simulation); web-based Farm Management Information Systems with features like Real Time Data Transfer; communication and data transfer with agricultural robots; communication and data transfer with tractor implemented through CAN Bus and ISO Bus.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
NOVATERRA	<i>NOVATERRA aims to reduce the use and negative impacts of CONTENTIOUS PLANT PROTECTION PRODUCTS (PPP) for integrated pest, disease and weed management in the two main Mediterranean crops in Europe: grapevines and olives. NOVATERRA will develop a pool of novel, integrated and sustainable strategies, technically and economically viable, resulting from three different approaches.</i>
ROBS4CROPS	<i>The vision of ROBS4CROPS is an agricultural sector where robots will replace humans in all heavy and unpleasant work. ROBS4CROPS will tackle technical challenges by creating a robotic farming solution that consists of three elements: smart implements, autonomous vehicles, and the farming controller. Existing agricultural implements and tractors will be upgraded so that they can function, together with existing agricultural robots, as parts of a robotic system.</i>
SmartDroplets (Horizon Europe)	<i>Project Coordinator: Smart Droplets' main objective is to advance both hardware and software capabilities to deliver a holistic system capable of translating large amounts of data into meaningful information and impactful spraying commands on the field. To demonstrate substantial impact on the Green Deal, Smart Droplets will implement Autonomous retrofit tractors with Direct Injection System (DIS) for intelligent spraying - avoiding exposure of farmers to hazardous chemicals.</i>
CRACKSENSE (Horizon Europe)	<i>CrackSense aims to address the problem of fruit cracking in citrus, pomegranate, table grapes and sweet cherries by developing and upscaling sensing technologies that will provide real-time sensor data through piloting activities. The sensor data collected will be upscaled into EU-wide data sets which will encompass Earth Observation Data (provided by Copernicus, e.g., Sentinel) and other data sets reflecting on environmental conditions.</i>
agROBOfood	<i>agROBOfood is dedicated to accelerate the digital transformation of the European agri-food sector through the adoption of robotic technologies. It will consolidate, extend and strengthen the current ecosystem by establishing a sustainable network of DIHs. This will boost the uptake of robotic solutions by the agri-food sector: a huge challenge requiring an inclusive approach involving all relevant European players.</i>

Administrative forms

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
RTK - GPS	High precision GNSS system (<2cm error)
Crop Circle	Vegetation index data sensor for NDVI,LAI
Agricultural robots	Zues customised ground vehicle and HUSKY commercial platform
UAV (DJI S-800 EVO)II & Spraying UAV platform	Flexible aerial platform for crop monitoring tasks with easy integration of additional sensors and cameras
MicaSense Rededge2	Vegetation index data sensor for NDVI,LAI

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999513706	INESC TEC - INSTITUTO DE ENGENHARIA DE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA

Short name: INESC TEC

Address

Street	RUA DR ROBERTO FRIAS CAMPUS DA FEUP
Town	PORTO
Postcode	4200 465
Country	Portugal
Webpage	www.inesctec.pt

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is unknown (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name

CRIIS - Centre for Robotics in Industry and Intelligent Systems

☐ not applicable

☒ Same as proposing organisation's address

Street

RUA DR ROBERTO FRIAS CAMPUS DA FEUP

Town

PORTO

Postcode

4200 465

Country

Portugal

Department 2

Department name

CTM - Centre for Telecommunications and Multimedia

☐ not applicable

☒ Same as proposing organisation's address

Street

RUA DR ROBERTO FRIAS CAMPUS DA FEUP

Town

PORTO

Postcode

4200 465

Country

Portugal

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Pedro

Last name*

Carvalho

E-Mail*

pedro.m.carvalho@inesctec.pt

Position in org.

Senior Researcher

Department

CTM - Centre for Telecommunications and Multimedia

☐ Same as organisation name

☒ Same as proposing organisation's address

Street

RUA DR ROBERTO FRIAS CAMPUS DA FEUP

Town

PORTO

Post code

4200 465

Country

Portugal

Website

Please enter website

Phone

+xxx xxxxxxxxx

Phone 2

+xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Aurora	Teixeira	aurora.l.teixeira@inesctec.pt	+xxx xxxxxxxxx
Marta	Barbas	mbarbas@inescporto.pt	+xxx xxxxxxxxx
Filipe	Santos	filipe.n.santos@inesctec.pt	+xxx xxxxxxxxx
Paula	Viana	paula.viana@inesctec.pt	+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Tatiana	Pinho	Woman	Portugal	tatiana.m.pinho@inesctec.pt	Category C Recognised	Team member	0000-0002-1403-7221	Orcid ID
Dr	Filipe	Santos	Man	Portugal	filipe.n.santos@inesctec.pt	Category B Senior resea	Team member	0000-0002-8486-6113	Orcid ID
Dr	Pedro	Carvalho	Man	Portugal	pedro.m.carvalho@inesctec.pt	Category B Senior resea	Leading	0000-0003-4983-4316	Orcid ID
Prof	Paula	Viana	Woman	Portugal	paula.viana@inesctec.pt	Category B Senior resea	Team member	0000-0001-8447-2360	Orcid ID
Prof	Mário	Cunha	Man	Portugal	mario.cunha@inesctec.pt	Category B Senior resea	Team member	0000-0002-8299-324X	Orcid ID
Prof	Maria Teresa	Andrade	Woman	Portugal	maria.t.andrade@inesctec.pt	Category B Senior resea	Team member	0000-0002-1363-5027	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Santos L C, Santos F N, Valente A, Sobreira H, Sarmento J, Petry M, 2022. Collision avoidance considering iterative Bézier based approach for steep slope terrains. IEEE Access, 10, 25005-25015. DOI: 10.1109/ACCESS.2022.3153496
Publication	Aguilar A S, dos Santos F N, Cunha J B, Sobreira H, Sousa A J, 2020. Localization and mapping for robots in agriculture and forestry: A survey. Robotics, 9(4), 97. DOI: 10.3390/robotics9040097
Publication	Aguilar A S, Santos F N D, Santos L C, Sousa A J, Boaventura-Cunha J, 2023. Topological map-based approach for localization and mapping memory optimization. Journal of Field Robotics. DOI: 10.1002/rob.22140
Publication	Pereira A, Carvalho P, Pereira N, Viana P, Corte-Real L, 2023. From a visual scene to a virtual representation: A cross-domain review. IEEE Access, 11, 57916-57933. DOI: 10.1109/ACCESS.2023.3283495
Publication	Pereira A, Carvalho P, Coelho G, Côte-Real L, 2020. Efficient CIEDE2000-based Color Similarity Decision for Computer Vision. IEEE Transactions on Circuits and Systems for Video Technology, 30(7), 2141-2154. DOI: 10.1109/TCSVT.2019.2914969

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
WATSON (Horizon Europe)	A holistic frameWork with Anticounterfeit and inTelligence-based technologieS that will assist food chain stakeOlders in rapidly identifying and prevenTing the spread of fraudulent practices (WATSON) is a large-scale project oriented to the development of tools and technology to help combat food fraud and promote traceability. INESC TEC is responsible for the wine pilot, including the development of image processing-based tools and IoT platforms for sensor data acquisition.
SCORPION (Horizon 2020) (Project Leader)	Cost effective robots for smart precision spraying (SCORPION)'s objective is to develop a safe and autonomous precision spraying tool integrated into a modular unmanned tractor (robotics platform) to increase spraying efficiency, while reducing human and animal exposure to pesticides, water usage and labour costs.
NOVATERRA (Horizon 2020)	Integrated Novel Strategies for Reducing the Use and Impact of Pesticides, Towards Sustainable Mediterranean Vineyards and Olive Groves (NOVATERRA). INESC TEC is responsible for the development of novel robot for application in viticulture.
DEMETER (Horizon 2020)	Precision Farming for Mediterranean Woody Crops (DEMETER) is a large-scale deployment of farmer-driven, interoperable smart farming IoT (Internet of Things) based platforms, delivered through a series of 20 pilots across 18 countries (15 EU countries). Involving 60 partners, DEMETER adopts a multi-actor approach across the value chain (demand and supply), with 25 deployment sites, 6,000 farmers and over 38,000 devices and sensors being deployed. INESC TEC is a pilot leader.
AgroRoboFood (Horizon 2020)	AgRoboFood aims to build a European ecosystem for the effective adoption of robotic technologies in the agri food sector, which in turn will become more efficient and competitive. The heart of the project is formed by Innovation Experiments, organized and monitored by Digital Innovation Hubs. In 7 Regional Clusters, Initial Innovation Experiments will demonstrate robotics innovations in agri food. INESC TEC is the Portuguese representative.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
-------------------------------------	--

Administrative forms

TRIBE	<i>TRIBE (Laboratory of Robotics and IoT for Smart Precision Agriculture and Forestry) has the mission of developing robotics, automation, and IoT based solutions, to improve the levels of smart precision agriculture and forestry, profitability, and automation in multiple alternative environments.</i>
-------	--

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999882694	UNIVERSITY OF LINCOLN

Short name: UNIVERSITY OF LINCOLN

Address

Street	Brayford Pool
Town	LINCOLN
Postcode	LN6 7TS
Country	United Kingdom

Webpage

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	16/02/1993 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	School of Computer Science	☐ not applicable
	☐ Same as proposing organisation's address	
Street	INB Brayford Pool Campus	
Town	Lincoln	
Postcode	LN6 7TS	
Country	United Kingdom	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Athanasios

Last name*

Polydoros

E-Mail*

apolydoros@lincoln.ac.uk

Position in org.

Senior Lecturer

Department

School of Computer Science

☐ Same as organisation name

☐ Same as proposing organisation's address

Street

INB 3120 Brayford Pool Campus

Town

Lincoln

Post code

LN6 7TS

Country

United Kingdom

Website

Please enter website

Phone

+xxx xxxxxxxxxx

Phone 2

+xxx xxxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Rebecca	Crookes	research@lincoln.ac.uk	+xxx xxxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Athanasios	Polydoros	Man	Greece	apolydoros@lincoln.ac.uk	Category C Recognised	Leading	0000-0002-4597-0567	Orcid ID
Dr	Patrick	Dickinson	Man	United Kingdom	pdickinson@lincoln.ac.uk	Category C Recognised	Team member	0000-0003-3830-8249	Orcid ID
Dr	Renata	Ntelia	Woman	Greece	Rntelia@lincoln.ac.uk	Category C Recognised	Team member	0000-0002-2996-6330	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Ravichandar H, Polydoros A S, Chernova S, Billard A, 2020. Recent advances in robot learning from demonstration. Annual review of control, robotics, and autonomous systems, 3, 297-330.
Publication	Polydoros A S, Nalpantidis L, Krüger V, 2015. Real-time deep learning of robotic manipulator inverse dynamics. In IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 3442-3448.
Publication	Spranger J, Buzatoiu R, Polydoros A, Nalpantidis L, Boukas E, 2018. Human-machine interface for remote training of robot tasks. In 2018 IEEE International Conference on Imaging Systems and Techniques (IST), 1-5.
Publication	Dickinson P, Cardwell A, Parke A, Gerling K, Murray J, 2021. Diegetic Tool Management in a Virtual Reality Training Simulation. In IEEE Virtual Reality and 3D User Interfaces (VR), 131-139.
Publication	Ntelia R E, 2023. Merits of Gamification: Transmediality and Spatiality of Walden, a Game. In Proceedings of the 18th International Conference on the Foundations of Digital Games, 1-6.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Robot Learning of harvesting tasks from visual..	'Robot Learning of harvesting tasks from visual demonstrations', funded by the EPSRC Centre for Doctoral Training in Agri-Food Robotics (AgriFoRwArdS). The project aims to increase the automation level of agricultural tasks with special focus on complex and challenging cases such as selective harvesting of fruits, crop husbandry, etc. Thus, the main objective is to increase the efficiency of the current agricultural robotic systems both in terms of task success rate and cycle time.
BACCHUS (Horizon 2020)	This project is developing an intelligent mobile robotic system that promises to reproduce hand harvesting operations, while at the same time take the manual legwork out by autonomously operating in four different levels. The envisioned system will be demonstrated and evaluated in the challenging vineyard environment inspecting diverse types of vines and harvesting bunches of grapes of different varieties in a human-like manner.
STRANDS	STRANDS enabled a robot to achieve robust and intelligent behavior in human environments through exploitation of long-term experience. The approach was based on understanding 3D space and how it changes over time, from milliseconds to months. The project developed control mechanisms which yield adaptive behavior in highly demanding, real-world security and care scenarios. UoL contributed with their unique expertise in long-term mapping and behaviour generation in integrated robotic systems.
Playing Maltese History	Playing Maltese History, funded by Malta Arts Council, involves the research for and development of a mobile AR game about the cultural history of Malta titled Valletta: Streets of History, released in March 2023. It was conceived to bridge the gap between historical research and detective games or to make the profound connections between the two practices explicit. The game invites the player to trace events in actual locations by means of geolocation mechanisms and AR elements.
Virtual Reality Games for People Using Wheelchairs	This project investigated design guidelines for VR games, to support accessibility for players with mobility impairments. The project was motivated by the predominant use of full-body interfaces for VR platforms, and perceived accessibility issues effecting players who use wheelchairs or have other mobility issues. Project outcomes included design guidelines for the developers of VR games to make products more accessible, and were published.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)
-------------------------------------	--

Administrative forms

<i>Robotic manipulator</i>	<i>A 6-DOF KUKA iisy robot will be used for implementing the proposed methods</i>
<i>Riseholm campus strawberry farm</i>	<i>Agricultural research facility with strawberry greenhouses for evaluating the system prototype in operational environment</i>

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999990655	DANMARKS TEKNISKE UNIVERSITET

Short name: TECHNICAL UNIVERSITY OF DENMARK DTU

Address

Street	ANKER ENGELUNDS VEJ 101
Town	KONGENS LYNGBY
Postcode	2800
Country	Denmark
Webpage	www.dtu.dk

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	13/01/2022 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Department of Electrical and Photonics Engineering	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Oersteds Plads, Building 343	
Town	Kongens Lyngby	
Postcode	2800	
Country	Denmark	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Prof.

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Lazaros

Last name*

Nalpantidis

E-Mail*

lanalpa@dtu.dk

Position in org.

Professor, Head of Group

Department

Department of Electrical and Photonics Engineering

☐ Same as organisation name

☐ Same as proposing organisation's address

Street

Elektrovej, Building 326

Town

Kongens Lyngby

Post code

2800

Country

Denmark

Website

Please enter website

Phone

+xxx xxxxxxxxxx

Phone 2

+xxx xxxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Ronja	Güldenring	ronjag@dtu.dk	+xxx xxxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Lazaros	Nalpantidis	Man	Greece	lanalpa@dtu.dk	Category A Top grade re	Leading	0000-0002-3620-4123	Orcid ID
Dr	Ronja	Güldenring	Woman	Germany	ronjag@dtu.dk	Category C Recognised	Team member	0000-0001-6430-7405	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Güldenring R, Andersen R E, Nalpantidis L, 2024. Zoom in on the plant: Fine-grained analysis of leaf, stem and vein instances. IEEE Robotics and Automation Letters, 1588–1595. (PUBLICATION + DATASET)
Publication	Li J, Güldenring R, Nalpantidis L, 2023. Real-time joint-stem prediction for agricultural robots in grasslands using multi-task learning. Agronomy, 13(9).
Publication	Güldenring R, van Evert F K, Nalpantidis L, 2023. RumexWeeds: A grassland dataset for agricultural robots. Journal of Field Robotics. (PUBLICATION + DATASET)
Publication	Güldenring R, Nalpantidis L, 2021. Self-supervised contrastive learning on agricultural images. Computers and Electronics in Agriculture, 191, 106510.
Publication	Güldenring R, Boukas E, Ravn O, Nalpantidis L, 2021. Few-leaf learning: Weed segmentation in grassland. Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). (Finalist for the Best Paper Award on Agri-Robotics)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
SAVA	The SAVA project addresses important environmental and safety challenges in agriculture, aiming to enhance agricultural operations through advanced technologies and a user-centered approach. The project will focus on the development and implementation of new sensor technologies for autonomous field robots and other machinery in agriculture. The aim is to improve the precision, range, and obstacle detection capabilities of sensor solutions available.
GALIRUMI (Horizon Europe)	The GALIRUMI project will deliver robot weeding for herbicide-free weed control in dairy farming, focusing on the broad-leaved dock (<i>Rumex obtusifolius</i> L.). The project will integrate off-the-shelf hardware components and mature software technologies (e.g. deep learning for weed detection) with high-precision Galileo positioning. The primary role of DTU in GALIRUMI is to provide the 2D and 3D perception modules for detection and localization of weeds.
DockWeeder	The DockWeeder project focused on detection of Rumex by means of novel 2D and 3D methods and destroying the roots of the weed by hot water treatment, thus supporting organic dairy farming.
RoBétArmé (EU Horizon)	The RoBétArmé project focuses on developing robotic systems for human-robot collaborative construction systems for shotcrete digitization and automation through advanced perception, cognition, mobility, and additive manufacturing skills.
CoCoBOT	CoCoBOT (Odense Robotics: Danish National Robotics Cluster) Collaborative construction site robot for assistive logistic tasks. Research and development activities for perception and collaborative behavior between humans and construction robots.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)
DTU ASTA (Autonomous Systems Test Arena)	DTU ASTA is a modern covered facility, equipped with a 16 cameras motion capture system (OptiTrack) where to experiment with multiple robotic systems. DTU ASTA allows for initial outdoor-scale tests of the developed robotic systems, before deployment in real outdoor agricultural settings.
Husky UGV robot (Clearpath Robotics)	Husky unmanned outdoor field robot with various mounted sensors (cameras, lidars, GPS) will become available for development and initial tests.

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
952674831	INSTITUTO NACIONAL DE INVESTIGAÇÃO AGRARIA E VETERINARIA

Short name: INIAV

Address

Street	AVENIDA DA REPUBLICA QUINTA DO MARQUES
Town	OEIRAS
Postcode	2780 157
Country	Portugal
Webpage	www.iniaiv.pt

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	01/01/1970 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	INIAV headquarters	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Quinta do Marquês, Av. da República	
Town	Oeiras	
Postcode	2780-154	
Country	Portugal	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Miguel

Last name*

Leão de Sousa

E-Mail*

miguel.leao@iniav.pt

Position in org.

Senior Researcher

Department

Innovation Campus of Alcobaça

☐ Same as organisation name

☐ Same as proposing organisation's address

Street

Estrada de Leiria

Town

Alcobaça 2460-059 Alcobaça

Post code

2460-059

Country

Portugal

Website

www.inia.pt

Phone

+351 262590680

Phone 2

+xxx xxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Teresa	Oliveira	teresa.oliveira@iniav.pt	+xxx xxxxxxxxx
Manuela	Berjano	manuela.berjano@iniav.pt	+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Miguel	Leão de Sousa	Man	Portugal	Miguel.leao@inia v.pt	Category C Recognised	Leading	0000-0002-7218-3745	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☒
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Baltazar A R, Santos F N, Leão de Sousa M, Moreira A P, Cunha J B, 2023. 2D LiDAR-based System for Canopy Sensing in Smart Spraying Applications. IEEE Access, 11, 43583-43591.
Publication	Leão de Sousa M, Gonçalves M, Fialho D, Ramos A, Lopes J P, Oliveira C M, De Melo-Abreu J P, 2022. Apple and Pear Model for Optimal Production and Fruit Grade in a Changing Environment. Horticulturae, 8, 873.
Publication	Leão de Sousa M, Gonçalves M, 2022. Effects of planting density on light interception and distribution, physiological and agronomic performance of 'Gala' apple orchards. Acta Horticulturae, 1346, 337-246.
Publication	Leão de Sousa M, Gonçalves M, 2022. Minimizing the effects of thermal stress by foliar nutrition, irrigation and kaolin applications in 'Gala' apple trees. Acta Horticulturae, 1333, 185-194.
Publication	Leão de Sousa M, De Melo e Abreu J P, 2015. Improved Training and Pruning Techniques Increased Productivity of 'Rocha' Pear. Acta Horticulturae, 1094, 213-222.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
DEMETER (Horizon 2020)	The Framework Programme for Research and Innovation. Digitising and transforming European industry and services: digital innovation hubs and platforms.
HIBA	Hub Iberia Agrotech: creación de un ecosistema Plurirregional para la Agrodigitalización através de los Digital Innovation Hub.
SFT-EDIH	Smart Sustainable Farms Foods and Trade European Digital Innovation Hub (Polo de inovação digital para o setor agroalimentar).
OPTIMAL	OPTIMAL (Optimização, Maçã, Alcobaça)
SPIN	Solução de Pulverização Inteligente

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Orchards	Orchards to test the technology
Sensors	Monitoring sensors to be used during the validation stages

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
880735945	SINOPSIS SOCIALLY INTEGRATED INNOVATION S.R.L.

Short name: SINOPSIS SOCIALLY INTEGRATED INNOVATION S.R.L.

Address

Street	CALEA VICTORIEI 155 TRONSON 6 BIROUL A30 B
Town	BUCURESTI
Postcode	010073
Country	Romania
Webpage	https://sinopsisinnovation.eu/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	11/09/2023 - yes
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Research and Innovation Development	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Calea Victoriei 155, Block D1, section 6	
Town	Bucharest	
Postcode	010073	
Country	Romania	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Dr

Gender

☐ Woman

☒ Man

☐ Non Binary

First name*

Georgios

Last name*

Chatzichristos

E-Mail*

george_hatzi@hotmail.com

Position in org.

CEO and director

Department

SINOPSIS SOCIALLY INTEGRATED INNOVATION S.R.L.

☒ Same as organisation name

☒ Same as proposing organisation's address

Street

CALEA VICTORIEI 155 TRONSON 6 BIROUL A30 BLOC D1 ETAJ 9

Town

BUCURESTI

Post code

010073

Country

Romania

Website

Please enter website

Phone

+40720414198

Phone 2

+xxx xxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Georgios	Chatzichristos	Man	Greece	george_hatzi@hotmail.com	Category B Senior resea	Leading	0000-0002-5004-5974	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☒
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☒
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Chatzichristos G, Olmedo L, Van Twuijver M, O'Shaughnessy M, 2023. Support Networks for Social Innovation: Insights from Ireland. Journal of Social Entrepreneurship. DOI: 10.1080/19420676.2023.2192229
Publication	Chatzichristos G, Perimenis A, 2022. Evaluating the social added value of LEADER: evidence from a marginalized rural region. Journal of Rural Studies, 94, 366-374. DOI: 10.1016/j.jrurstud.2022.07.016
Publication	Chatzichristos G, Hennebry B, 2021. Social innovation in rural governance: A comparative case study across the marginalised rural Europe. Journal of Rural Studies, 99, 193-203. DOI: 10.1016/j.jrurstud.2021.06.004
Publication	Chatzichristos G, Nagopoulos N, Poulimas M, 2021. Neo-endogenous rural development: a path towards reviving rural Europe. Rural Sociology, 86(4), 911-937. DOI: 10.1111/ruso.12380
Publication	Chatzichristos G, Nagopoulos N, 2021. Social entrepreneurs as institutional entrepreneurs: evidence from a comparative case study. Social Enterprise Journal. DOI: 10.1108/SEJ-12-2020-0137

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
RurAction (Horizon 2020)	Social Entrepreneurship in Structurally Weak Rural Regions: Analysing Innovative Troubleshooters in Action.
AURORA	AURORA will demonstrate transformative solutions to empower climate resilience towards public health status in the EU Boreal Region.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
883857890	GREEN & DIGITAL IDIOTIKI KEFALAIOUCHIKI ETAIREIA

Short name: GREEN & DIGITAL P.C.

Address

Street	ARCHIPELAGOUS 17
Town	GLYFADA
Postcode	165 61
Country	Greece
Webpage	green-digital.eu

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	11/01/2023 - yes
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

No department involved

Department name

Name of the department/institute carrying out the work.

☒ not applicable

☐ Same as proposing organisation's address

Street

Please enter street name and number.

Town

Please enter the name of the town.

Postcode

Area code.

Country

Please select a country

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Mr

Gender

Woman

Man

Non Binary

First name*

Christos

Last name*

Kiokakis

E-Mail*

info@green-digital.eu

Position in org.

Manager

Department

GREEN & DIGITAL IDIOTIKI KEFALAIOUCHIKI ETAIREIA

Same as organisation name

Same as proposing organisation's address

Street

ARCHIPELAGOUS 17

Town

GLYFADA

Post code

165 61

Country

Greece

Website

Please enter website

Phone

+xxx xxxxxxxxxx

Phone 2

+xxx xxxxxxxxxx

Other contact persons

First Name	Last Name	E-mail	Phone
Anastasia	Kannavou	kannavou@gmail.com	+xxx xxxxxxxxxx

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Anastasia	Kannavou	Woman	Greece	kannavou@gmail.com	Category C Recognised	Leading	0000-0001-5291-3547	Orcid ID
Mr	Lazaros	Kiokakis	Man	Greece	kiokakis@gmail.com	Category D First stage r	Team member	0009-0007-4930-4856	Orcid ID

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
STELLA (Horizon Europe)	WP Leader (Policy Analysis and Recommendations). STELLA is developing a real-time pest surveillance system with three components: an early warning system using forecasting models and IoT sensors, a pest detection system with drones, satellites, and a smartphone app, and a pest response system for data-driven recommendations. It will be tested in six pilots across five EU countries, targeting eight significant diseases. The project includes training for stakeholders in sustainable crop protection
PRUDENT (Horizon Europe)	WP Leader (Policy Analysis and Recommendations). The vision of the PRUDENT is to study and test 'green nudges' in various farming and forestry systems across Europe, using innovative web and mobile apps. The project seeks to develop social innovations, business models, and policy recommendations to facilitate this transition, involving stakeholders at multiple levels in co-creation activities for a comprehensive understanding and effective implementation of sustainable practices.
The Cohesion Policy through the investment obj...	Study entitled "The Cohesion Policy through the investment objective about a greener Europe" for the European Parliament. The aim of this study was to examine the Partnership Agreements of Greece and Cyprus, in the context of the cohesion policy, in order to identify the projects and actions planned to be implemented under investment objective 2: "A more green Europe", during the programming period 2021-2027.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
964514457	ASSOCIACAO DOS JOVENS AGRICULTORES DE PORTUGAL

Short name: AJAP

Address

Street	RUA DOM PEDRO V 108-2
Town	LISBOA
Postcode	1269-128
Country	Portugal
Webpage	ajap.pt

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	14/03/1983 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Technician	☐ not applicable
	☒ Same as proposing organisation's address	
Street	RUA DOM PEDRO V 108-2	
Town	LISBOA	
Postcode	1269-128	
Country	Portugal	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title	Dr	Gender	☐ Woman	☒ Man	☐ Non Binary
First name*	João	Last name*	Mira		
E-Mail*	joao.mira@ajap.pt				
Position in org.	Technician				
Department	Technician			☐	Same as organisation name
	☒ Same as proposing organisation's address				
Street	RUA DOM PEDRO V 108-2				
Town	LISBOA	Post code	1269-128		
Country	Portugal				
Website	www.ajap.pt				
Phone	+351 213244970	Phone 2	+xxx xxxxxxxxx		

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mrs	Maria José	Piçarra	Woman	Portugal	maria.picarra@ajap.pt		Team member		
Mrs	Célia	Sousa	Woman	Portugal	celia.sousa@ajap.pt		Team member		
Dr	João	Mira	Man	Portugal	joao.mira@ajap.pt		Leading		
Mrs	Teresa	Peixoto	Woman	Portugal	teresa.peixoto@ajap.pt		Team member		

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	Magazine "Jovens Agricultores" - Trimester publication with the aim of disseminating technical information to young farmers and the agricultural sector in Portugal
Other achievement	Terra 4.0 – TV Programme Young Farmers and Young Rural Entrepreneurs success stories and good practices, share and transfer qualifying information following articulation with opinion makers and researchers
Other achievement	Young Rural Entrepreneur: Renewal of the Rural World in Portugal - Publication that aims to highlight the figure of the JER, in the context of the Portuguese rural world
Other achievement	Manual "Impacts of Climate Change on the Rural World" Manual "Adaptation to Climate Change in the Rural World" Manual "Climate Change Mitigation for the Rural World"
Other achievement	Implementation of the Rural Laboratory and training activities: Circular Economy

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
LIVING LAB OF AGRICULTURE 4.0	Digital platform targeted to young farmers in Portugal, dedicated to the project Empowerment and Promotion of Competitiveness of Young Farmer and Young Rural Entrepreneur.
PAGE	Agricultural and Food Landscapes with Generations of Innovative Women.
EqualRural	The project's objective general to enhance the participation and representation of women in all agricultural sector positions, including those related to leadership and decision making.
TANGO Circular	Project focused on the Circular Economy.
FRUITGRADING	Developed under the Seventh Framework Programme (FP7) It consisted in developing non-invasive and non-destructive technology to measure fruit parameters, aiming at a quick classification according to different parameters (quality, ripeness, sugars, texture, etc.) for different markets/uses and choosing the best moments to harvest and efficiently manage storage time.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Mozambique Delegation	The Maputo Delegation of AJAP is a platform for activities to be carried out in various Mozambican provinces.
Bruxells Delegation	The Permanent Representation Delegation in Brussels aims at monitoring and raising the awareness of Community instances on the Common Agricultural Policy (CAP).
Portuguese Delegations	AJAP has a technical team installed in delegations in all districts of Portugal, composed of 40 field technicians who have a car, portable computer, scanner, among other computer equipment, guaranteeing local technical assistance.
Sao Tome and Principe Delegation	Technical assistance and professional training
Brazil Delegation	Technical assistance and professional training

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
932139349	Federación de Asociaciones de Mujeres Rurales- FADEMUR

Short name: Federación de Asociaciones de Mujeres Rurales- FADEMUR

Address

Street	AGUSTÍN DE BETANCOURT, 17- 3º
Town	MADRID
Postcode	28003
Country	Spain
Webpage	www.fademur.es

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	11/01/2022 - no
SME self-assessment	unknown
SME validation	unknown

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Research and Innovation Development	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Agustin De Betancourt, 17. 6º	
Town	Madrid	
Postcode	28003	
Country	Spain	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title	Ms	Gender	☒ Woman	☐ Man	☐ Non Binary
First name*	Paula	Last name*	Álvarez		
E-Mail*	palvarez@fademur.es				
Position in org.	Press officer				
Department	Federación de Asociaciones de Mujeres Rurales- FADEMUR			☒	Same as organisation name
	☒ Same as proposing organisation's address				
Street	AGUSTÍN DE BETANCOURT, 17- 3º				
Town	MADRID	Post code	28003		
Country	Spain				
Website	Please enter website				
Phone	+xxx xxxxxxxxxx	Phone 2	+xxx xxxxxxxxxx		

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Administrative forms

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☒
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☒
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Alimentos Solidarios	During the confinement decreed in 2020 due to Covid-19, FADEMUR detected the economic suffering of different groups. To mitigate the most direct effects of the pandemic on them, rural women began to develop the first donation campaign based on a strategic alliance with Red Eléctrica de España (REE). They called the initiative and, under this name, they made daily donations of products purchased to farms with difficulties in marketing their food.
Rural Proximity Services Cooperatives	Fademur has been developing a continuity program since 2006 thanks to the Ministry of Health, Social Services and Equality in its call for programs of general interest charged to the personal income tax allocation. This state program is developed in 14 Autonomous Communities and is based on the training of women in what are new employment opportunities in rural areas, related to dependent people and their care.
Cultivando Igualdad	The program arises after almost 20 years of experience working with rural women. FADEMUR has a long experience on special vulnerability that rural women have in the face of gender violence. To this end, FADEMUR developed the Cultivating Equality program, whose fundamental pillars are training and awareness-raising for both women and society in general and for professionals who care for victims and their environments.
Rural Platform for Agri-Food Supply and Services	The Rural Platform for Agri-Food Supply and Services of Fademur, Cashrural, was born as a pilot experience subsidized by the Ministry of the Environment and Rural and Marine Environment and co-financed by the EAFRD and addresses a series of needs detected by producers in the sector. agricultural, artisans and service cooperatives that work in rural areas.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☐ Yes

☒ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999695387	MARTEL GMBH

Short name: MARTEL GMBH

Address

Street	UBERLANDSTRASSE 111
Town	DUBENDORF
Postcode	8600
Country	Switzerland
Webpage	www.martel-innovate.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2020 - yes
SME self-assessment	31/12/2020 - yes
SME validation	31/12/2008 - yes

Administrative forms

Departments carrying out the proposed work

No department involved

Department name

Name of the department/institute carrying out the work.

☒ not applicable

☐ Same as proposing organisation's address

Street

Please enter street name and number.

Town

Please enter the name of the town.

Postcode

Area code.

Country

Please select a country

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	<i>Boosts the consolidation of the European Computing Continuum ecosystem building on the valuable activities and impact generated so far by within the existing EUCloudEdgeIoT ecosystem, while providing a forward-looking and bold vision in new areas and directions that have not been explored so far. Martel leads the outreach activities and contributes on the R&I roadmapping and ecosystems engagement.</i>
Service	<i>Martel is/has been leading and/or directly contributing to the organisation of several European and International events for promotion and impact creation of many research and innovation projects. This includes organization of events with international reach, several brokerage and networking events for the EC through CSAs Martel implements. Martel has extensive experience in organizing, promoting and conducting online workshops with live social media coverage and over 200 participants.</i>
Service	<i>Martel has been directly contributing to policy and strategic recommendations and road mapping activities (both technical and business) in several projects. This work has not only been valuable to establish contacts to a number of research communities and innovators in Europe and worldwide, but also to establish procedures and tools that can be used at the service of communication, community building and stakeholders engagement on a broad perspective within the project context.</i>
Service	<i>Martel has been in charge of technical contributions, authoring and editing of many deliverables, reports, newsletters, blogs and white papers produced within the context of EC projects Martel has been involved in. These deliverables are of various nature: some more technical, some more visionary, some very much geared at project, quality, risks, data management aspects, while others are dedicated to dissemination, communication, impact and marketing.</i>
Service	<i>Provide training programmes in various organisations and individual Innovators, related to EC funded instruments and tools ad also related to innovative technologies in the domains of Cloud Computing, AI and data analytics, IoT and smart cities, FIWARE usage and development, Standardization and harmonization of data models and Docker solutions. For all the training programmes Martel offers complete documentation and training materials that are designed to be completely modular within each field.</i>

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
INCODE (Horizon Europe)	<i>Provides solutions as key enablers for an open and trusted cloud-native programming platform poised to tame the emerging dynamicity of distributed and heterogeneous private edge infrastructures. Martel leads the communication, synergies creation and exploitation activities, the assessment of the benefits in terms of applicability and migration from existing cloud based or provider-locked solutions, and the Ethics management aspects.</i>
WeNet (Horizon 2020)	<i>Aims to develop the culture, science and engineering, methodologies, algorithms, social interaction protocols and an online platform which will empower machine mediated diversity-aware people interactions. Martel is strongly involved in the impact creation activities, supporting the Open Calls management and the dissemination and communication activities, as well as community building efforts.</i>
6G-XR	<i>6G-XR develops an evolvable experimental infrastructure for the duration of the SNS programme and beyond that covers demonstrating the performance of key B5G/6G candidate technologies, components, and architectures to keep the infrastructures valid now, in mid-term and in long-term. Martel leads the communication and dissemination activities of the projects as well the implementation of the planned Open Calls.</i>

Administrative forms

TERMINET (Horizon 2020)	Provide a novel next generation reference architecture based on cutting-edge technologies such as SDN, multiple-access edge computing, and virtualisation for next generation IoT, while introducing new, intelligent IoT devices for low-latency, market-oriented use cases. Martel leads the work on next generation IoT Ecosystem Expansion and Empowerment with the end goal to attract and involve Open Call 3rd parties in the project methodology.
NexusForum (Horizon Europe)	Boosts the consolidation of the European Computing Continuum ecosystem building on the valuable activities and impact generated so far by within the existing EUCloudEdgeIoT ecosystem, while providing a forward-looking and bold vision in new areas and directions that have not been explored so far. Martel leads the outreach activities and contributes on the R&I roadmapping and ecosystems engagement.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☐ Yes

☒ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
998802114	FORSCHUNGSINSTITUT FUR BIOLOGISCHEN LANDBAU STIFTUNG

Short name: FIBL RESEARCH INSTITUTE OF ORGANIC AGRICULTURE

Address

Street	ACKERSTRASSE 113
Town	FRICK
Postcode	5070
Country	Switzerland
Webpage	www.fibl.org

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019 - no
SME self-assessment	unknown
SME validation	25/08/2008 - no

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	Department Food System Sciences	☐ not applicable
	☒ Same as proposing organisation's address	
Street	ACKERSTRASSE 113	
Town	FRICK	
Postcode	5070	
Country	Switzerland	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Olivier	Ejderyan	Man	Switzerland	olivier.ejderyan@fibl.org	Category B Senior resea	Leading	0000-0002-2382-2758	Orcid ID
Dr	Sylvain	Quiédeville	Man	France	sylvain.quiedeville@fibl.org	Category B Senior resea	Team member	0000-0001-8746-5086	Orcid ID
Dr	Raphael	Charles	Man	Switzerland	Raphael.charles@fibl.org	Category A Top grade re	Team member	0000-0001-6563-7764	Orcid ID
Ms	Beatrice	Steinemann	Woman		Beatrice.Steinemann@fibl.org	Category B Senior resea	Team member		

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Pearce B J, Ejderyan O, 2020. Joint problem framing as reflexive practice: Honing a transdisciplinary skill. <i>Sustainability Science</i> , 15 (3), 683-698. https://doi.org/10.1007/s11625-019-00744-2
Publication	Fantasia S, Mota M, Lamy F, Marchand D, Sonnard R, Delabays N, Zufferey V, Steinemann B, Degruene F, Heger T, Rienth M, 2022. The effects of alternative herbicide-free cover cropping systems on soil health, vine performance, berry quality and vineyard biodiversity in a climate change scenario in Switzerland. In: <i>Proceedings of the 14th International Terroir congress and 2nd ClimWine symposium Terclim 2022. July 3-8, 2022, Bordeaux, France.</i>
Publication	Devaux-Spatarakis A, Quiédeville S, 2018. Evaluating the impacts of agricultural innovations. In: Faure G, Chiffolleau Y, Goulet F, Temple L, Touzard J M (Eds.) <i>Innovation and development in agricultural and food systems. Éditions Quae, Versailles Cedex, France, chapter 14, 251-268.</i>
Publication	Ejderyan O, Quiédeville S, 2022. Promouvoir la digitalisation en milieu rural à l'aide de living labs: l'exemple du projet DESIRA. <i>Economie et Société Rurales</i> , 117-119.
Publication	Alvial-Palavicino C, Ejderyan O, Vienni-Baptista B, 2023. Building pathways between transdisciplinarity and transformation: Lessons from practice. <i>GAIA - Ecological Perspectives for Science and Society</i> , 32(1), 131-137. doi:10.14512/gaia.32.1.10

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
DESIRA (Horizon 2020)	The DESIRA project aims to improve the capacity of society and political bodies to respond to the challenges that digitalisation generates in agriculture, forestry and rural areas. The project approached all activities considering digitalisation as a process of social transformation driven by digital technologies.
WATSON (Horizon Europe)	WATSON will develop a set of tools and systems that can detect and prevent fraudulent practices across the entire food supply chain, increasing transparency through improved track-and-trace mechanisms for accurate, time-relevant and untampered information on the food product throughout its whole journey.
PATH2DEA (Horizon Europe)	PATH2DEA is committed to unlocking digitalisation's catalysing power to foster European agriculture's transition towards enhanced sustainability. It builds on farmers' competences and views and match them with the rich repertoire of digital solutions for agriculture, aimed at tailoring digital technologies to users' needs and fostering wide-range adoption of digital agroecological farming in the EU and associated countries.
Biosonnenblumenöl with Migros Aid Fund	Wertschöpfungskette "Biosonnenblumenöl" aus der Schweiz mit neuen Impulsen entwickeln und strategisch ausbauen (Migros Aid Fund). The aim of the project is to significantly expand the scope of organic sunflower cultivation by the testing and implementation of new cultivation methods for weed management. Testing a robot-assisted method for the late sowing of diversity-enhancing undersown and artificial meadow stands in sunflowers.
Use of robots for efficient weed control	Use of robots for efficient weed control (Federal Office for Agriculture). This project aimed to identify and verify new methods (such as heat, electricity, sandblasting technology or mechanical tools) that can be automated with the aid of autonomous robots.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure or equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes ☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

PIC	Legal name
999973971	ECOLE POLYTECHNIQUE FEDERALE DE LAUSANNE

Short name: EPFL

Address

Street	BATIMENT CE 3316 STATION 1
Town	LAUSANNE
Postcode	1015
Country	Switzerland
Webpage	www.epfl.ch

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	17/05/2016 - no
SME self-assessment	04/10/1991 - no
SME validation	04/10/1991 - no

Administrative forms

Departments carrying out the proposed work

Department 1

Department name	School of Engineering, Institute of Mechanical Engineering	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Route Cantonal	
Town	Lausanne	
Postcode	1015	
Country	Switzerland	

Links with other participants

Type of link	Participant
--------------	-------------

Administrative forms

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Nana	Obayashi	Woman	Japan	Nana.obayashi@epfl.ch	Category A Top grade re	Team member		
Mr	Max	Polzin	Man	Germany	Max.polzin@epfl.ch	Category A Top grade re	Team member		

Administrative forms

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

Administrative forms

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	Junge K, Pires C, Hughes J, 2023. Lab2Field transfer of a robotic raspberry harvester enabled by a soft sensorized physical twin. <i>Communications Engineering</i> , 2(1), 40.
Publication	Birrell S, Hughes J, Cai J Y, Iida F, 2020. A field-tested robotic harvesting system for iceberg lettuce. <i>Journal of Field Robotics</i> , 37(2), 225-245.
Publication	Armanini C, Junge K, Johnson P H, Whitfield C, Renda F, Calisti M, Hughes J, 2024. Soft robotics for farm to fork: applications in agriculture & farming. <i>Bioinspiration & Biomimetics</i> .
Publication	Hughes J, Scimeca L, Ifrim I, Maiolino P, Iida F, 2018. Achieving robotically peeled lettuce. <i>IEEE Robotics and Automation Letters</i> , 3(4), 4337-4342.
Publication	Polzin M, Hughes J, 2023. Automation in steep terrain agriculture: an optimal controller to prevent tipping and slipping of tethered robots on slopes. <i>Advanced Robotics</i> , 37(15), 987-998.

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Robotic raspberry harvesting	Robotic raspberry harvesting utilizing a robotic real world simulator for demonstration and learning of the compliant control.
VEGEBOT	Project Vegebot, developing a robotic lettuce harvesting system, including the requisite computer vision, learning and manipulators.
Steep slope viticulture	Steep slope viticulture in the Swiss Lavaux region, developing novel robots and controllers for these extreme environments.
CROSS	Review and analysis of the role of soft robotic technologies in the Agri-food industry and the accompanying ethical and societal issues, as a part of the CROSS project.
Physical plant simulation	Physical plant simulator to evaluate and simulate crop conditions for different geonotypes/phenotypes.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Panther Robotic Base	Robotic base which can operate in steep and complex terrains
UV and hyperspectral cameras	Cameras suited for agricultural tasks and operations
Soft robotic arms	Lightweight, 1m length soft arms which can be deployed for robotic operation

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below?

☒ Yes

☐ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas to be covered** and addressed via concrete measures and targets are:
 - o work-life balance and organisational culture;
 - o gender balance in leadership and decision-making;
 - o gender equality in recruitment and career progression;
 - o integration of the gender dimension into research and teaching content;
 - o measures against gender-based violence including sexual harassment.

Administrative forms

Proposal ID 101182954

Acronym AGROBOOST

3 - Budget

No.	Name of beneficiary	Country	Role	Personnel costs/€	Subcontracting costs/€	Purchase costs - Travel and subsistence/€	Purchase costs - Equipment/€	Purchase costs - Other goods, works and services/€	Financial support to third parties/€ (actual cost)	Internally invoiced goods and services/€ (Unit costs-usual accounting practices)	Indirect costs/€	Total eligible costs	Funding rate	Maximum EU contribution to eligible costs	Requested EU contribution to eligible costs/€	Max grant amount	Income generated by the action	Financial contributions	Own resources	Total estimated income
1	University College Dublin,	IE	Coordinator	464 040	0	17 000	30 000	19 000	500 000	0	132 510.00	1 162 550.00	100	1 162 550.00	1 162 550.00	1 162 550.00	0.00	0.00	0.00	1 162 550.00
2	Ethnicon Metsovion Polytechnion	EL	Partner	255 300	0	15 000	0	9 700	0	0	70 000.00	350 000.00	100	350 000.00	350 000.00	350 000.00	0.00	0.00	0.00	350 000.00
3	Netcompany-intrasoft Sa	LU	Partner	283 500	0	17 000	0	6 500	0	0	76 750.00	383 750.00	100	383 750.00	383 750.00	383 750.00	0.00	0.00	0.00	383 750.00
4	Wisefour Innovation And	CY	Partner	172 000	0	17 000	0	4 000	0	0	48 250.00	241 250.00	100	241 250.00	241 250.00	241 250.00	0.00	0.00	0.00	241 250.00
5	Katholieke Universiteit Leuven	BE	Partner	258 281	0	17 000	0	20 000	0	0	73 820.25	369 101.25	100	369 101.25	369 101.25	369 101.25	0.00	0.00	0.00	369 101.25
6	Heriot-watt University	UK	Partner	144 946	0	17 000	0	0	0	0	40 486.50	202 432.50	100	202 432.50	202 432.50	202 432.50	0.00	0.00	0.00	202 432.50
7	Geoponiko Panepistimion Athinon	EL	Partner	216 300	0	20 000	0	5 000	0	0	60 325.00	301 625.00	100	301 625.00	301 625.00	301 625.00	0.00	0.00	0.00	301 625.00
8	Inesc Tec - Instituto De Engenhariade	PT	Partner	280 000	0	17 000	1 000	90 000	0	0	97 000.00	485 000.00	100	485 000.00	485 000.00	485 000.00	0.00	0.00	0.00	485 000.00
9	University Of Lincoln	UK	Partner	251 576	0	15 600	3 240	8 640	0	0	69 764.00	348 820.00	100	348 820.00	348 820.00	348 820.00	0.00	0.00	0.00	348 820.00
10	Danmarks Tekniske Universitet	DK	Partner	255 000	0	20 000	5 000	11 000	0	0	72 750.00	363 750.00	100	363 750.00	363 750.00	363 750.00	0.00	0.00	0.00	363 750.00

Administrative forms

Proposal ID **101182954**
Acronym **AGROBOOST**

11	Instituto Nacional De Investigacao	PT	Partner	55 200	2 000	5 000	0	3 000	0	0	15 800.00	81 000.00	100	81 000.00	81 000.00	81 000.00	0.00	0.00	0.00	81 000.00
12	Sinopsis Socially Integrated	RO	Partner	176 000	0	17 000	0	5 000	0	0	49 500.00	247 500.00	100	247 500.00	247 500.00	247 500.00	0.00	0.00	0.00	247 500.00
13	Green & Digital Idiotiki Kefalaiouschiki	EL	Partner	156 400	0	17 000	0	0	0	0	43 350.00	216 750.00	100	216 750.00	216 750.00	216 750.00	0.00	0.00	0.00	216 750.00
14	Associacao Dos Jovens Agricultores	PT	Partner	95 007	0	7 000	0	2 000	0	0	26 001.75	130 008.75	100	130 008.75	130 008.75	130 008.75	0.00	0.00	0.00	130 008.75
15	Federación De Asociaciones De Mujeres	ES	Partner	81 600	0	11 000	0	0	0	0	23 150.00	115 750.00	100	115 750.00	115 750.00	115 750.00	0.00	0.00	0.00	115 750.00
16	Martel Gmbh	CH	Associated	0	0	0	0	0	0	0	0.00	0.00	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00
17	Forschungsinsitut Fur Biologischen	CH	Associated	0	0	0	0	0	0	0	0.00	0.00	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00
18	Ecole Polytechnique Federale De	CH	Associated	0	0	0	0	0	0	0	0.00	0.00	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	TOTAL			3 145 150	2 000	229 600	39 240	183 840	500 000	0	899 457.50	4 999 287.50		4 999 287.50	4 999 287.50	4 999 287.50	0.00	0.00	0.00	4 999 287.50

Administrative forms

Proposal ID 101182954
Acronym AGROBOOST

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos		Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	☐ Yes ☒ No	
Does this activity involve the use of human embryos?	☐ Yes ☒ No	
2. Humans		Page
Does this activity involve human participants?	☐ Yes ☒ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	☐ Yes ☒ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014) ? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	☐ Yes ☒ No	
3. Human Cells / Tissues (not covered by section 1)		Page
Does this activity involve the use of human cells or tissues?	☐ Yes ☒ No	
4. Personal Data		Page
Does this activity involve processing of personal data?	☐ Yes ☒ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	☐ Yes ☒ No	
Is it planned to export personal data from the EU to non-EU countries?	☐ Yes ☒ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country?	☐ Yes ☒ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	☐ Yes ☒ No	
5. Animals		Page
Does this activity involve animals?	☐ Yes ☒ No	
6. Non-EU Countries		Page
Will some of the activities be carried out in non-EU countries?	☒ Yes ☐ No	39-40
UK and Switzerland will perform two pilot validation campaigns in WP5.		
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	☐ Yes ☒ No	
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	☐ Yes ☒ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	☐ Yes ☒ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	☐ Yes ☒ No	

Administrative forms

Proposal ID 101182954
Acronym AGROBOOST

Does this activity involve [low and/or lower middle income countries](#), (if yes, detail the benefit-sharing actions planned in the self-assessment) ☐ Yes ☒ No

Could the situation in the country put the individuals taking part in the activity at risk? ☐ Yes ☒ No

7. Environment, Health and Safety

Page

Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ? ☐ Yes ☒ No

Does this activity deal with endangered fauna and/or flora / protected areas? ☐ Yes ☒ No

Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ? ☐ Yes ☒ No

8. Artificial Intelligence

Page

Does this activity involve the development, deployment and/or use of Artificial Intelligence-based systems? ☒ Yes ☐ No 14

9. Other Ethics Issues

Page

Are there any other ethics issues that should be taken into consideration? ☐ Yes ☒ No

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines [How to Complete your Ethics Self-Assessment](#) ☐

Administrative forms

Proposal ID **101182954**

Acronym **AGROBOOST**

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

The project involves carrying out data collection, including personal data and metadata (in the context of the validation campaigns and the surveys). All processing of personal data will be conducted in accordance with the provisions of: a) the GDPR (Regulation (EU) 2016/679), b) the Universal Declaration of Human Rights and the Convention 108 for the Protection of Individuals with Regard to Automatic Processing of Personal Data, and c) the national laws applying its provisions, including those governing the acquisition of valid consent. Data managed during the project will be processed only under the following preconditions which need to be met: (i) When the data subject has given her/his consent; (ii) When the processing is necessary for the performance of or the entering into a contract; (iii) When processing is necessary for compliance with a legal obligation; (iv) When processing is necessary in order to protect the vital interests of the data subject. Personal data managed within the project will be anonymized and stored in a form which does not permit identification of users (Task 1.4, page 35).

Remaining characters

3881

Compliance with ethical principles and relevant legislations

The project involves carrying out data collection, including personal data and metadata (in the context of the validation campaigns and the surveys). All processing of personal data will be conducted in accordance with the provisions of: a) the GDPR (Regulation (EU) 2016/679), b) the Universal Declaration of Human Rights and the Convention 108 for the Protection of Individuals with Regard to Automatic Processing of Personal Data, and c) the national laws applying its provisions, including those governing the acquisition of valid consent. Data managed during the project will be processed only under the following preconditions which need to be met: (i) When the data subject has given her/his consent; (ii) When the processing is necessary for the performance of or the entering into a contract; (iii) When processing is necessary for compliance with a legal obligation; (iv) When processing is necessary in order to protect the vital interests of the data subject. Personal data managed within the project will be anonymized and stored in a form which does not permit identification of users (Task 1.4, page 35 and Task 2.4, page 36)

Remaining characters

3860

Administrative forms

Proposal ID 101182954
Acronym AGROBOOST

Security issues table

1. EU Classified Information (EUCI) ²		Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	☐ Yes ☒ No	
Does this activity involve non-EU countries which need to have access to EUCI?	☐ Yes ☒ No	
2. Misuse		Page
Does this activity have the potential for misuse of results?	☐ Yes ☒ No	
3. Other Security Issues		Page
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	☐ Yes ☒ No	

Security self-assessment

Please specify: (Maximum number of characters allowed: 5000)

Remaining characters 5000

²According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, “European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States”.

³Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

⁴EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture

Type of Action: Research and Innovation Action (RIA)

AGROB00ST Agricultural Ground-breaking sOlutions based on roBotics and augmented reality for boOsting sOcial sustainability and competitivenessS while increasing the safeTy of workers

List of participants #@APP-FORM-HERIAIA@#

PART#	PARTICIPANT ORGANISATION NAME	TYPE	ACRONYM	COUNTRY
1 (Co)	UNIVERSITY COLLEGE DUBLIN	RTO	UCD	IE
2	NATIONAL TECHNICAL UNIVERSITY OF ATHENS	RTO	NTU	EL
3	KATHOLIEKE UNIVERSITEIT LEUVEN	RTO	KUL	BE
4	ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE	RTO	EPF	CH
5	AGRICULTURAL UNIVERSITY OF ATHENS	RTO	AUA	EL
6	INSTITUTO DE ENGENHARIA DE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIÊNCIA	RTO	INE	PT
7	UNIVERSITY OF LINCOLN	RTO	LNC	UK
8	NATIONAL INSTITUTE FOR AGRARIAN AND VETERINARIAN RESEARCH	RTO	NVR	PT
9	DANMARKS TEKNISKE UNIVERSITET	RTO	DTU	DK
10	HERIOT-WATT UNIVERSITY	RTO	HWU	UK
11	FORSCHUNGSINSTITUT FÜR BIOLOGISCHEN LANDBAU STIFTUNG	RTO	FIB	CH
12	NETCOMPANY-INTRASOFT S.A.	LE	NCI	LU
13	SINOPSIS SOCIALLY INTEGRATED INNOVATION SRL	SME	SSI	RO
14	WISEFOUR INNOVATION AND DEVELOPMENT ORGANISATION LTD	SME	WIS	CY
15	MARTEL INNOVATE GMBH	SME	MAR	CH
16	GREEN & DIGITAL IKE	SME	G&D	EL
17	FEDERACIÓN DE ASOCIACIONES DE MUJERES RURALES	NGO	FAD	ES
18	ASSOCIAÇÃO DOS JOVENS AGRICULTORES DE PORTUGAL	NGO	AJP	PT

1 Excellence #@REL-EVA-RE@#

In several branches and regions, the agricultural sector faces difficulties due to lack and/or aging of workforce, making it imperative to explore innovative technologies. Managing just 11% of all farm holdings in the European Union (EU), farmers aged under 40 face a considerable challenge, with female young farmers representing a mere 3%¹. Encouraging a greater number of young individuals to enter the farming profession proves to be a significant hurdle. Agriculture is currently employing half the global workforce, being among the top three hazardous sectors, with 170,000 annual fatalities or double the risk of other industries according to the International Labour Organization (ILO)². In fact, despite declining fatal accident rates elsewhere, agriculture maintains high mortality rates, and workers face severe injuries from machinery and pesticides. Under-reporting suggests the actual occupational health challenges may be worse than official data show. Preliminary EU data for 2023 indicates a 6.6% year-on-year decline in the EU's agricultural labour productivity index, following previous growth from 2019 to 2022. This drop is attributed to a 7.9% decrease in factor income and a further 1.4% reduction in the volume of agricultural labour³. All these statistics clearly show that there is a great need for improving working conditions in the agriculture sector. Digital and data technologies can play a crucial role in revolutionizing agriculture within the European Union (EU), offering solutions to challenges such as labour shortages, productivity and enhancing overall working conditions. The agricultural sector can not only mitigate the impact of labour shortages but also make farm-related jobs more appealing, especially to younger generations and women through the deployment of digital and data technologies. Technologies such as robotics and Augmented Reality (AR) can offer multifaceted benefits, ranging from increased safety for workers to mitigating the challenges of labour shortages and improving the overall productivity. Robotics can offer automation of tasks while AR solutions can provide numerous opportunities to facilitate the use of digital tools. AR can help train and educate farmers, providing real-time instructions and contextual information, for example by overlaying information about crops, animals and machinery onto the farmers' field of vision, enabling them to make more informed decisions. The implementation of precision agriculture and automation in agriculture is essential to overcome challenges such as labour shortages, enhance working conditions, and improve overall efficiency. Precision agriculture leverages advanced technologies like GPS, sensors, robotics, Artificial Intelligence and data analytics to optimize the use of resources, ensuring precise application of water, fertilizers, and pesticides. Automation, through robotics and machinery, addresses labour shortages by automating repetitive and strenuous tasks, freeing up human workers for more strategic roles. This shift allows human workers to focus on more strategic and complex aspects of farming, contributing to a safer and healthier work environment. AR solutions can also

¹ https://agriculture.ec.europa.eu/common-agricultural-policy/income-support/young-farmers_en

² https://www.ilo.org/safework/areasofwork/hazardous-work/WCMS_110188/lang--en/index.htm

³ <https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20231215-4>

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

provide remote assistance, a crucial aspect for businesses in remote areas, particularly those in rural regions. The integration of such technologies can greatly assist in alleviating the drudgery often associated with agricultural work as they can take on monotonous and strenuous tasks, freeing up human labour for more intellectually engaging activities. The use of such technologies can also contribute to lower environmental impacts and production costs, aligning with sustainable practices and making agriculture more environmentally friendly and economically viable. In the face of a shortage of agricultural labour, particularly in certain EU regions, the integration of such technologies becomes a strategic solution to fill the gaps left by labour shortages.

AGROBOOST is a pilot driven, multi-disciplinary project that aims to transform the agricultural sector into a more attractive, sustainable and efficient industry by harnessing the power of advanced robotics, AR, AI and automation revealing the societal and economic benefits for all the people working in agriculture, especially women, young professionals and farmers with moving disabilities. Through the development of cutting-edge, ground-breaking solutions, the project will improve the safety and appeal of agricultural work, reduce environmental impacts, enhance productivity and address labour shortages. **AGROBOOST** will also emphasize on the development of flexible business models, concrete system analysis tools and training curricula to ensure the widespread adoption and the rapid uptake of these technology innovations. Six distinct and diverse pilot validation campaigns will be demonstrated, covering multiple agricultural practices and sectors including (1) tree pruning and flower thinning, (2) mushroom selective harvesting, (3) automated weeding, (4) strawberries picking, (5) animal husbandry and livestock farm management and (6) crop monitoring. Through the engagement of a diverse range of stakeholders and the incorporation of social sciences perspectives, **AGROBOOST** will create a more inclusive and forward-looking agricultural landscape, securing a promising future for the industry.

AGROBOOST will focus its research efforts along 5 complementary pillars: (a) Innovation and Technology Integration, undertaking the development and seamless integration of cutting-edge technologies in the agricultural domain; (b) Sustainability and Environmental Responsibility, emphasizing on environmental sustainability in order to minimize the ecological footprint associated with agricultural practices; (c) Inclusive and Socially Responsible Agriculture, with strategic focus on cultivating a socially inclusive and responsible agricultural landscape; (d) Business Model Development and System Analysis, developing robust business models and a unique system analysis tool to analyse the consequences for farmers and rural communities; (e) Adherence to Legal and Ethical Standards, aligning the project activities with the pertinent EU legislation along with the ethical standards and the societal values.

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

1.1.1 Objectives

Obj.1 – DESIGN and DEVELOP ground-breaking solutions that enhance working conditions and safety of workers in unhealthy working environments, lower environmental impacts and production costs while increasing product quality and safeguarding inclusivity in agriculture.

Pertinence to the work programme topic: From the call: “It will therefore also directly and/ or indirectly contribute to the enhancement of the sustainability performance of the sector, including social sustainability, and competitiveness in agriculture through research and innovation which will support the further deployment of digital and data technologies as key enablers [...] Enhanced working conditions in agriculture (including increased safety of workers and reduced drudgery) through innovative digital solutions exploiting the potential of augmented reality [...] Lowered environmental impacts and productions costs and increased product quality in and through the use of digital technologies, through robotics and augmented reality in particular [...] Development of robotic solutions for tasks, for which there is a high interest/ need to support and/ or replace the human work force, not only because of an interest to improve productivity, but to ensure production in an environmentally and socially sustainable way [...]”

Means to Achieve: **AGROBOOST** undertakes a multifaceted approach through its tasks to fulfil the objective of designing and developing groundbreaking solutions for agriculture. Socioeconomics methodologies and tools will be employed to understand stakeholder needs, analyse working conditions, and address the legal and ethical compliance of the developed technologies, while ensuring the inclusivity of the solutions [WP2]. AR and VR as well as robotic solutions will be developed for tasks like pruning, thinning, weeding, picking, harvesting, livestock production, health, welfare, as well as training, directly contributing to enhanced working conditions and safety [WP3]. The project will leverage automation and AI to foster environmentally and socially responsible practices, integrating Deep Learning, Machine Learning and human-robot collaboration [WP4]. Additionally, it will involve the implementation of six pilot validation campaigns across diverse applications, ensuring real-world testing and user acceptance evaluation [WP5].

Measurable Outcomes, Success Indicators and Verification Means

- Accidents reduction $\geq 20\%$ showcasing a tangible improvement in workplace safety and a proactive approach to risk mitigation – Comparison before and after the deployment of the proposed emerging technologies in agriculture [D5.7].
- Reduction by $\geq 20\%$ the absenteeism related to farmers’ injuries or health issues caused by unsafe working conditions – Comparison before and after the deployment of proposed emerging technologies in the agricultural domain [D5.7].
- Improvement by $\geq 30\%$ the workplace ergonomics – Evaluation and quantification of improvements in workplace ergonomics through the deployment of AR and robotic technologies, reducing physical strain and discomfort for workers that participate in the pilot campaigns [D2.3].
- Reduction by $\geq 20\%$ of the water resources used in agricultural activities, $\geq 20\%$ in overall carbon footprint associated with farming operations and $\geq 40\%$ in the use of fertilizers and pesticides per unit of production – Thorough assessment through the collection of data during and after the deployment of the novel technologies and comparative

analysis providing insights into their effectiveness in lowering the overall environmental impact of agricultural practices [D5.7].

- Reduction by $\geq 30\%$ in the time needed for specific agricultural tasks (e.g., harvesting, weeding, etc.) through robotics and automation – Field trials and comparative analyses before and after the implementation of the proposed innovative solutions. Time-motion studies, equipped with precision timekeeping tools, to accurately capture the duration of each task under traditional methods and compare them with the time taken when using robotics, AR and automation [D5.7].
- Increase $\geq 15\%$ in the overall farm profitability, considering the impact of cost reductions and improved product quality but also the affordability of the proposed solutions in a 5-year lifespan – Detailed analysis of operational expenses, comparing pre-implementation costs with those incurred after adopting the proposed technologies. Improvements in product quality, e.g., increased yield and enhanced market value, factored into the revenue side of the equation, while also considering the affordability aspect (initial investment, maintenance costs, accessibility of the proposed solutions for farmers, etc.) [D5.7].
- Increase by $\geq 35\%$ of the inclusivity of the developed technologies across different demographic groups, with a specific focus on youth, women, and persons with disabilities – Surveys and feedback from users, targeting youth, women, and persons with disabilities and assessment of their satisfaction levels and perceived inclusivity due to the use of the proposed technologies. Collection of success stories and testimonials from users within targeted demographic groups, highlighting their positive experiences [D2.3].

Reference to Workplan: WP2, WP3, WP4, WP5

Obj.2 – REVITALIZE working conditions by developing integrated AR and robotic solutions, ensuring increased safety, failure avoidance and improved performance, with a specific focus on addressing unhealthy working conditions.

Pertinence to the work programme topic: From the call: “Development of augmented reality-based solutions to improve working conditions, safety and failure avoidance, [...] Development of robotic solutions to improve unhealthy working conditions, where applicable [...] Development of robotic solutions for tasks, for which there is a high interest/ need to support and/ or replace the human work force, not only because of an interest to improve productivity, but to ensure production in an environmentally and socially sustainable way [...] Development of a tool for system analyses of the consequences for farmers and rural communities of enhancing working conditions through automation and of replacing human work force with robotic systems [...]”

Means to Achieve: AGROBOOST will focus on replacing unattractive and risky agricultural tasks that are very challenging for human labour. Several activities are foreseen ensuring that the developed solutions align with the real-world challenges faced by agricultural workers [T2.1–T2.4] building upon a thorough user needs analysis that will assist in the deep understanding of working conditions. The development of autonomous robotic systems and AR-based solutions for tasks like pruning, thinning, weeding, harvesting, livestock management and training, will directly address the global concerns for increased safety, failures avoidance and enhanced overall performance in agriculture. The integration of AR and robotics [T3.1–T3.5] will provide innovative tools to mitigate the challenges posed by unhealthy working conditions. Their effectiveness will be validated during the pilot validation campaigns, allowing for real-world testing of these integrated solutions in diverse agricultural settings [T5.4–5.10].

Measurable Outcomes, Success Indicators and Verification Means

- Deviation from intended action of robotic tasks $\leq 5\%$, ensuring that the executed actions closely align with the intended path or action – Rigorous testing, feedback and continuous calibration of robotic systems will be performed detailed in the respective reports [D3.3].
- Energy consumption reduction per unit of work performed $\geq 20\%$, due to the use of advanced robotic sensors that increase precision and accuracy. – Report including baseline measurements of energy consumption per unit of work before and after the implementation of advanced robotic technologies [D3.3].
- Latency in synchronizing real-time data ≤ 100 ms, for seamless integration of augmented reality data with the physical environment – Report including baseline measurements of energy consumption per unit of work before and after the implementation of advanced robotic technologies [D3.3].
- User interaction latency ≤ 50 ms, for exceptional responsiveness, targeting users experience minimal delays when interacting with the augmented reality system. – Implementation of timestamping mechanisms and use of built-in tools for latency measurement. [D3.3].
- System failure rates and downtime reduction $\geq 25\%$ enhancing the reliability and operational efficiency of AR and robotic solutions in agriculture – Documentation on the number of system failures based on historical data along with continuous monitoring mechanisms to track system failures and downtime events [D3.3].

Reference to Workplan: WP2, WP3, WP5

Obj.3 – EMPOWER sustainable agriculture through automation and enhanced AI capabilities by developing robotic solutions for tasks crucial to supporting or replacing human labour, ensuring environmentally and socially responsible practices and optimizing performance and efficiency of digital technologies in agriculture.

Pertinence to the work programme topic: From the call: “Reduced share of risky or unattractive actions/tasks to be performed by workers through automation-based solutions [...] Mitigated shortage of work force in agriculture in some sub-sectors through automation-based solutions [...] Automation is increasingly used in agriculture; frequently, the cost-effectiveness of innovative digital and data technologies still presents a bottleneck to their use in the sector, particularly in fields where their application is not primarily relevant for increasing process efficiency and effectiveness”

Means to Achieve: AGROBOOST will exploit the potential of robust, trustworthy and fair AI techniques such as Deep Learning and Machine Learning algorithms for seamless human-robot collaboration to enhance environmental and social responsibility in agricultural practices, contributing to the development of automated solutions that support or replace human labour while ensuring sustainability [T4.1–T4.5]. Specific emphasis will be given to the use of these technologies towards reskilling initiatives and workforce training programs [T6.4] highlighting the potential impact of automation on job displacement and guaranteeing a smooth transition for affected workers. Last, the core objective of the proposed AI-fuelled robotic solutions is placed around the replacement of unattractive and risky tasks that are very challenging for human labour. These automated and AI-enhanced solutions will be extensively tested in diverse agricultural settings, contributing to the optimization of performance through digital technologies [T5.1-T5.10].

Measurable Outcomes, Success Indicators and Verification Means

- Overall increase of tasks automation $\geq 30\%$ reducing reliance on human labour for tasks that are very risky and/or unattractive. – Report on the methodology, data collection procedures and analysis technique for the level of automation achieved before and after the use of AI-based robotic solutions [D4.3].
- System reliability and operational uptime $\geq 90\%$, ensuring that automated systems consistently perform tasks with minimal downtime. – Establishment of a baseline reliability before any improvements are made, considering historical performance, frequency of breakdowns, duration of downtime, etc. and deployment of monitoring mechanisms to track the performance of automated systems in real-time [D4.3].
- Algorithm accuracy $\geq 95\%$ for all AI algorithms used in agricultural decision-making. – Documentation of the entire evaluation process, including datasets used, training methodologies, evaluation results, etc. that will report the achieved accuracy and any adjustments made during the iterative improvement process. [D4.3].
- User acceptance rate $\geq 85\%$ for AI-generated recommendations – Documentation of user feedback, survey results, etc. reporting on the user acceptance rate and the factors influencing acceptance or resistance [D4.3].

Reference to Workplan: WP2, WP4, WP5

Obj.4 – VALIDATE and DEMONSTRATE the effectiveness of the proposed technology innovations through six validation campaigns in real life settings, in five different European regions with the active engagement of relevant agricultural stakeholders and third party projects, promoting the rapid adoption of the developed technologies.

Pertinence to the work programme topic: From the call: “Digital and data technologies can facilitate the work in agriculture, enhance working conditions and mitigate the challenge of a lack of work force, by which some branches and regions are affected [...] Digital and data technologies can increase the effectiveness and efficiency of applications, including for instance through a higher level of precision, and thus increase the sustainability and competitiveness of the sector [...] Automation is increasingly used in agriculture; frequently, the cost-effectiveness of innovative digital and data technologies still presents a bottleneck to their use in the sector [...] Robotics tasks to be fostered might be directly related to agricultural production, such as harvesting, weeding, crop monitoring, animal husbandry or indirectly related, such as logistics/ farm management [...] Projects are expected to develop training material allowing the targeted end users and multipliers to easily deploy and promote the new technologies.”

Means to Achieve: AGROBOOST will demonstrate the effectiveness of its proposed technologies through six validation campaigns in real-life settings across six different EU regions [T5.1-T5.10]. Each pilot will address a unique yet complementary scenario, considering different cultural environments and diverse operational procedures, and has been selected for its high societal, environmental and financial impact on the EU and the agricultural ecosystem. Benchmarking activities will take place before the pilots’ deployment to familiarise the end-users with the overall framework and assess the progress of the developed technologies [T5.11]. Replicable activities (in different countries), user acceptance and lessons learnt derived from the pilot phase will be documented, providing material to the policy recommendations activities.

Measurable Outcomes, Success Indicators and Verification Means

- ≥ 6 validation campaigns with ≥ 3 use case scenarios for each one of them with ≥ 2 benchmarking activities in each pilot – Based on the co-creation method and incorporating the multi-stakeholder approach, the use cases will start early on the project, whereas the scenarios will be drafted in order to meet the needs and requirements of the stakeholders, demonstrating the high impact of the proposed solutions [D5.7].
- ≥ 8 successful validations through the open calls, as part of the project’s funding and implementation strategy, involving EU-based entities and organizations selected based on specific criteria set by the open call committee and managed through internal evaluation, contracting, and mentoring until project completion [D5.7].
- ≥ 500 external stakeholders participating actively and providing feedback on the pilot validation campaigns – A registration list will be provided with all the external participants in each pilot organised by the project in the 5 different EU countries along with the report on feedback from the participants [D5.7].
- $\geq 95\%$ of the external stakeholders participating in the project’s demonstration and training activities provide positive feedback – Surveys, in-depth interviews, focus groups and questionnaires will be filled in by relevant stakeholders to receive feedback and assess the ground-breaking solutions. Rate satisfaction score will be analysed along with the results from the benchmarking activities [D5.7].

Reference to Workplan: WP5, WP6

Obj.5 – CONSOLIDATE international and European links, raise awareness, promote multi-stakeholder cooperation, ensure the technology transfer of the project’s results and their rapid uptake and collaborate with EC services to

contribute to the long-term sustainability of safe and enhanced working conditions in agriculture and ensure compliance with relevant EU and national regulations.

Pertinence to the work programme topic: From the call: “Development of business models for the use of the developed innovative technologies under consideration of various farm structures and inter-farm linkages as well as of various biogeographic and socio-economic framing conditions [...] Proposals must implement the ‘multi-actor approach’ including a range of actors to ensure that knowledge and needs from various stakeholder groups, including farmers, farm workers, farm advisors and scientists are taken into consideration [...] Projects are expected to take into consideration the results of other related Horizon 2020/ Europe projects as well as of other relevant EU-funded projects and initiatives [...] should take into account relevant (forthcoming) EU legislation, in particular linked to the horizontal Act on AI, as well as the legislation related to liability and machinery.”

Means to Achieve: **AGROBOOST** will consolidate international and European links, raise awareness, promote multi-stakeholder cooperation, ensure the technology transfer of project results, and facilitate their rapid uptake [T6.1-T6.5]. The project will collaborate with EC services to contribute to the long-term sustainability of safe and enhanced working conditions in agriculture, ensuring compliance with relevant EU and national regulations. Through strategic branding, dissemination, and communication activities, **AGROBOOST** will aim to foster collaboration with relevant EU projects and initiatives while actively engaging with the community. Additionally, innovation management, business model development, and commercial pathways will be established to facilitate the transfer of technologies and knowledge. Replicable activities, knowledge transfer, and training curricula will be emphasized, ensuring a broad and sustainable impact. Standardization activities and policy recommendations will be developed to enhance working conditions in agriculture in alignment with regulatory frameworks. The goal is to build a robust ecosystem that supports the continued advancement and implementation of safe and enhanced working conditions in agriculture beyond the project's duration.

Measurable Outcomes, Success Indicators and Verification Means

- Fulfilment of dissemination, communication and exploitation KPIs as presented in Section 2.2 - Analytics tools will be used to track website traffic, capture visitors in the project’s website, users engaged in social media channels (via social media monitoring tools), users enrolled in the training activities, stakeholders enrolled in the validation campaigns, etc. Email marketing platforms will be employed to track engagement metrics along with online survey tools and contact forms on the project's website and social media accounts [D6.4].
- Liaison activities with ≥ 15 EU funded projects and initiatives - Keeping a record of the collaborations established by the project with other EU projects and initiatives. The record will include the name of the project/initiative, the nature of collaboration, the timeline of collaboration, and the impact of the collaboration on achieving project goals. The record will be periodically reviewed and updated by the project management team [D6.4].
- Replicable results for ≥ 5 additional European regions with different cultural, societal and economic contexts based on best practices and lessons learnt – Feasibility studies examining the applicability of the proposed innovations and the transferability of the overall framework targeting sustainable development in EU agriculture [D6.4].
- Development of ≥ 10 business models for the use of the developed technologies for various farm structures and biogeographic and socio-economic framing conditions – Surveys and reports showcasing the increase in the adoption of the proposed technologies on the farms participating and engaged in the pilot validation campaigns, the farms they are collaborating with, the economic impact of the developed business models, considering factors such as increased productivity, reduced costs, and enhanced profitability [D6.4].
- Publication of ≥ 5 white papers for policy recommendations - Publication of White Papers and dissemination through the project’s website and social media accounts [D6.4].

Reference to Workplan: WP6

1.1.2 Research and innovation maturity



Robotics and Augmented Reality Solutions for Enhanced Working Conditions [WP3]

Current state-of-the-art: Robotics and AR solutions can enhance working conditions in many aspects, especially in cases where the user should be separated from the robot, for the sake of safety and cannot directly observe the robot’s motions and activities, AR can improve user experience and safety by offering solutions for challenges like collision detection, emergency shutdown mechanisms and real-time hazard identification. Moreover, AR can overspread real-time safety instructions or hazard warnings onto the worker's field of view to ensure obedience to safety rules and formalities⁴. AR has been proposed in several human-robot collaboration applications e.g., for the creation of an AR-based interaction interface that controls a teleoperated agricultural robot sprayer⁵ or for safety operation and management of robotic tractors⁶. Combining robotics and AR can make on-the-job training facile by providing immersive simulations or step-by-step instructions and guidance so that the workers can acquire new skills⁷. Nevertheless, despite the recent progressions, the extensive use of robotics and AR, in agriculture for enhancing working conditions has not reached a great maturity⁸. It is common for farmers and agricultural workers

⁴ Su, Y.-P.; et al., Integrating Virtual, Mixed, and Augmented Reality into Remote Robotic Applications: A Brief Review of Extended Reality-Enhanced Robotic Systems for Intuitive Telemanipulation and Telemanufacturing Tasks in Hazardous Conditions. Appl. Sci. 2023, 13, 12129.

⁵ Adamides, G., et al., (2014). A reality-based interaction interface for an agricultural teleoperated robot sprayer. In P. Gonzalez-de-Santos, & A. Ribeiro (Eds.), International Conference on Robotics and Associated High-technologies and Equipment for Agriculture and Forestry.

⁶ Huuskonen, J., et al., (2019). Augmented Reality for Supervising Multirobot System in Agricultural Field Operation. IFAC 52-30, 367–372.

⁷ E. Marino, et al., (2021). An Augmented Reality inspection tool to support workers in I4.0 environments, Computers in Industry, 127, 103412.

⁸ Javaid, M., et al. (2022). Enhancing smart farming through the applications of Agriculture 4.0 technologies. International Journal of Intelligent Networks, 3, 150-164.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

to require guidance in managing the complexities associated with robotics and AR technologies. Closing the skills disparity and offering cost training initiatives are necessary measures to facilitate the seamless incorporation of these innovations across different agricultural sectors.

Advances beyond the state-of-the art: **AGROBOOST** will develop ground-breaking robotics and AR solutions to mitigate the risk of hazardous agricultural tasks such as pesticides application, tractor operation, lift and/or carrying of heavy objects, livestock handling, pruning among others. For this purpose, a novel autonomous platform will be developed with its level of autonomy being increased through a VR environment developed with high-quality visual data and deep learning techniques targeting pruning and thinning activities. In addition, robotic system for mechanical weeding will be developed equipped with advanced AR solutions to assist in the remote handling of the weeding machinery. New AR-enabled control algorithms will be designed and developed to optimize the harvesting process. These algorithms will be based on new computer vision algorithms for object recognition and tracking such as YOLO-NAS, CNNs and DeepSORT, coupled with navigation algorithms integrating LiDAR and GPS to guide the robotic harvesting equipment to selectively target ripe crops. All these AR-enabled robotic technology advancements will prioritize the ease of use by human operators. In light of this, a pioneering VR environment will be developed enabling the telepresence of the human operator and assisting in the training of the robotic system. Human-robot collaboration will be strengthened, empowering farm workers to use technologies to handle hazardous tasks.



Automation to Mitigate Workforce Shortages and Improve Productivity [WP3]

Current state-of-the-art: The EU agricultural sector confronts significant challenges due to workforce shortages, a situation that has intensified recently as more people find the sector unattractive. According to a report from the EU Parliament⁹, farming employment across the EU-27 has consistently declined, plummeting from 13.1 million Annual Work Units (AWU) in 2003 to 9.1 million AWU in 2018, marking a substantial 30% reduction. The European Commission's latest agricultural outlook¹⁰ forecasts a projected annual decrease of approximately 0.7% in agricultural labor, measured in AWU, from 2022 to 2032. This extends the ongoing decline in EU agricultural labor since 2007, primarily influenced by structural changes within the EU agri-food industry and increasing employment opportunities in alternative sectors. Research¹¹ in the agricultural sector has shown that automation, in forms of robotic solutions, clearly helps mitigate work shortages and improve sustainable growth. Automation in agriculture can be even more effective for reducing work shortages when the tasks that are automated are labor-intensive, risky, and/or unattractive. Automation of labor-intensive tasks such picking, harvesting and weeding allows a vast number of previously excluded workers (women, disabled people, young people) to now participate in the agriculture sector by only requiring tech skills from them. However, in the agriculture sector most automated solutions are still in prototype or development phases¹² and aren't mature enough to become commercially available. Also, worries about how automation might affect occupational safety and workers health exist as for example the new expectations from workers may result in workers having less time to rest and more mental fatigue. Moreover, the affordability of technologies is still a worry¹³, for farmers particularly in fields where the main goal isn't just about improving efficiency as the wide range of practices makes it hard to implement automation solutions that work across the board. Thus, there is a need to create more affordable and effective automation solutions.

Advances beyond the state-of-the art: **AGROBOOST**'s automation solutions will be strategically designed so that they can be applied in a vast amount of agriculture cases, considering the challenges posed by farm sizes, crop varieties and locations. Automated pruning, thinning, weeding, and harvesting solutions will be developed and validated extensively to ensure that the automations improve productivity and have a significant impact on inclusivity and workforce shortage mitigation. Moreover, **AGROBOOST** will target worker groups that are severely underemployed in the agriculture sector such as younger generations, women, and disabled people. **AGROBOOST** will perform dedicated training, using digital technologies such as AR and VR, to these groups so they can gain the newly required tech skills. Finally, **AGROBOOST** aims to offer automation solutions that are affordable so that they can have a wide impact on the sector and be implemented by many end-users. It will also seek to reduce burdens on farmers, specifically those with small scale operations by finding ways to cut upfront costs and maintenance expenses.



Artificial Intelligence Capabilities in Agro-Robotics [WP3]

Current state-of-the-art: Robots powered by artificial intelligence (AI), navigate fields and execute agricultural tasks such as weeding, crop monitoring, harvesting and multi-purpose farm management operations with minimal human intervention. This technology enhances operational efficiency, reduces labour costs, and enables continuous operations. Robots employing a combination of sensors, cameras, radar, and AI for navigation between points, enabling these vehicles to perform agricultural tasks with significant power output at low speeds, marking a significant advancement in autonomous agricultural operations. Consequently, to enhance navigation under such circumstances, robots leverage both natural cues (like generating trajectories based on row spacing and airflow

⁹ https://eprints.glos.ac.uk/7629/1/IPOL_STU%282019%29629209_EN.pdf

¹⁰ https://agriculture.ec.europa.eu/data-and-analysis/markets/outlook/medium-term_en

¹¹ Marinoudi, V., *et al.* (2019). Robotics and labour in agriculture. A context consideration. *Biosystems Engineering*, 184, 111-121.

¹² Lincoln, J. M., *et al.*, (2023). Emerging technology in agriculture: Opportunities and considerations for occupational safety and health researchers. *Journal of Safety Research*.

¹³ Lowenberg-DeBoer, J., *et al.* (2020). Economics of robots and automation in field crop production. *Precision Agriculture*, 21, 278-299.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

direction) and artificial cues (such as RFID tags and wireless sensors). Various SLAM algorithms and path-planning strategies tailored for agricultural robots are currently developed and implemented like Extended Kalman Filter (EKF) SLAM, FastSLAM, Dijkstra's algorithm, etc. In modern algorithms, the SLAM challenge is typically addressed upfront through scan matching. To estimate the robot's pose during mapping, localization algorithms, including visual odometry and LiDAR odometry, are employed. Local path planning becomes crucial for navigating around obstacles with unpredictable movements, utilizing 3D LiDAR data to project points over a plane. Artificial intelligence tasks in scouting and mapping, such as classification, segmentation, and object detection, are vital components, with popular methods including Support Vector Machine and Convolutional Neural Network for classification and segmentation, while object detection commonly employs models like YOLOv2, YOLOv3, and SSD MobileNet contributing to the effective identification of plant types within images and LiDAR data. One notable area of advancement is in weed detection and management, where AI-powered computer vision systems can identify and classify weeds in real time. The control systems of robotic weeders employ ML and deep learning (DL) techniques, allowing for selective weed removal. This not only reduces the need for herbicides but also minimizes potential damage to crops¹⁴. Harvesting automation has seen remarkable progress with the integration of AI-enabled robotic systems. Computer vision algorithms identify ripe produce and robotic grippers perform harvesting tasks with extremely high precision¹⁵.

Advances beyond the state-of-the-art: In **AGROBOOST**, the LiDAR SLAM algorithms will be customized to better suit multiple farm settings. Instead of Kalman filters, factor graphs - as the leading-edge technique for integrating position estimations - from multiple sensors will be employed. This approach will allow for a more cohesive integration of various sensors. Point clouds containing vegetation indices, obtained from depth cameras and assigned to individual points, will be developed in the designed robotic platform prospectively. The autonomous navigation of the robots, will offer supplementary information, enhancing the understanding and perception of the environment to be mapped. The efficiency of multi-robot systems employing fleet navigation methods will be further enhanced through the integration of AI solutions and supervisory management strategies. Consequently, **AGROBOOST** pilots will be aimed at generating synthetic data for training neural networks or AI-based tools to automate the labelling process. Moreover, **AGROBOOST**'s training process will be enhanced by incorporating self-attention mechanisms such as transformers, which will assign varying importance to different parts of the input data.



Technology Acceptance and Inclusivity in the Agricultural Sector [WP2]

Current state-of-the-art: Technology Acceptance Models (TAMs) are the most common models in smart agriculture primarily assessing the factors influencing farmers' acceptance of new emerging technologies¹⁶. They gauge farmers' perceptions regarding the usefulness and ease of use of technologies that can be used in agriculture and greatly assist in their daily lives. TAMs help predict and explain farmers' intentions to adopt or reject technological innovations through a deeper understanding of these perceptions. The review of literature indicates that the primary factors influencing acceptance rates revolve around the effects on productivity and expenses. Farmers tend to embrace technology when it leads to reduced overall farm costs and minimizes resource requirements. Additionally, significant considerations involve the provision of useful information and guidance by technological solutions to end-users, frequently integrated with personalized and localized approaches¹⁷. Thus, users perceive the most significant barriers as the infrastructure demands and related expenses associated with implementing technological solutions, followed by challenges or intricacies in their utilization¹⁸. These findings suggest that users lacking resources may struggle to adopt these solutions effectively. Integrating gender perspectives into acceptance models is imperative for promoting inclusivity and addressing social inequalities in technology adoption. Men and women farmers face diverse obstacles in farming activities, exacerbated by elements such as gender expectations, socio-cultural contexts, religious beliefs and ethnic backgrounds¹⁹. In particular, women shoulder heavier workloads due to their domestic and caregiving duties. One significant factor contributing to this gap is that agricultural tools are primarily designed to accommodate men's physical attributes and tasks, making them uncomfortable or impractical for women to use²⁰. Similarly, young people encounter obstacles including restricted access to land and capital and limited participation in decision-making processes.

Advances beyond the state-of-the-art: **AGROBOOST** will develop a more expansive and inclusive understanding of how agricultural stakeholders engage with technological innovations. Diverging from the conventional reliance on quantitative metrics seen in TAMs, a holistic analysis that incorporates both quantitative and qualitative data will be prioritized. This approach will facilitate a thorough assessment that not only identifies statistical trends but also acknowledge the nuanced intricacies of user perceptions and experiences in adopting agricultural technologies, with

¹⁴ Zhang, W., et al., (2022). Review of Current Robotic Approaches for Precision Weed Management. *Curr Robot Rep* 3, 139–151.

¹⁵ Saleem, M.H., et al., (2021). Automation in Agriculture by Machine and Deep Learning Techniques: A Review of Recent Developments. *Precision Agric*, 22, 2053–2091.

¹⁶ Sayruamyat, S., et al., (2020). Acceptance and readiness of Thai farmers toward digital technology. *Smart Innov. Syst. Technol.* 165, 75.

¹⁷ Jain, M., et al., (2018). FarmChat: a conversational agent to answer farmer queries. *Proc. ACM Mob. Wearable Ubiquitous Technol.* 2.

¹⁸ Khan, N., et al., (2021). Current progress and prospects of agriculture technology: Gateway to sustainable agriculture. *Sustainability*, 13(9).

¹⁹ Huyer, S., et al., (2021). Expanding Opportunities: A Framework for Gender and Socially Inclusive Climate Resilient Agriculture. *Frontiers in Climate*, 3, 718240.

²⁰ Nichols, C. E. (2024). Being a woman with the “skills of a man”: negotiating gender in the 21st century US Corn Belt. *Agriculture and human values*, 1-16.

due consideration given to gender and minority perspectives. **AGROBOOST** will also align its efforts with the pursuit of socially inclusive and participatory technologies. The initiative will also explore innovative approaches for developing socially inclusive agricultural tools that consider factors such as gender, age and disability. In addition, **AGROBOOST**'s research will focus on the intersectionality of marginalization and agriculture, employing interdisciplinary methods to evaluate the effectiveness of interventions such as training programs.

1.1.3 **AGROBOOST positioning and TRL advances**

Technology Offerings, Target TRL Level & Means to Reach It (Starting TRL " Foreseen TRL)

Autonomous Robotic Systems for Pruning & Thinning Tasks [INE, T3.1] – The robotic Modular-E solution developed by **INE** in **NOVATERRA** H2020 project (TRL6) will be extended, incorporating a specialized robotic module, referred to as Prune-I, equipped with dexterous manipulators and advanced high-resolution cameras. This enhancement will increase its TRL reaching TRL8, showcasing a substantial advancement in functionality and precision for operations in the management of permanent woody crops. The Prune-I will employ advanced hardware like GPU, TPU and FPGA to deploy Deep Learning models for interpreting the environment and tree structure. It will also use VineSLAM, incorporating GNSS, LiDAR, RADAR, etc. for navigation and will also integrate a 5G infrastructure for seamless communication, allowing real-time visualization of robot operations and remote support. **(TRL6 → TRL8)**

AR-based System for Automated Mechanical Weeding [AUA, T3.2] – A state-of-the-art AR system for automated mechanical weeding, based on **LIN**'s AR solutions (TRL6) and **AUA**'s vision system (TRL6), developed within **GOOD** HEU project. The system will integrate a vision system, an autonomous navigation system and weeding implements along with the current AR solutions, (a) enabling vision detection, (b) facilitating autonomous navigation in fields and (c) allowing for precise destroying of weed without crop damaging. Finally, the AR solutions will be fine-tuned to allow end-user to have the same view as the system, allowing non-experts to easily operate, troubleshoot and perform minor repairs. Thus, this – applicable in various crops – system will be tested and validated in real-commercial conditions reaching TRL8 before the end of the project. **(TRL6 → TRL8)**

AR-enabled Control Algorithms for Selective Harvesting [UCD, T3.3] – The development process for the AR-based recognition system for selective harvesting will build upon the control algorithms already developed for inform decisions making in education and predictive planning capabilities within the context of **ARETE** and **DARKO** H2020 projects. During **AGROBOOST**, novel Machine Learning algorithms will be employed to enable the system to learn and adapt based on historical data, optimizing harvesting efficiency and accuracy over time. In addition, a user-friendly interface will be developed that will allow operators to interact seamlessly with the AR-enabled control system providing real-time feedback on crop selection and harvesting parameters. **(TRL5 → TRL7)**

Gamified VR Environment for Robotic Systems [LIN, T3.4] – The technological solution will build upon **LIN**'s prior work in fruit harvesting, notably in **BACCHUS** H2020 project, employing robot learning from demonstration methods in a gamified VR environment to master manipulators for active inspection and selective harvesting of high value crops (strawberries), previously evaluated in controlled lab conditions (TRL4). In the **AgroBoost** project, the TRL of the solution will be elevated to TRL7, by showcasing the operational capabilities of the system prototype in real field conditions. Control software for the robotic system will be further developed using ROS2, and the VR solution will use enterprise-level mixed-reality headsets (Varjo XR4) with software developed in Unity engine. **(TRL4 → TRL7)**

AR-powered Operational & Training for Sustainable Agriculture [LIN, T3.5] – This module will be based on the development of an AR-powered platform to support and train farmers in sustainable agricultural practices. It will integrate existing agricultural management systems, databases, and sensor networks to provide real-time information on crop health, soil conditions and other relevant parameters based on best practices and lessons learnt from **LIN**'s **FLOBOT** and **STEP2DYNA** H2020 projects. Visualization feature will be further enhanced allowing real-time data to be overlaid on fields through AR devices. Interactive AR training modules will also be designed to offer immersive learning experiences, reaching TRL8 through user feedback and field testing at the pilot campaigns. **(TRL5 → TRL8)**

Deep Learning for Complex Scene Recognition [UCD, T4.1] – **UCD** has successfully created an AI-based computer vision growth monitoring system (TRL4) that uses deep learning for complex scene recognition in the context of the **MLSysOps** HEU project. This innovative system, serving as one of the foundations of the **AGROBOOST** project, will undergo substantial enhancement through the integration of a specialized robotic arm and an advanced computer vision system tailored for the specific requirements of *Agaricus bisporus* mushrooms. This augmentation aims to elevate the technology readiness (TRL7), signifying a big advancement in the adaptability and applicability of the solution in the context of mushroom harvesting. **(TRL4 → TRL7)**

Animal-Robot Interaction for Poultry Management [KUL, HWU, T4.2] – Flock-scale monitoring, initially proven in a simulated environment by **KUL** and **HWU** (TRL4) during **DEEPFIELD** H2020 project, will progress to real-environment trials in the **AGROBOOST** project (TRL7). Also, the already developed mobile quadruped, presently undergoing human-robot interaction tests in a simulated environment (TRL4), will extend to animal-robot interactions and undergo real-commercial testing conditions (TRL7). In more detail, the quadruped robot will be equipped with a robotic arm and RGB-D sensors for poultry evaluation and egg manipulation, offering superior locomotion, featuring a robust localization system to minimize drifting on varied terrain. The robot will employ model predictive controllers for optimal maneuvering in confined spaces. **(TRL4 → TRL7)**

Human-Robot Collaboration for Safe Working Conditions [LIN, T4.3] – **LIN** has achieved significant success in creating robot control algorithms (TRL4) through participation in the **CORTECH** H2020 project. The **AGROBOOST** project will seek to elevate these algorithms (TRL7), focusing on optimizing efficiency while ensuring the safety of human workers within a collaborative framework. Machine learning models, trained in a gamified virtual reality environment through user

demonstrations, will be developed prioritizing safety in interactions between humans and robots. The integration of self-tuned machine learning methods, including Gaussian Kernel Density Estimators, will dynamically adapt to varying operational conditions, enhancing safety further. (TRL4 → TRL7)

Automation and AI Innovations in Pathway Planning and Navigation Control [DTU, T3.3] – The pathway planning and the navigation control of the robotic systems in this project will capitalize upon the experience and knowhow gathered from deploying autonomous robots in agricultural environments, as in the [GALIRUMI](#) H2020 project, as well as other challenging dynamic environments, such as construction in the [RobetArme](#) HEU project. In AGROBOOST, custom Deep Reinforcement Learning techniques will be explored to enable robots to autonomously navigate through complex agricultural environments. These algorithms will leverage AI to adapt to changing conditions and optimize routes in real-time, while further improving their performance as relevant experience is accumulated. (TRL3 → TRL7)

XAI-enabled Decision Support System for Socially Responsible Agricultural Practices [NCI, T4.5] – The framework of the Decision Support System has been implemented in [WATSON](#) and [ALLIANCE](#) HEU projects providing suitable recommendations for potential food fraud incidents based on a diverse set of data sources. Moreover, the explainability features have been tested during the [AI4PublicPolicy](#) HEU project. In AGROBOOST, Deep Learning techniques will be developed coupled with Knowledge Graphs to enhance the decidability process related to the transparency, validity and trust of AI systems producing recommendations for environmentally and socially responsible agricultural practices. Last, the underlying AI mechanisms will analyse vast datasets from all pilot activities including weather patterns, soil conditions, crop health, social impact indicators, etc. and assist farmers into gaining a deeper understanding of the environmental, social and economic implications of their choices. (TRL4 → TRL8)

Monitoring System for Analyses of the Consequences for Farmers and Rural Communities [NCI, T5.11] – The core mechanisms of this dynamic system have been already developed during the [Data4Food2030](#) HEU project targeting the monitoring of the data economy for food systems, thus starting at TRL4. During AGROBOOST, it will be modified incorporating indicators related to changes in working conditions, workforce dynamics, environmental sustainability, and economic outcomes capturing the dynamics of how the proposed technologies influence the livelihoods of farmers and the broader rural community. In addition, sophisticated data analytics and ML algorithms will be employed for adaptive strategies and informed decision-making based on the evolving consequences of technological interventions, reaching TRL7 before the end of the project. (TRL4 → TRL7) #SPRJ-OBJ-PO\$#

1.2 Methodology #@CON-MET-CM@# #@COM-PLC-CP@#

1.2.1 Concept and approach

AGROBOOST will leverage cutting-edge technologies such as Robotics, Augmented and Virtual reality, Machine Learning and Artificial Intelligence to significantly enhance working conditions in agriculture. Its main goal is to transform the agricultural landscape, fostering safer, sustainable and efficient practices through the deployment of robotic solutions combined with AR/VR. Towards this direction, **AGROBOOST** will engage with stakeholders at all levels, including farmers, farming associations and competent authorities, and develop an inclusive approach that addresses the needs and concerns of all the involved actors. Training programs will also be established to ensure that all stakeholders are equipped with the necessary skills and knowledge to effectively use these new technologies.

1.2.2 Overall methodology

The project methodology will provide a systematic and structured approach to the research, development and deployment process ensuring that the results will be rigorous, credible, accurate and reliable, as depicted in Figure 1. From a conceptual point of view, the project can be split into four (4) evolving phases that follow an iterative multi-step development approach offering rapid development when moving from phase to phase and from iteration to iteration. Each phase relates to the respective WPs and the foreseen activities to meet the project's objectives. The need to inject agility into this well-designed methodology arises from the fact that **AGROBOOST** envisions ground-breaking solutions for farmers and workers in the agriculture, thus the innovation dimension needs to be reflected in the actual development and demonstration phases. The project methodology is based upon the identification of the needs of modern farmers, agricultural technology developers, digital innovation hubs, research institutions, governmental agricultural agencies, policymakers, local communities and environmental organisations to increase safety of workers and enhance the working condition in agriculture through the use of emerging technologies.

PHASE ① – THE LANDSCAPE: This phase is dedicated to the project set up, driving its research and innovation potential towards effective technologies for sustainable and inclusive agricultural practices, including robotic solutions, augmented reality applications and AI innovations. It starts with the identification of research challenges and the deeper understanding of the end users' needs [T2.1], following a co-creation approach that brings together stakeholders from the academia, industry, public governance and civil society. In parallel, a thorough analysis will be performed to identify existing challenges, conditions and shortages related to labour within the agricultural sector, including aspects such as safety, health and overall job satisfaction [T2.2]. Socioeconomic drivers that influence farmers' behaviour and attitudes in relation to the adoption of emerging technologies will also be explored [T2.3], emphasising on the ecological, social and economic benefits of implementing sustainable agricultural practices. A continuous monitoring process will be established in close collaboration with the technical partners to assess the privacy, ethical, social and legal impact of the proposed innovations [T2.4]. Last, strategies that ensure the inclusion

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

and equitable access of young farmers, women involved in agriculture and farmers with disabilities in the adoption of advanced technologies will be developed [T2.5], addressing the barriers that these specific groups might face.

PHASE ② – THE DEVELOPMENT: This phase entails all the technical activities of the project accommodating the

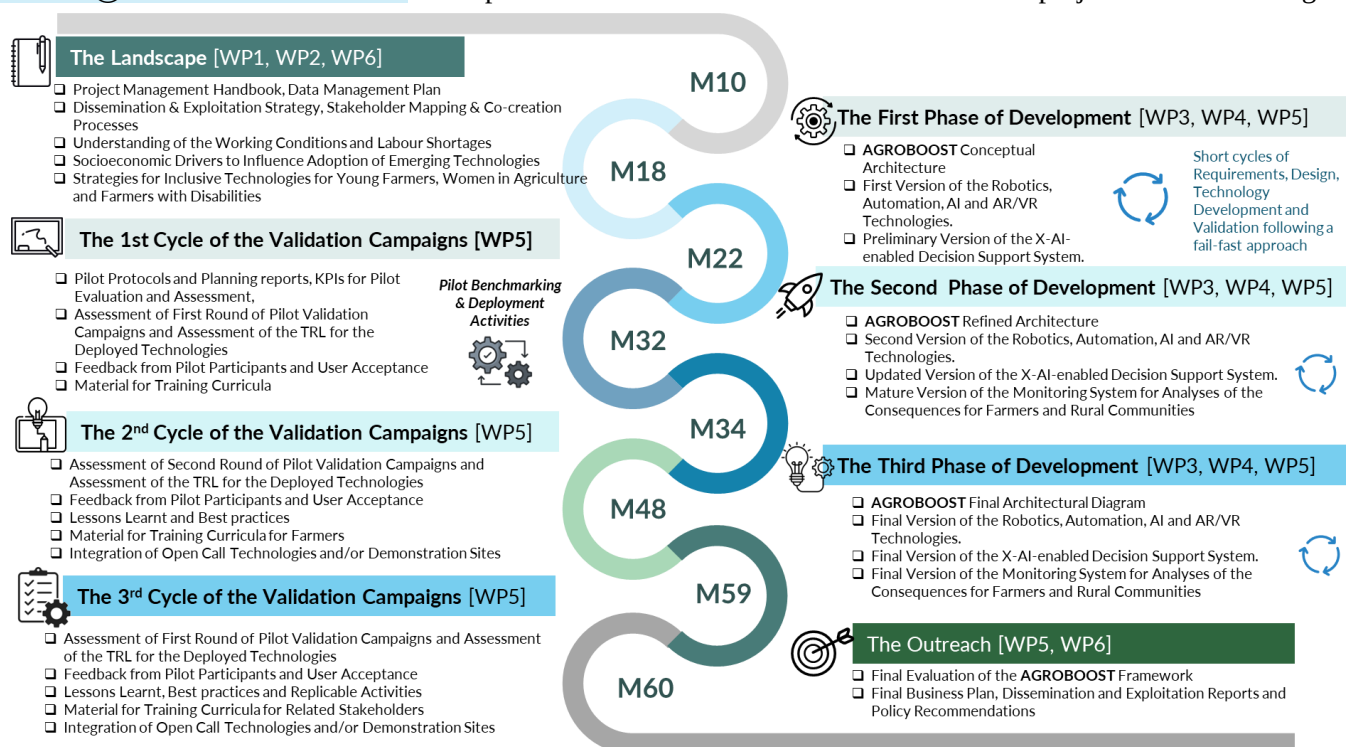


Figure 1: The AGROBOOST methodology roadmap.

foreseen innovations required to enhance the working conditions of workers in agriculture and increase their performance while ensuring production in an environmentally and socially sustainable way. Emerging technologies targeting the increased safety of the workers in agriculture and the improved performance of robotic system will be developed including Augmented/Virtual Reality and Robotics will be employed for pruning and thinning tasks [T3.1], automated mechanical weeding [T3.2], selective harvesting [T3.3] coupled with a VR environment for robotic systems [T3.4] and AR-powered training curricula [T3.5]. In parallel, advanced AI algorithms and automation will be explored for environmentally responsible practices leveraging Deep Learning for complex scene recognition [T4.1] necessary to a range of tasks in agriculture, human-robot collaboration for poultry management [T4.2] and for enhanced of safety of the working conditions [T4.3] linked with well-established innovation for pathway planning and navigation control. Insightful recommendations for sustainable practices to be followed will be provided to the robot operators through an XAI-enables Decision Support System [T4.4].

PHASE ③ – THE VALIDATION: The AGROBOOST framework and its underlying robotic and AR innovations will be demonstrated in six (6) diverse use cases in six different countries. The validation activities will be based on clearly defined short- and long-term realistic and verifiable goals [T5.1] and will be prepared according to the plan produced in close collaboration with the involved stakeholders and based on the overall system architecture [T5.2]. In order to strengthen the innovative solutions offered, third party solutions will also be integrated within the use cases through Open Calls [T5.3]. The foreseen use cases include (a) Intelligent Autonomous Robotic System for Harvesting of Strawberries Crops [T5.4]; (b) Fully Autonomous, AR-enabled Mushroom Harvesting Robots [T5.5]; (c) Orchards 6.0 with AI-based Robots for Tree Pruning and Flower Thinning [T5.6]; (d) Mechanical Autonomous Selective Weeder [T5.7]; (e) Autonomous Management of Poultry Production, Health, and Welfare [T5.8]; (f) Pilot 6: Automated Work on the Ground in Steep Vineyards [T5.9], and they will be performed in four iterations providing feedback to the technical partners on the usability and functionality of the developed technologies. At their beginning, the use cases will include benchmarking activities to assess their progress and identify areas for improvement. At the end of each pilot iteration, a thorough assessment process will be conducted, evaluating the user acceptance of the proposed solution and the TRL of the developed technology components [T5.10]. Each pilot will provide insights to the monitoring system that will analyse the use of robotic technologies and AR for rural communities [T5.11].

PHASE ④ – THE OUTREACH: The last phase focuses on four key aspects that ensure extroversion, namely (a) communication and dissemination activities to ensure wide visibility not only by relevant stakeholders but also by a broader audience. For this purpose, a well-designed strategy consisting of clear and tailored messages will be created communicating the project's results [T6.1]. To attract more stakeholders and maximise its impact, the project will liaise with relevant EU funded projects, sharing the knowledge obtained throughout the project [T6.2]; (b) exploitation and financial sustainability of AGROBOOST innovations during and after the project's duration, will be ensured through the continuous monitoring of the IPRs and the innovation potential of the proposed solutions [T6.3] along with a detailed analysis of market's needs for each of the proposed technologies coupled with novel

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

business models for the use of the developed technologies under consideration of the various farm structures; (c) capacity building activities and training curricula for farmers that will enhance their understanding on these technologies for sustainable agricultural practices and promote the adoption of emerging technologies for enhanced working conditions [T6.4]. Moreover, feasibility studies will be performed for different EU countries with varying socioeconomic criteria revealing the replicability potential of the proposed solutions. Last, (d) policy recommendations will be drafted targeting the various stakeholders involved in the agricultural sector, towards the rapid uptake of robotics, AR, and AI, for increased safety and improved performance in agriculture [T6.5].

As part of the environmental analysis of the solutions **AGROBOOST** will ensure that it is fully compliant with the **Do No Significant Harm (DNSH) Principle** since its foreseen activities substantially contribute in a positive way and/or do not significantly harm any of the six environmental objectives set out in Art. 9 and 17 of the EU Sustainable Finance Taxonomy Regulation²¹. **AGROBOOST** will apply a life-cycle oriented approach to ensure that the proposed solutions demonstrate substantial life cycle GHG emission savings compared to the best performing alternative technology. Lifecycle GHG emissions and net emissions will be calculated using Recommendation 2013/179/EU or, alternatively, using ETSI ES 203 199, ISO 14067:2018 or ISO 14064-2:2019. In this sense, **AGROBOOST** will fully analyse and consider as part of the technology developments, the climate adaptation and mitigation criteria related to the application of ICT technologies and the Circular Economy principles of the related equipment. Specific focus will be paid on the data storage, management, control, display, and transmission through data centres and edge computing following the EU Taxonomy Compass for the ICT Sector²².

1.2.3 High level architecture

AGROBOOST is built upon a modular architecture integrating cutting-edge. At its core the architecture includes the advanced technological solutions that will be developed during the project and validated through the pilot validation campaigns.

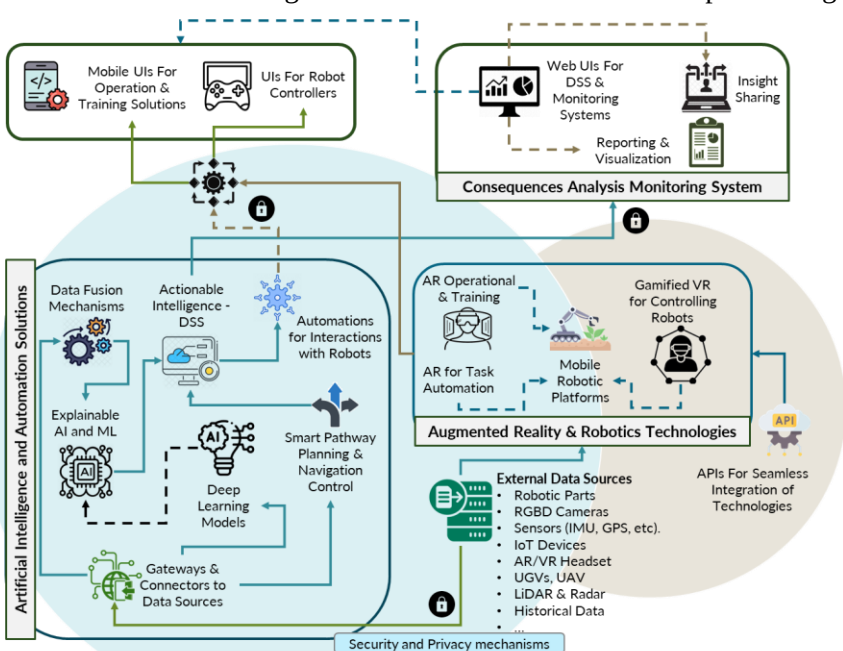


Figure 2: **AGROBOOST** conceptual architecture.

advancements. The interconnectedness of these subsystems results in a unifying and intelligent agricultural platform that addresses key challenges in the sector, ultimately contributing to the enhancement of working conditions.

1.2.4 Technology offerings

1.2.4.1 Autonomous Robotic Systems for Pruning & Thinning Tasks [T3.1]

The autonomous platform will be developed using an existent and field-tested robotic prototype and adding the necessary control and mechanical actuator features. The platform's degree of autonomy will be incrementally increased, first through a VR environment and learning by demonstration, to increase flexibility and fasten the deployment in the field. Powered by high-quality visual data and deep learning architectures, the decision-making module will be further improved. Through the support of high-speed communications (e.g., 5G), the decision support system will enable the real-time access to visual information augmented with relevant information, as well as input capture. Given the complexity of the tasks, development will prioritise the remote access to information and platform control, fostering usage by the farmer's and critical data acquisition for autonomy increase.

1.2.4.2 AR-based System for Automated Mechanical Weeding [T3.2]

An autonomous robotic mechanical weeding system suitable will be developed consisting of a HUSKY UGV, a lightweight mechanical weeding implementation, a vision module and AR solutions. The weeding process will rely

²¹ https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en

²² https://ec.europa.eu/sustainable-finance-taxonomy/activities/activity_en.htm?reference=8.1

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

on a multi-sensor simultaneous localization and mapping system. By fusing inertial, wheel odometry, range, and imaging together through a state estimator, the system will address sensor uncertainties, communication delays, and failures, thus providing robust local positioning. The EU's Galileo GNSS and ENGOS SBAS will be used for all GPS-dependent tasks. Moreover, a weed detection model will be developed using colour cameras (RGB) and the latest AI architectures, such as transformers and CNNs. A second computer vision algorithm for canopy density characterization will also be developed for optimizing weeder actuation movement, thus minimizing the risk of plant mechanical injury. Finally, AR solutions incorporating the developed machine vision models will be integrated to facilitate and enable non-experts to adopt robotic technologies more quickly by making wider decision making more transparent and explainable, thus enhancing the end user confidence in the machinery to make the correct decision.

1.2.4.3 AR-enabled Control Algorithms for Selective Harvesting [T3.3]

An AR head-mounted display system that captures images of mushrooms and displays the predicted maturity label in real-time by utilising cutting-edge deep learning technology (i.e., YOLOv7) will be developed for the rapid detection and segmentation of mushroom maturity in a farm environment. An object detector for the maturity identification of mushrooms will be designed and tested using different datasets obtained from the robotic system's RGBD sensors for in-field validation. Finally, an AR application framework for the real-time detection of the maturity levels of mushrooms in the farm environment will be developed. The primary novelty of this approach lies in the integration of AR technology with object detection for the real-time identification of mushroom harvesting maturity. While object detection has been used extensively for various applications, its combination with AR for mushroom harvesting assessment is a pioneering concept. This integration will offer a visually intuitive and contextually informative platform, bridging the gap between digital predictions and real-world scenarios.

1.2.4.4 Gamified VR Environment for Robotic Systems [T3.4]

The VR environment that will be developed will prioritize the ease of use by human operators while training the robotic manipulator to perform harvesting tasks. It will project to its user various configurations of strawberry clusters alongside with the location of the specific fruit that must be harvested from the cluster. A VR headset and controllers with hand tracking and gesture recognition will be used for recording the trajectories and grasping poses generated from human operators during harvesting of the virtual strawberries. This process will be gamified by measuring task efficiency related scores such as harvested fruits per minute, keeping performance records and creating various levels of difficulty based on the strawberry's position within the cluster. The gamification will aim to make the repetitive process of providing training data the robot more pleasant for the humans while providing enough demonstrations to efficiently train the robotic system. VR technologies will not only be applied for training of the robotic system in harvesting tasks, but also for remote control and monitoring of its operation by enabling the telepresence of the human operator. This will be achieved by capturing, appropriately processing and broadcasting in real time the feed obtained from the robotic system's RGBD sensors to the headset of the VR human operator.

1.2.4.5 AR-powered Operational & Training for Sustainable Agriculture [T3.5]

The AR training system will provide on-site information and training for human operators, in the relevant agricultural environment, to help support sustainable agriculture. The system will be deployed on a mobile platform such as a phone, tablet, or AR headset, and will make use of location-based data to display context specific information. The AR system will use appropriate inputs to determine and track the user location, potentially including GPS or other positioning systems, and/or marker-based recognition. The system will display real-time data about the location, provided by online services across Wifi or mobile data connections, as well as contextualised outputs from the decision support system developed in T4.5, which will also be provided via an online data connection. The system will also provide training to help improve users' knowledge of sustainable agriculture. This training will comprise both passive and active components, including relevant information summaries about sustainable practices, and also interactive knowledge tests to help assess individual users' knowledge gaps, and provide more targeted information.

1.2.4.6 Deep Learning for Complex Scene Recognition [T4.1]

Implementing deep learning for complex scene recognition in a mushroom harvesting robot will involve several tasks which will enable the robot to comprehend and interpret its environment effectively. A diverse dataset of images or videos representing various scenes encountered during mushroom harvesting in the greenhouse will be collected and they will be labelled to indicate different objects, structures, and environmental conditions relevant to mushroom harvesting (e.g., mushrooms, composite, obstacles, tools). The collected data will be pre-processed to enhance its quality and suitability for deep learning algorithms, including resizing, normalization and augmentation to increase the diversity of the training data. After that, various deep learning algorithms will be tested based on a suitable architecture for scene recognition. Using the labelled dataset, the selected deep learning models will be tested. The trained model will be evaluated on a separate validation dataset to assess its performance. Metrics such as accuracy, precision, recall and F1-score will be used to quantify the model's effectiveness in recognizing complex scenes relevant to harvesting. The trained DL model will be integrated into the mushroom harvesting robot's control system.

1.2.4.4 Animal-Robot Interaction for Poultry Management [T4.2]

For the animal monitoring aspect of the technology a state-of-the-art highly efficient computer vision-based animal detector and tracker using rotated bounding box will be developed. This tool will be suitable for group animal behaviour monitoring because of its compact rotated box (RBB) which can eliminate pixels from background or nearby animals. For multi-objects tracking task, the angle information will be used to predict the position likelihood in the next frame. For pose estimation and segmentation tasks, a compact bounding box will exclude disturbances

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

from background and pen mate. To translate this technology to that of poultry production, a complete pipeline will be developed that finds candidate features reflecting the emergence of abnormal behaviours in the flock. Confident localization will be ensured through proprioceptive sensors²³, complemented by ceiling cameras and a unique artificial tag for a global view. Egg detection will utilize instance segmentation through the robot's vision system, with enhancements from the ceiling vision system if necessary. The robot will employ model predictive controllers for optimal manoeuvring in confined spaces. Manoeuvres will consider scene interpretation and asset tracking will follow established guidelines²⁴ for safe motion based on prior research efforts.

1.2.4.5 Human-Robot Collaboration for Safe Working Conditions [T4.3]

AGROBOOST will develop sophisticated robot control algorithms tailored for agricultural operations, with a dual commitment to optimizing efficiency and ensuring the safety of human workers within a collaborative human-robot framework. This innovative approach will be tailored to empower non-experts, such as farmers unfamiliar with robotics or machine learning, enabling them to easily program specific tasks for the robots. These algorithms will be designed not only to enhance robotic performance but also to prioritize the inherent safety of interactions between humans and robots. User demonstrations in a gamified virtual reality environment will serve as a valuable training ground for machine learning models, enabling them to comprehend and adapt to diverse scenarios, ultimately contributing to a safer working environment. The incorporation of self-tuned machine learning methods, such as Gaussian KDE²⁵, will provide an additional layer of safety by dynamically adjusting to varying operational conditions. Formulation of ML controllers as time-invariant dynamical systems, with analytical proofs of stability, will establish a rigorous foundation for ensuring the reliability of the robotic system across different circumstances.

1.2.4.6 Automation and AI Innovations in Pathway Planning and Navigation Control [T4.4]

The pathway planning and navigation control software will consider both modular pipelines implementing individual perception, localization, planning and control modules, as well as end-to-end approaches that optimize the combined task, from raw perception to control signals, as a single deep learning task. Well-established planning approaches for global planning will be combined with Deep Reinforcement Learning, more precisely Proximal Policy Optimization for local planning, providing real-time corrections. It will allow the robot to learn from its experiences, improving its performance over time. Through the application of PPO, the robots will be able to navigate effectively in complex agricultural environments. They will demonstrate adaptability to novel situations, such as changing weather conditions or unexpected obstacles. Moreover, they will exhibit continuous performance improvement over time, becoming more efficient and effective the more they navigate. This combination of global and local planning, along with the use of modular and end-to-end approaches, will result in novel custom solutions for agricultural robotics.

1.2.4.7 XAI-enabled Decision Support System for Socially Responsible Agricultural Practices [T4.5]

This Decision Support System will be developed as a user-friendly, cloud-based platform that will seamlessly integrate existing farm management systems with the proposed technologies. The system will harness Machine Learning algorithms and ground-breaking explainability techniques to analyse vast datasets such as weather patterns, soil conditions, crop health, water and energy use, historical yield data, etc. Through the user-friendly interface, farmers will be able to provide specific requirements in order for the system to generate personalized, transparent recommendations. The explainability feature will ensure that farmers not only receive recommendations tailored to their unique circumstances but also understand the underlying logic and data driving each recommendation produced. Regular updates will allow for adaptive decision-making assisting farmers to align their agricultural practices with social responsibility. The development of the system will prioritize accessibility, and ease of integration, fostering adoption and ensuring that farmers can incorporate socially responsible practices into their daily operations. The recommendation produced will be finally conveyed to the farmers through web or through the AR equipment.

1.2.4.8 Monitoring System for Analyses of the Consequences for Farmers and Rural Communities [T5.10]

The specific platform will be designed to assess the environmental, economic and social impacts resulting from the integration of emerging technologies such as robotics, Augmented Reality, Automation and Artificial Intelligence. It will leverage a combination of (predictive) data analytics, social sciences and community engagement, to continuously monitor and evaluate the consequences of the technology adoption by the farmers. Moreover, it will employ key performance indicators to measure changes in farmer livelihoods, community well-being and environmental sustainability. In order to further engage the farmers, it will also incorporate feedback mechanisms and qualitative data-driven, in-depth assessments to capture and prioritize barriers and insights from the farmers, empowering community members in the decision-making process, promoting equitable outcomes and enabling policymakers, researchers and communities to collaboratively address challenges and optimize benefits. The tentative indicators that will be monitored include farmers' income growth due to the use of the proposed technologies, operational costs for farmers, measurement of inclusivity and participation of diverse and vulnerable groups in community activities through the social inclusion index, the carbon footprint associated with agricultural practices,

²³ Quail, C., *et al.* "Legged Robots in the Agricultural Context: Analysing Their Traverse Capabilities and Performance." IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society (2023): 01-07.

²⁴ Villacrés, J., *et al.* Apple orchard production estimation using deep learning strategies: A comparison of tracking-by-detection algorithms (2023) Computers and Electronics in Agriculture, 204, art. no. 107513.

²⁵ Heer, J. "Fast & accurate gaussian kernel density estimation," In *2021 IEEE Visualization Conference (VIS)* (pp. 11-15). IEEE.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

improvement in the efficient water use for agricultural activities, community resilience in the face of external challenges, such as climate events or economic fluctuations, soil health and fertility, etc.

Technical and Societal Robustness of AI: AGROBOOST will provide technically robust and trustworthy AI systems focusing on three key aspects: (a) **Transparency of the AI models** through interpretability and explainability features used not only for the design of the AI models model but also for the description of the data used in the phase of model conceptualization. For this purpose, an explainability-by-design approach will be followed during the activity of the architectural design promoting transparency in AI models. Explainability of the AI will be targeted via well-established XAI techniques (e.g., DeepLift²⁶, LIME²⁷, SHAP²⁸) to interpret models and boost trustworthiness. (b) **Reliability of the AI models**, identifying in early stage any potential vulnerabilities that the AI model might have and implementing appropriate technical solutions that will guarantee that the overall system will not be manipulated by an adversary or fail²⁹. (c) **Protection of data in AI models**, through the application of appropriate control points in cases where sensitive data are involved. On top of the abovementioned activities, the project will also leverage the partners' background results and expertise in AI and their participation in relevant EU funded projects such as [AI4PP](#), [CEDAR](#) and [STAR](#). AGROBOOST will ensure compliance with applicable regulations, and align its activities in consideration of relevant guidelines, such as “Ethics by Design in Algorithmic systems” and the EU HLEG Guidelines on AI. Specific technical and organisational measures will be implemented to safeguard the fundamental rights of the citizens, such as strong access control mechanisms, data anonymisation, data minimisation, audit trails and logging, data retention policies and data privacy impact assessment (DPIA) to name a few. In order to demonstrate a sufficient understanding of the demands of AI Ethics by Design, as described in the EU's “Ethics By Design and Ethics of Use Approaches for Artificial Intelligence³⁰” an external advisory board for the ethics issues will be set up which, along with **WIS**, will employ the specific steps outlined in the AI Ethics by Design guidelines, in order to develop an ethically compliant platform also with the Artificial Intelligence Act (AI Act). For this purpose, AGROBOOST work plan has foreseen a dedicated Task, namely **T2.4**, which will be active throughout the whole duration of the project leveraging the Assessment List for Trustworthy Artificial Intelligence³¹ (ALTAI) framework to evaluate the trustworthiness of the AI systems developed. Moreover, a Fundamental Rights Impact Assessment³² (FRIA) will be carried out by **WIS**, considering some of the established methodologies.

1.2.5 Validation campaigns

The AGROBOOST solution will be tested and validated in 6 real-world use cases, which will showcase the effectiveness of the proposed technological solutions in enhancing the working condition in agriculture, addressing labour shortages and ensure production in an environmentally and socially sustainable way.

1.2.5.1 Pilot 1 – Intelligent Autonomous Robotic System for Harvesting in Strawberry Crops

 **Background, Motivation and Challenges:** Farming tasks such as strawberry harvesting, are considered unattractive since they must be performed in harsh labour conditions within greenhouses³³. Therefore, the automation of fruit harvesting by robotic systems in conjunction with VR/AR technologies can reduce the burden and risk for the worker's, make them more attractive for young workers and deal with the work force shortage.

Although, there has been effort during the last decade to introduce robotic systems for fruit harvesting³⁴, the automation level of such tasks is low. This happens due to the large variety of conditions that the robot may encounter during its operation, and the pre-programming for all those conditions is extremely difficult or even impossible. The aspect of robotic manipulation of fruits is underserved despite the challenges that exist on how to harvest fruits efficiently without damaging them using robotic systems.

AGROBOOST Proposed Solution: AGROBOOST will propose a technological solution to address challenges in agricultural harvesting through the development of an intelligent autonomous robotic system that learns harvesting tasks from human demonstrations in a gamified virtual reality (VR) environment. After training, the robotic system will be able to generalize tasks to different environments, operating autonomously or in shared autonomy with human control through VR. The VR simulation will allow users to manually pick products using natural body movements, with motion capture data used to train the robot. To enhance engagement, a gamified version of the simulation will be created. Novel Machine Learning methods will be employed for trajectory modelling with stability guarantees. The system will learn tasks from non-experts, eliminating hyper-parameter fine-tuning. The robot,



Figure 3: The strawberry harvesting robot developed by LIN, in action.

²⁶ Avanti Shrikumar, *et al.*, “Learning important features through propagating activation differences”, In Proc. of ICML'17, Vol (70), 2017.

²⁷ Marco Tulio Ribeiro, *et al.*, “Why Should I Trust You?: Explaining the Predictions of Any Classifier”, In Proc. KDD '16, 1135–1144.

²⁸ Scott M. Lundberg, *et al.*, A unified approach to interpreting model predictions. In Proc. of NIPS'17, USA, 4768–4777.

²⁹ Hamon, R., Junklewitz, H. and Sanchez Martin, J., “[Robustness and Explainability of Artificial Intelligence](#)”, EUR 30040 EN, 2020,

³⁰ [Ethics By Design and Ethics of Use Approaches for Artificial Intelligence](#)

³¹ [The Assessment List for Trustworthy Artificial Intelligence](#)

³² [Fundamental Rights Impact Assessment](#) (FRIA)

³³ De Preter, A., *et al.*, (2018). Development of a robot for harvesting strawberries. IFAC-PapersOnLine, 51(17), 14-19.

³⁴ Zhou, H., *et al.*, (2022). Intelligent robots for fruit harvesting: Recent developments and future challenges. *Precision Agriculture*, 23(5).

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

supervised by a remote human operator, will use mixed-reality headsets and haptic gloves for corrections. The solution builds on LIN's previous work on fruit harvesting by demonstrating the prototype in the field with added AR/VR technologies. ROS2 will be used for control software and Varjo XR4 mixed-reality headsets will be employed in the VR solution. Off-the-shelf components, including a collaborative manipulator, soft gripper and RGBD cameras, make up the robotic system. Evaluation will take place in LIN's existing infrastructure, including strawberry greenhouses and a 5G communication network.

The proposed solution integrates Machine Learning, robot control and virtual reality technologies. The use of virtual reality and gamification will ease the burden of providing demonstrations to the system, improving working conditions by enabling remote task execution and monitoring. Gamification of robot training tasks can attract a younger workforce and streamline the introduction of task demonstrations. Additionally, working conditions are enhanced as the robot's operator can supervise and perform harvesting tasks remotely rather than being physically present in the greenhouse or field.

Expected Outcomes: The proposed solutions aim to benefit the agricultural industry by improving harvesting efficiency and productivity while prioritizing the well-being of the workforce. Automation of fruit harvesting tasks through remote robot operation, facilitated by VR technology, will reduce the physical burden on workers and enhance safety by minimizing the risk of injuries and exposure to extreme conditions in greenhouses. The approach involves learning harvesting tasks from humans instead of explicit programming, increasing system efficiency and enabling the handling of challenging scenarios. This solution will allow for continuous remote supervision through telepresence and addresses workflow shortages without requiring expertise in robotics or AI.

KPIs to be measured: The goal is to achieve a **90%** success rate for fully autonomous harvesting by the robotic system, even in challenging scenarios like dense fruit clusters. Additionally, there is an aim for human shared-autonomous harvesting with a **100%** success rate, where the robotic system successfully performs strawberry harvesting tasks under human supervision through telepresence. Furthermore, there is a target to measure an **80%** reduction in both mental and physical effort for harvesting tasks, using activity trackers and heart rate monitors.

1.2.5.2 Pilot 2 – Fully Autonomous, Vision-Guided Mushroom Harvesting Robot

Background, Motivation and Challenges: The mushroom industry faces a unique challenge with mushrooms growing at a remarkable rate of 4% per hour, making timely harvesting crucial to prevent overgrowth. Labor shortages, particularly in skilled migrant workers, compound the issue. The Irish mushroom industry, contributing almost 50% of the country's edible horticultural output³⁵, produces around 68,000 tons annually, with a majority exported to the UK. Skilled harvesters can pick up to one ton of mushrooms per week, emphasizing the need for expertise. The vulnerability of mushrooms to damage adds complexity to the process. Recruiting and retaining foreign national workers is challenging and expensive. Therefore, employing harvesting robots as a precision agriculture solution seems to be promising. These robots offer a more controllable, and customizable approach to mushroom harvesting, overcoming labour shortages and reducing costs.



Figure 4: UCD's robotic solution equipped with an AI-based computer vision growth monitoring system.

A recent study³⁶ found that one-third of workers in mushroom farming experienced injuries, with concerns about health and safety linked to workplace factors. Risks included poorly maintained facilities and hazardous production tasks, such as exposure to pesticides and physical strain. Back pain was a common issue, particularly attributed to work organization. Women expressed heightened worry about slips and falls. Considering these challenges, the study suggests that developing a harvesting mushroom robot is beneficial for improving working conditions in the industry.

AGROBOOST Proposed Solution: AGROBOOST will revolutionize the mushroom farming industry by creating a robotic solution for selective and gentle harvesting. It will design and manufacture a robot manipulator and end-effector, tailored to the unique characteristics of mushrooms such as high moisture content and brittleness. The solution will focus on biomechanical properties, incorporating DL models for accurate recognition and intelligent control. A ground-breaking algorithm for smart end-effector control will be developed, enabling selective harvesting with considerations for mushroom positioning, obstacle avoidance and human-machine collaboration. A concrete evaluation framework to assess the robot's performance in terms of picking efficiency, damage rate, missed picking rate and economic considerations, will also be developed.

The deployment of the robot will reduce the need for manual labour, mitigating risks and challenges associated with labour-intensive and monotonous tasks in agriculture. It will create opportunities for new jobs in robot maintenance, programming, and supervision, fostering skill development in the agricultural workforce. Additionally, the robot's automation will address workforce shortages by consistently performing tasks traditionally done by humans, offering flexibility in different shifts and schedules.

³⁵ <https://naturerising.ie/irish-mushroom-industry/>

³⁶ Sexsmith, K., et. al., (2021). Latino/a Farmworkers' Concerns about Safety and Health in the Mushroom Industry. J. Agromed., 27, 182.

Expected Outcomes: The implementation of the mushroom harvesting robot will significantly improve working conditions in agriculture by addressing worker safety aspects and alleviating strenuous tasks. The designed robot is expected to reduce the risk of work-related injuries by automating hazardous manual labour, minimize physical strain on human workers, enhance overall well-being and thus increase productivity. The robot's precision in identifying and selectively harvesting mature mushrooms will reduce accidental damage to surrounding crops. Furthermore, by adhering to strict standards, the robot will minimize waste and optimize crop yield, ensuring consistent and uniform quality in the harvested mushrooms.

KPIs to be measured: KPIs for the developed mushroom harvesting robot include a targeted **30%** reduction in harvesting time compared to traditional handpicking methods. The robot aims to achieve a **25%** boost in yield per labour hour through continuous and efficient operation. The goal is a significant **40%** reduction in workplace accidents associated with manual harvesting activities, enhancing overall safety. Precision in targeted harvesting is expected to lead to a **20%** reduction in resource usage, encompassing water and pesticides. Operational costs are targeted to decrease by **15%**, factoring in reductions in labour expenses and resource inputs. Creation or transformation of **10** new roles/jobs related to robot supervision, maintenance and technical support is envisaged.

1.2.5.3 Pilot 3 – Orchards 6.0 with AI based robots for tree pruning and flower thinning



Background, Motivation and Challenges: In 2017, apple orchards in the EU covered a total area of 473,500 hectares, with Poland alone accounting for 34.0% of this area (160,800 ha). Italy and Romania each contributed one-quarter, with 55,800 ha and 55,100 ha, respectively. Crucial tasks like pruning and flower thinning, dependent on human labour and chemicals, hold significant economic importance by contributing to fruit tree health and affecting the quantity and quality of harvested fruits. Labor shortages in traditional agriculture may require exploring alternative methods like mechanized pruning technologies, crucial to maintaining production levels, especially in countries facing labour scarcity. Advanced technology in agriculture needs more precise pruning technologies. Developments in robotics and AI could automate pruning, preserving horticultural principles. Thinning, is vital for orchard management, influencing fruit quality, tree health and productivity. It optimizes resources for remaining fruits, enhancing size and quality, meeting market expectations.

Thinning is labour-intensive, requiring careful manual selection. Labor shortages pose challenges, affecting costs and potentially delaying operations. Precision fruit farming aims for an ideal number of fruits per tree, demanding timely and skilled labour. Automated thinning solutions face development challenges. Climate changes impact thinning timing and effectiveness, requiring adaptive strategies. Managing waste from thinning operations for minimal ecological impact is crucial.

Both pruning and thinning are seasonal, labour-intensive tasks heavily dependent on skilled human labour, posing challenges due to harsh conditions, seasonal timing, chemical hazards, and negative psychological impacts. Despite efforts to introduce robotic systems in agriculture, automation levels for pruning and thinning remain low due to the complexity of conditions encountered.

AGROBOOST Proposed Solution: The proposed technological solution aims to develop an intelligent autonomous robotic system that will be able to learn the required pruning and thinning tasks from representative datasets and human inputs. This will be done through the use of existing data and second, through the AR powered decision support system. The system will include algorithms and models for additional processing of the acquired visual data. Inputs from skilled labour will be captured through a mobile device, displaying visual data augmented with related relevant information. This strategy, with incremental application, will reduce the burden of bulk annotation of acquired data. After training, the robotic system should be able to generalise the harvesting task to various environment variations. Furthermore, it will be able to operate both fully autonomously and within a shared autonomy/decision support control scheme where the human operators will provide control input, when necessary, remotely and based on the augmented visual data.

This solution will significantly improve the working conditions since: (1) task execution will be performed by a robotic platform avoiding unnecessary risks to the human operator; (2) monitoring of the tasks can take place remotely, increasing flexibility and reducing reaction times; (3) a skilled operator will monitor and, if necessary, remotely control multiple robotic platforms, thus minimising the negative impact of scarce skilled human labour.

Expected Outcomes: The automation of these agricultural tasks aims to address the scarcity of highly skilled and seasonal human labour, transforming an unattractive job into an efficient and automated process. It will also improve working conditions for manual workers, minimizing task burden and safety risks by enabling remote robot operation with minimal user input. This approach will reduce the reliance on chemical products, particularly in flower thinning, contributing to improved fruit and environmental quality. The incorporation of gamification, AR and robotics aims to increase the appeal and interest in farming jobs, making them more engaging and technologically advanced.

KPIs to be measured: The KPIs for this pilot include achieving a fully autonomous flower thinning with a **70%** success rate, while human shared-autonomous flower thinning is expected to reach a **100%** success rate. For pruning tasks, the goal is to achieve fully autonomous pruning with a **60%** success rate and attain a **100%** success rate for



Figure 5: The apple orchard and the INE's Modular-E robots in steep slope vineyards

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

human shared-autonomous pruning. A target has been set to reduce mental and physical effort for harvesting tasks by **80%**, emphasizing improvements in ease of manual labour associated with orchard management.

1.2.5.4 Pilot 4 – Mechanical Autonomous Selective Weeder

Background, Motivation and Challenges: The popularity of organic vegetable cultivation is rising as consumers express growing concerns about chemical consumption in fresh produce. This concern includes pesticides and herbicides. The global demand for organic fruits and vegetables has consequently increased, but limited supply persists due to high production costs³⁷. In 2024, the Organic Fruits and Vegetables Market is estimated at 46.19 billion USD, and it is projected to grow to 75.11 billion USD by 2029, with a Compound Annual Growth Rate (CAGR) of 10.21% during the forecast period (2024-2029)³⁸.

However, organic farming is considered labour intensive and is up to 50% higher compared to traditional farming³⁹ due to the minimal usage of chemicals and the reduced efficacy of bio-pesticides compared to the traditional ones. Weeding, which can no longer take place through the use of chemicals; instead, it has to be performed mechanically, is often mentioned as one of the main sources of labour in organic farming and, consequently one of the costliest.

Organic weeding costs, particularly for almonds and strawberries, can be significantly higher, up to 600% more for almonds and up to 2087 USD per acre for strawberries, compared to conventional methods⁴⁰. While inter-row weeding (between rows) is relatively straightforward, intra-row weeding (within the row) requires intelligent solutions to selectively destroy weeds. Labor shortages exacerbate the challenge, as finding personnel for this repetitive task is difficult. Even with workers, training is necessary to accurately differentiate between weeds and crops, slowing down the process. Technological solutions for organic farming are lacking, partly because many organic farms are not yet large enough for desired equipment depreciation periods. Additionally, small tractor sizes on most organic farms limit the use of heavy weeding implements.

AGROBOOST Proposed Solution: A vision system has already been developed by AUA for automating mechanical weeding in organic broccoli cultivation. During the project, a mechanical autonomous selective weeder will be developed consisting of a vision system, an autonomous navigation system, weeding implements, and AR solutions. The vision module will be mounted on a lightweight mechanical weeding implement capable of being mounted and carried by lightweight, small vehicles. In our case, it will be carried by a Husky robot with a max payload capacity of 75 kg. The autonomous navigation system will also be developed enabling the robot to navigate organic fields autonomously. Also, the lightweight selective mechanical weeding implement will be developed and paired with the vision system and adapted to work on various crops and weeds. Both vision models and actuation of the implements is currently performed using Python while the navigation of the Husky robot is conducted using ROS. The AR solutions will allow the end-user to see the field as the robot sees it, using the same computer vision algorithm. At the same time, they will allow non-experts to operate and perform minor repairs.



Figure 6: AUA's Husky robot with the vision module mounted.

The introduction of a mechanical autonomous selective weeder is poised to relieve agricultural workers of strenuous tasks, offering a safer alternative by minimizing exposure to hazards associated with manual labour. This innovative solution is anticipated to decrease production costs in organic farming by reducing the reliance on manual labour, making organic agriculture more appealing to farmers. If proven effective, the implement could be adopted by conventional farmers seeking to cut chemical costs and mitigate agriculture's environmental impact. Furthermore, the versatility of the vision device allows it to be mounted on implements usable by both conventional and autonomous tractors, potentially reducing initial costs for farmers embracing robotic solutions.

Expected Outcomes: The autonomous weeder will automate one of the most labour-intensive processes in agriculture, reducing the need for human workers to engage in risky or unattractive tasks. The robotic implementation will lessen the physical and mental strain on human workers by consistently handling weeding while minimizing the risk of slips and falls, especially in challenging conditions. The weeding implement will address workforce shortages by performing tasks traditionally done by humans, offering a solution in regions facing difficulties in finding and retaining skilled workers. The robotic implementation will be valuable in situations of seasonal labour shortages, allowing farmers to allocate labour efficiently among various tasks.

KPIs to be measured: Weeding efficiency will be measured by ensuring a minimum accuracy of **95%** in weed detection and removal. The solution's impact on reducing manual labour will be assessed by targeting a **50%** decrease in the average manual labour required for weeding 0.1 hectares, compared to traditional manual mechanical weeding practices. Additionally, an average satisfaction level of at least **80%** for the robotic weeding solution is set.

³⁷ R., Mohanapriya., et. al. (2022). Quality Enhancement of Fruits and Vegetables through Organic Cultivation, Agricultural Reviews.

³⁸ <https://www.mordorintelligence.com/industry-reports/global-organic-fruits-and-vegetables-market>

³⁹ Jansen, Kees. "Labour, livelihoods and the quality of life in organic agriculture in Europe." *Biological agriculture & horticulture* 17.3 (2000).

⁴⁰ Klonsky, Karen. "Comparison of production costs and resource use for organic and conventional production systems." *American Journal of Agricultural Economics* 94.2 (2012): 314-321.

1.2.5.5 Pilot 5 – Autonomus Management of Poultry Production Health and Welfare

Background, Motivation and Challenges: The intensive poultry production in the EU is facing criticism for limiting the egg-laying period to under 60 weeks and having a significant portion (45% in 2021) of hens reared in caged systems, affecting their well-being. The EU is transitioning to more welfare-friendly laying systems aligned with the Farm to Fork Strategy and Green Deal. Cage-free production poses challenges like feather pecking and abnormal egg laying, requiring careful daily farmer observations. Farmers currently address issues in poultry production through thorough daily inspections. This involves monitoring various aspects such as floor eggs, body condition, flock growth variation, respiration rate, plumage condition, signs of pecking, and the overall health of various body parts, including droppings, eyes, skin, beak, legs, feet, and claws. Additionally, farmers pay attention to external parasites, feed and water consumption and litter quality. Automation through robotic systems for poultry inspection can enhance precision in animal welfare management.

In recent years, attempts to develop robotic solutions for poultry management have seen varying degrees of success, primarily focusing on the broiler (meat bird) sector due to its perceived ease of operation. Challenges in the three-dimensional environment of laying hen housing systems have hindered the development of effective robotic solutions. While flock-level monitoring is now considered separately from individual animal monitoring, integration has not been achieved. Additionally, labour-intensive tasks like floor-egg and dead-bird picking lack automated solutions, with robotic applications limited to post-poultry building egg processing.

AGROBOOST Proposed Solution: In AGROBOOST, an automated flock-level monitoring and stimulation solution for poultry management will be developed. The livestock husbandry robotic solution will (a) include an integrated automated system for continuous and objective monitoring of laying hen health and welfare. This involves augmenting the environment to prevent negative behaviours and operates at both flock and individual animal levels; (b) a mobile platform will be designed to pick up objects like eggs and dead birds from the litter, aiming to reduce the time farmers spend in the barn and enhance the value of their visits. The solution will incorporate fixed flock-level PTZ-camera monitoring and a mobile quadrupedal robotic platform equipped with manipulators. Cutting-edge machine learning techniques support both fixed and mobile solutions, with the quadruped addressing manoeuvrability challenges faced by traditional wheeled vehicles. The pilot will also include a quadruped legged robot with a robotic arm, equipped with RGB-D and thermal sensors for poultry evaluation and egg manipulation. Integrating the system with VR/AR in poultry production environments is planned, providing farmers with VR-goggles for information on challenging-to-measure animal conditions.



Figure 7: KUL's quadruped legged robot with a robotic arm, equipped with RGB-D and thermal sensors.

The proposed poultry monitoring solution goes beyond the current state of the art in poultry technology by combining fixed and mobile robotic solutions towards poultry house management. The main advantage of the system will be the capacity to act locally when a deviation at the flock-level is identified, which act to prevent the emergence of health and welfare related problems before they become serious. The linking of the system with AR, so that farm workers can get objective understanding of the condition of their animals will eventually lead to an increase in the appeal of working in the sector, assisting farm workers in being more precise while taking the drudgery out of their daily tasks.

Expected Outcomes: The solution will represent a transformative leap forward for the industry, fostering cage-free systems that will minimize health and welfare concerns while simultaneously improving efficiency and well-being for farm workers. The combination of fixed and mobile robotic technologies will allow for health monitoring, early detection of issues and timely intervention, either autonomously or with the aid of VR technology. This approach will not only enhance worker conditions by reducing exposure to challenging environments but will also significantly improve animal welfare by minimizing stress and enabling precise procedures like floor egg and dead bird picking.

KPIs to be measured: The success of robotic task execution will be measured by the mobile robotic system autonomously achieving an egg-picking success rate of **80-90%**, evaluated in both lab and commercial production-like environments. Also, the project aims to demonstrate improved animal welfare, targeting a **20-50%** enhancement through the integration of the robotic system. Additionally, the efficiency of farm work is targeted to increase by **50%**, tracked by monitoring the frequency of workers entering the poultry building with and without the integrated system. Last, the well-being of workers be enhanced by **50%**, assessed through questionnaires.

1.2.5.6 Pilot 6 – Robotisation of understock weeding and mowing in steep slope viticulture

Background, Motivation and Challenges: Organic wine production has increased significantly in the past years and the market is expected to continue to grow until 2030. The trend has been supported by consumer demand associating organic wine with health and environmental benefits⁴¹ and is expected to continue with anticipated policies in European countries and the EU to restrict synthetic chemicals use. Organic

⁴¹ D. Moscovici, *et al.*, (2022) Consumer preferences for organic wine - Global analysis of people and place, Journal of Cleaner Production.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

viticulture requires more mechanical tasks to be performed either through human labour or machinery. Some of these tasks, such as work around the understock, including mowing, weeding or aerating the soil are physically demanding and must be performed regularly all throughout the year. Because of the overall labour shortage in viticulture⁴², finding qualified labour is especially difficult. There are ongoing efforts to robotize some of these tasks but such robots have been mainly used in large vineyards. They are not tailored to work on smaller plots or steep slope and terraces that characterize production areas for some premium terroir wines.

Mowing around the understock is an essential practice in viticulture. It reduces competition between the vines and growing grass as well as other weeds for essential nutrient as well as for water, while maintaining a minimal cover that enables the management of soil moisture effectively. It might also contribute to pest control by reducing the risk of transfer of pest from tall grass to the grapes and leaves. Furthermore, it contributes to soil health by adding organic matter as the clippings decompose. Targeted robotized mowing can help finding an optimal balance between managing competition and benefitting from the grass cover.

AGROBOOST Proposed Solution: The proposed technological will explore soft robotics solutions⁴³ and develop light weight robots, which can also morph into drones and be deployed as ‘rovers’ around the understock for mowing and potentially perform weeding and soil aeration tasks. They can then morph into drones to overcome walls of the vineyards, and capture data above the canopy. These robots have minimal environmental disturbance or soil erosion due to their lightweight structures and can operate for long periods of time due to their high efficiency. The development of a fleet of light weight, autonomous compliant robots, will lead to precision mowing and other targeted interventions in around the understock. A learnt controller will be demonstrated through VR enabling the robots to robustly operate in these challenging environments.



Figure 8: EPF's morphing compliant robot in action.

This solution will significantly improve the working conditions since it will reduce workers need to perform tasks on the ground in steep or constrained terrain, thus reducing fatigue and potential musculoskeletal conditions. The proposed innovations will be tested and demonstrated in Switzerland on plots in steep terraced vineyards of the UNESCO world heritage region of Lavaux on the shore of Lake Geneva. Because of its world cultural heritage status, the stonewalls delimiting the terraced plots cannot be removed. Working on this terrain with heavy robots is not possible. The Lavaux vineyards will provide an ideal testing ground for the development of lightweight compliant robots that can morph into drones which could provide a game changing solution to local vineyards.

Expected Outcomes: The morphing compliant robot will demonstrate a significant potential to reduce human intervention on the ground, enhancing efficiency and minimizing the need for manual labour in certain agricultural tasks. This reduction in human involvement contributes to improved working conditions by alleviating the physical burden on agricultural workers and reducing the risk of fatigue and injuries associated with repetitive or strenuous tasks. Moreover, their use will minimise the environmental impact of agricultural practices. Specifically, in the context of weeding, the robots can employ precision techniques that target specific areas, thus reducing the need for widespread herbicide and pesticide spraying.

KPIs to be measured: Effective mowing around the understock **80%** with autonomous moving in slope and between terraces achieved at **70%**. Human intervention around the understock will be reduced by 45% demonstrating a significant reduction of operation costs of at least **30%**.

Business and/ or governance models for the use of developed solutions: Business models in agro-robotics will involve startups and SMEs specializing in fruit harvesting, offering intelligent robotic systems to farmers through subscription, lease-based model (with the farmer making use of them only during crucial periods, e.g., when crops are sensitive to weeds) or pay-by-use (per area covered) models, reducing the initial capital investment for farmers and ensuring ongoing support and upgrades. These systems, adaptable to various agricultural tasks through a learning-from-demonstration approach, can be easily transferred and reused. Agro-robotics companies will also be able to provide operator training in AR/VR technologies, develop field infrastructure for efficient data transfer and telepresence control during harvesting. Moreover, a governance-based model like industry certification can be established for harvesting robots to ensure compliance with safety, quality and environmental standards.

Social innovation aspects derived by the use of technologies in the pilots: The social innovation achieved through (a) the attraction of young farmers due to the gamification elements for performing the harvesting tasks will be enhanced along with the inclusion of farmers with disabilities since they will be able to monitor and execute tasks remotely through a VR environment; (b) training and skill development programs can be developed to familiarise farmers, farm workers and relevant stakeholders with robotic technology with community-specific needs and cultural contexts; (c) robot's automation can alleviate physical strain and reduce dependence on physical labour, thus be particularly beneficial for women in agriculture and farmers with moving disabilities; and (d) older farmers will be

⁴² (2023) EURES Report on labour shortages and surpluses 2022. European Labour Authority

⁴³ Armanini, C., Junge, K., Johnson, P. H., Whitfield, C., Renda, F., Calisti, M., & Hughes, J. (2024). Soft robotics for farm to fork: applications in agriculture & farming. Bioinspiration & Biomimetics.

able to stay longer in the sector (should they like to) addressing key problems through the use of emerging robotics and AR technologies.

1.2.6 Coordination and implementation of the Open Calls (FSTP)

Two Open Calls to third parties will be organized for attracting EU associates to implement their products and use cases using the **AGROBOOST** technologies and available solutions. These Open Calls will be the key to establish the envisioned ecosystem around European actors, SMEs, startups, as well as, around research and innovation communities, initiatives and players and adopters of the **AGROBOOST** solutions, helping to put the foundations for the project's ecosystem's sustainability, and to disseminate further the planned and expected outcomes. In this respect, two Open Calls (FSTP - cascade funding) are planned, which will be coordinated by **MAR** that is widely recognized for its experience in handling such activities, in collaboration with key partners comprising the Open Calls Committee (OCC). Following the planned work and the expected deliverables, the implementation of the third-party projects will be within M12-M18 for the first Open Call and M34-M40 for the second Open Call, with a budget of 60.000€ each. **UCD**, the coordinator, will be the Open Calls treasurer, holding the overall 500.000 € budget. It is expected that the planned activities will achieve a **TRL7-8** up to the end of their implementation and that they will bring additional use cases, testbeds and applications, beyond the planned ones under the project's piloting settings, in the same or additional geographical areas. Our third-party partners will be having access to (i) training on the material and specifications, (ii) our tech experts, (iii) peer-to-peer knowledge and experience sharing, (iv) use cases that best fit their app/services, and (v) workshops/webinars promoted by the project. In addition, the non-funded applicants of the Open Calls will be invited to participate in the public activities and events of the project. Open Calls are further detailed in the Annex.

1.2.7 Links to other national and international projects and relevant associations

Project and Relevant to AGROBOOST Activities	Partners
ROBS4CROPS (H2020, GA: 101016807) – ROBS4CROPS will build upon the existing agricultural machinery, standards and best practices and deliver a flexible and modular, fully autonomous system that will significantly reduce farmers' dependency on hired labour, increase safety and reduce the overall carbon footprint of food production. AGROBOOST will incorporate best practices and technical innovations in the robotics and advance the robotics solutions targeting enhanced working conditions and inclusivity for disable people and women in agriculture.	AUA
agROBOfood (H2020, GA: 825395) - agROBOfood aims to accelerate the sector's digital transformation through the adoption of robotic technologies, establishing a sustainable network of digital innovation hubs. AGROBOOST will examine seamless integration with the technological advancements.	AUA, INE
DEMETER (H2020, GA: 857202) – DEMETER was a large-scale project that analysed data obtained from a wide range of actors (production sectors and systems) to provide an integrated interoperable data model enabling optimal resource management in the European agri-food sector. AGROBOOST will capitalise on the mechanisms of the developed intelligent Decision Support Systems and further enhance them to match the needs of workers for enhanced working conditions in agriculture.	INE, NCI, NVR
NOVATERRA (H2020, GA: 101000554) – NOVATERRA will investigate novel ways to reduce the use of contentious plant protection products in crops through smart farming technologies. AGROBOOST will benefit from NOVATERRA's expertise in sustainable farming practices and integrate these innovative approaches into the project's framework.	INE, AUA
WATSON (HEU, GA: 101084265) – WATSON will develop a set of tools and systems that can detect and prevent fraudulent practices across the entire food supply chain, increasing transparency through improved track-and-trace mechanisms for accurate, time-relevant and untampered information on the food product throughout its whole journey. AGROBOOST will examine the foundational AI algorithms developed for the Early Warning System, enhancing the functionality of the XAI-enabled Decision Support System for socially responsible agricultural practices.	UCD, FIB, NTU, INE, NCI,
PRUDENT (HEU, GA: 101134051) – PRUDENT will study and test “green nudges” in various farming systems across Europe, using innovative technologies. The project seeks to develop social innovations, business models and policy recommendations to facilitate this transition, involving stakeholders at multiple levels in co-creation activities for a comprehensive understanding and effective implementation of sustainable practices. AGROBOOST will share insights, finding and innovations fostering knowledge exchange and explore opportunities for joint research activities and policy advocacy efforts.	AUA, G&D

1.2.8 Interdisciplinarity of AGROBOOST

One of the biggest challenges facing humanity is achieving a sustainable balance between economic development, environmental conservation, and social equity. To create safe working environments and improve the competitiveness in agriculture, a holistic, multidimensional, interdisciplinary (synthesising and harmonising links between disciplines into a coordinated and coherent manner), transdisciplinary (integrating sciences in a humanities context and transcending their traditional boundaries)⁴⁴ and systemic approach for better understanding agro-robotics and the needs of farmers is a prerequisite for fostering transition towards sustainability needs. Major technological

⁴⁴ Choi B.C.K., Pak, A.W.P. (2007). Multidisciplinarity, interdisciplinarity, and transdisciplinarity in health research, services, education and policy: Clinical & Investigative Medicine, 30(6), 224. <https://doi.org/10.25011/cim.v30i6.2950>

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

advances in computing power, robotics, automation, AR/VR and AI/ML have provided new tools to improve the working conditions of workers in agriculture, leading to comprehending issues related to agricultural production. Under this scope, **AGROBOOST** aims to effectively bring together diverse actors to trigger social, cognitive and scientific dynamics and develop a solution that will thoroughly address all the issue's aspects. The project recognises the difficulties that may arise from trans-disciplinary collaboration (e.g., different language and terminology among different disciplines, varying methods of work, different levels of scientific understanding and cognitive structures, etc.⁴⁵) and will adopt a proactive approach towards them by following the fundamentals of the Methodology in Interdisciplinary Research (MIR) framework.

1.2.9 Integration of social sciences and humanities

Integrating social sciences and humanities (SSH) disciplines into the research process ensures a holistic approach that efficiently considers the ethical implications, legal frameworks and the human dimension of the use of emerging technologies for enhanced working conditions. This collaboration not only strengthens the scientific foundation of the proposed approach but also addresses the broader social, ethical and legal implications of its methods. To maximize the returns to society, **AGROBOOST** has been organized in a way that Socio-Economic Sciences and Humanities (SSH) are in perfect balance with research and innovation activities, with **SSI** leading the related activities of WP2, assisted by **FAD**, a Spanish women association in agriculture and **AJP**, the Portuguese association of young farmers. Towards this holistic approach, the project will base its research around the following pillars: (a) understanding the intricate working conditions prevalent in agriculture [T2.2], which will serve as a foundational step and lay the groundwork for informed decision-making and policy formulation [T6.5]; (b) delving into the acceptance and adoption of technology in agriculture [T2.3] for sustainable practices; (c) assessing the social added value of the project in the agricultural sector and therefore its overall contribution to the inclusiveness and sustainability of rural communities [T2.5]. Thus, the technologies and methodology of **AGROBOOST** are designed with an inherent SSH dimension and feature activities directly aimed at working “with and for European society”, addressing the economic, environmental and social problems that will enable the adoption of the proposed solutions.

1.2.10 Consideration of gender dimension in research and responsible R&I

Gender consideration: Considering that the gender dimension in the proposed research is paramount, given the disproportionate impact of agricultural technologies and practices on women, who often play integral roles in farming and rural communities, **AGROBOOST** acknowledges the necessity to consider the gender dimension in achieving project excellence. Possessing necessary experience gained in this field in previous projects, **SSI** will guide the consideration of gender aspects in the project's activities. The project's global methodology will incorporate the sex/gender dimension as a quality control mechanism. Gender aspects will be examined systematically throughout the project development, including co-design methods and local population surveys. From the viewpoint of the gender dimension, **AGROBOOST** stands out as a project in which very specific socio-economical aspects of women labour in a traditionally men-dominated area of agriculture will be investigated in the activities linked with all pilot validation campaigns, capitalising on the research performed for inclusive technologies for women in agriculture [T2.5] and the participation of **FAD**, the Spanish women association in agriculture, in the project. All the related gender aspects will be considered under T1.4 activities. Gender mainstreaming is not considered a separate activity or module, but rather it is integrated within all phases and activities of the project. This mainstreaming includes the following principles: (i) consider gender issues when collecting, analysing and presenting all statistics and data; (ii) introduce gender related planning criteria as a standard; (iii) make sure that gender-balance underpins all participatory processes (i.e., elicitation of user needs [T2.1], pilot evaluation workshops [T5.10], capacity building and training activities [T6.4]), (iv) use gender-sensitive indicators to evaluate impacts of **AGROBOOST** innovations, (v) build targeted objectives for including gender equality and women's empowerment into the plans and budgets, (vi) establish links with the EU platform of women scientists to promote gender equality in research.

1.2.11 Open science practices

The EC has set off the ambition of the [European Open Science Cloud \(EOSC\)](#), a trusted environment for hosting and processing research data thereby enhancing scientific endeavours across the EU, where **NCI** has a leading technical role and thus will lead the support of these initiatives and other related activities aimed at integrating the project's results within the framework of open science. Resonating with the principle 'as open as possible', **FIDEL** wholeheartedly supports the goals of the [Open Science Policy under Horizon Europe](#) and thus, appropriate open science actions will be implemented as an integral part of its proposed methodology as described below. In its attempt to achieve increased rigour, accountability, transparency and reproducibility of research through sharing knowledge and data as early as possible in the research and development processes with all relevant actors, **AGROBOOST** will incorporate open science principles/practices as integral parts of its overall work-plan. Knowledge and data sharing among all relevant actors, i.e., researchers across universities, farmers, research centres and industry, as well as its dissemination to other stakeholders, including public authorities and policymakers and inquiring broad public in general, will take place regularly, as early as possible, namely towards stimulating creativity and trust in the R&I process. As supported by EC, open access will be assured, and thus, all scientific publications generated from **AGROBOOST**'s developments and outcomes, as well as other relevant material, will be appropriately disseminated and made available to the community, free of charge, online, for any user.

⁴⁵ <https://link.springer.com/article/10.1007/s11229-016-1236-4>

1.2.12 Research data management and management of other research outputs

AGROBOOST is committed to a comprehensive and strategic management of research data generated and collected throughout the project's duration. A well-defined Data Management Plan (DMP) will be crafted during the early stages of the proposal, with detailed and updated versions to be reported in deliverables D1.2 (M06) – D1.4 (M60), incorporating the output of T1.4. Adhering to the Horizon Europe Data Management Plan Template and utilizing tools such as the [Argos](#) online machine-actionable tool for DMPs and the [Amnesia](#) service for sensitive data anonymization from the OpenAIRE Open Science Lens, the **AGROBOOST** DMP will address critical aspects, including data summary, FAIR (Findable, Accessible, Interoperable, Reusable) principles, resource allocation, data security, and ethical considerations. Compliance with the General Data Protection Regulation (EU 2016/679) and other relevant guidelines will be ensured. More specifically, in the pursuit of making research data FAIR, **AGROBOOST** will implement measures such as: a) **Findability**: Utilizing trusted open-access data repositories like Zenodo and Open Data Repository, incorporating persistent identifiers (e.g., DOI), and providing rich metadata using standard descriptive formats like DataCite Metadata Schema. b) **Accessibility**: Ensuring prompt open availability of data, accounting for embargo periods for intellectual property protection, restricted access for verification, and providing necessary software documentation/reference. c) **Interoperability**: Adopting widely known and standard file formats, employing controlled vocabularies for metadata formulation (e.g., UNESCO thesauri), and adhering to common protocols (e.g., Open Archives Initiative Protocol for Metadata Harvesting) for (meta)data interoperability. d) **Reusability**: Implementing common types of licenses for data sharing/reuse (e.g., Creative Commons, Open Data Commons), using open tools for data interpretation/reuse, providing comprehensive documentation (e.g., readme files, codebooks, data cleaning information), and applying relevant data quality assurance processes. This strategic approach ensures the optimal utilization and accessibility of **AGROBOOST**'s research data while upholding ethical standards and legal requirements. #@CON-MET-CM@# #@COM-PL-CP@# #REL-EVA-RE\$#

2 Impact #@IMP-ACT-IA@#

2.1 AGROBOOST pathways towards impact

2.1.1 Contribution to the expected outcomes and impacts set out in the Work Programme and Destination

AGROBOOST Pathway Towards Outcomes and Impacts
AGROBOOST Expected Outcomes
OUT1 – Enhanced working conditions in agriculture (including increased safety of workers and reduced drudgery) through innovative digital solutions exploiting the potential of augmented reality. AGROBOOST Pathway to Outcomes: AGROBOOST aims to significantly enhance working conditions in agriculture by harnessing the potential of robotics technologies, Augmented Reality and Artificial Intelligence, automating labour-intensive tasks, reducing physical strain on workers and increasing overall productivity. The development of AR applications [WP3] will provide real-time training and guidance to workers in the agricultural sector, offering step-by-step instructions and safety guidelines thus, enhancing their skills and reducing the risk of accidents. This approach will not only contribute to increased safety but will also address the imperative need for continuous skill development among agricultural workers. Several innovative AR features for hazard identification and safety alerts will be integrated highlighting potential hazards in the agricultural environment, such as machinery accidents, exposure to chemicals (pesticides, herbicides, fertilizers, etc.), injuries due to ergonomic strains, risks of injuries due to kicks, bites, etc., exposure to high levels of noise, heat-related illnesses due to prolonged exposure to high temperatures, biological hazards (e.g., exposure to bacteria, viruses, fungi, etc.), promoting a safer workplace and preventing accidents. Additionally, AGROBOOST emphasizes the integration of AR with robotic solutions [WP4] to automate repetitive and physically demanding tasks, further reducing physical strain on workers and improving overall productivity. This integration will assist workers to visualise and control automated processes, reducing the need for manual labour in outdoor environments. Last, the project has included a dedicated in its WP structure to further analyse the safety concerns of workers in agriculture, the drudgery associated with different tasks and how they can be efficiently addressed [T2.2]. This analysis will complement the development activities thus providing a holistic perspective to the project's results. Outcomes Metrics: OUT1.1 ≥ 20% accidents reduction; OUT1.2 ≥ 20% reduction of the workers absenteeism related to farmers' injuries or health issues caused by unsafe working conditions; OUT1.3 ≥ 60% improvement of the workplace ergonomics; OUT1.4 ≥ 25% reduction of the costs associated with safety incidents, healthcare expenses and operational disruptions; OUT1.5 ≥ 95% of the participants in the pilot activities provide positive feedback. Project's Objectives: Obj.1, Obj.4 Related WPs: WP2, WP3, WP5
OUT2 – Lowered environmental impacts and productions costs and increased product quality in and through the use of digital technologies, through robotics and augmented reality in particular. AGROBOOST Pathway to Outcomes: The deployment of the AGROBOOST technological innovations will allow for precise, efficient and sustainable farm management practices, reducing the need for resource-intensive processes and minimizing environmental impacts. For this purpose, a Decision Support System will be developed [T4.5] that will assist farmers towards the optimization of the application of fertilizers, pesticides and water resources, promoting precision agriculture techniques that enhance sustainability and reducing the environmental footprint of farming operations. AR application will be leveraged [WP3] to enhance the decision-making process for farmers, providing real-time insights into crop health, soil conditions and environmental factors and allowing them to optimize the use of resources. Moreover, farmers through the innovative AR interfaces will be able to identify and address issues promptly, thus improving the crop yields and its overall product quality while also minimizing the food loss. The proposed integration of AR into training

curricula [T6.4] will also contribute to increased operational efficiency, reducing the production costs and enhancing the overall sustainability of agricultural practices.

Outcomes Metrics: **OUT2.1** ≥ 20% reduction of the water resources used in agricultural activities; **OUT2.2** ≥ 20% reduction in overall carbon footprint associated with farming operations; **OUT2.3** ≥ 40% in the use of fertilizers and pesticides per unit of production; **OUT2.4** ≥ 30% reduction in the time needed for specific agricultural tasks (e.g., harvesting, weeding, etc.) through robotics and automation; **OUT2.5** ≥ 15% in the overall farm profitability, considering the impact of cost reductions and improved product quality but also the affordability of the proposed solutions; **OUT2.6** ≥ 25% reduction of system failure rates and downtime; **OUT2.6** ≥ 20% reduction of the energy consumed per unit of work performed; **OUT2.7** ≥ 25% reduction of the costs associated with operational disruptions; **OUT2.8** ≥ 15% reduction in agricultural waste (in particular food loss reduction through precise and automated harvesting that assists in minimizing damage to crops and reduce losses during harvest). || **Project's Objectives:** Obj.1, Obj.2 || **Related WPs:** WP3, WP4, WP5

OUT3 – Reduced share of risky or unattractive actions/tasks to be performed by workers through automation-based solutions.

AGROBOOST Pathway to Outcomes: **AGROBOOST** will start its activities with a thorough analysis of the tasks identified as risky or unattractive within the agricultural sector [T2.2] in order to deeper understand the current working conditions and highlight priority areas for intervention through the automation-based solutions, robotics and AR technologies. This analysis will contribute to the design and development of ground-breaking robotic solutions tailored to automate the identified high-risk or unattractive tasks such as the ones addressed in the pilot validation campaigns [T5.4 – T5.8]. Hazardous tasks like pesticide application or repetitive actions such as weeding will be delegated to robotic systems equipped with precision sensors and machine learning algorithms. The introduction of robotic solutions will not only reduce the exposure of workers to unsafe conditions but will also eliminate physically demanding tasks, increasing agricultural workers' job satisfaction. Moreover, user-friendly AR interfaces for workers [T3.4] will be developed providing them the opportunity to remotely perform dangerous tasks, supervising the automated systems. The whole process will ensure that workers remain actively engaged in the agricultural processes, steering away from unattractive tasks and transitioning towards more supervisory and skill-intensive roles. These interaction tools will promote a positive shift in the nature of agricultural work, making it more appealing and reducing the reliance on manual labour for hazardous or repetitive actions.

Outcomes Metrics: **OUT3.1** ≥ 30% increase of overall tasks automation; **OUT3.2** ≥ 85% increase of the user acceptance for the AI-generated recommendations; **OUT3.3** ≥ 35% decrease in tasks that require manual handling by workers; **OUT3.4** ≥ 80% reduction in both mental and physical effort for harvesting tasks; **OUT3.5** ≥ 50% decrease in the average manual labour required for weeding 0.1 hectares, compared to traditional manual mechanical weeding practices; **OUT3.6** ≥ 10 new roles or jobs related to robot supervision, maintenance and technical support; **OUT3.7** ≥ 30% reduction in harvesting time compared to traditional methods; and **OUT1.1**, **OUT1.2**, **OUT1.3** and **OUT1.5**. || **Project's Objectives:** Obj.1, Obj.3, Obj.5 || **Related WPs:** WP2, WP3, WP4, WP5

OUT4 – Mitigated shortage of work force in agriculture in some sub-sectors through automation-based solutions.

AGROBOOST Pathway to Outcomes: **AGROBOOST** will develop and deploy robotic technologies that can seamlessly integrate into tasks traditionally carried out by human labour [WP4], particularly in areas facing acute shortages. These technologies will automate repetitive and physically demanding tasks such as harvesting, weeding, flock-level and crop monitoring, alleviating the strain on the available workforce and enhancing the overall productivity. Through the use of AR, AI and advanced robotics, **AGROBOOST** will enhance the efficiency of agricultural operations. AR applications will provide intuitive interfaces for workers to interact with robotic systems, reducing the learning curve and making automation accessible to a broader range of individuals [WP3]. Moreover, novel solutions will be developed based on these technologies that cannot only replace manual labour but also improve the working conditions in agriculture. For instance, the deployment of robotic systems equipped with safety features and failure-avoidance mechanisms will contribute to increased safety for workers. **AGROBOOST** is built upon the need for a holistic approach to address workforce shortages, considering factors such as training, adaptability and integration into existing agricultural practices. For this purpose, the viability and effectiveness of automation-based solutions in mitigating workforce challenges will be demonstrated [WP5] focusing on sub-sectors where labour scarcity is a critical issue. Through the development of scalable and adaptable technologies, **AGROBOOST** aims to offer sustainable, long-term solutions that contribute to the resilience and competitiveness of agriculture while fostering a positive impact on working conditions.

Outcomes Metrics: **OUT4.1** ≥ 30% increase in overall labour productivity; **OUT4.2** ≥ 30% decrease in the number of unfilled job positions; **OUT4.3** ≥ 35% increase of the inclusivity across different demographic groups, with a specific focus on youth, women and persons with disabilities; **OUT4.4** ≥ 75% increase in the retention rate of skilled workers within targeted sub-sectors; **OUT4.5** ≥ 35% increase in the allocation of the workforce towards higher-skilled roles; **OUT1.3**, **OUT1.5** and **OUT3.6**. || **Project's Objectives:** Obj.4, Obj.5 || **Related WPs:** WP2, WP4, WP6

AGROBOOST Expected Impacts

IMP1 – innovative governance models enabling sustainability and resilience notably to achieve better informed decision-making processes, societal engagement and innovation;

AGROBOOST Pathway to Impacts: **AGROBOOST** recognises that transformative changes in agriculture require collaborative and adaptive governance structures. For this purpose, novel governance models will be established leveraging cutting-edge technologies, such as AI, data analytics, robotics and decentralized platforms. The development of smart decision support systems [T4.5] that integrate real-time data from various sources will enable farmers and policymakers to make informed decisions based on accurate and up-to-date information, enhancing transparency and trust among stakeholders. This approach will streamline decision-making processes, ensuring that all relevant parties have access to the

same data, fostering collaboration and consensus-building. In addition, **AGROBOOST** will prioritize societal engagement through participatory models that will involve farmers (young and women), farm workers, advisors, technology providers and scientists in decision-making processes through a multi-actor approach [T2.1]. The integration of social sciences and humanities disciplines will ensure that the diverse needs and requirements of different stakeholders are considered [T2.3]. This inclusive governance model will empower local communities and promote the co-creation of solutions that align agricultural practices with societal expectations and values [T2.4]. Through this approach, **AGROBOOST** will encourage continuous innovation and adaptive strategies that enhance sustainability and resilience in agriculture [T6.5].

Related Outcomes' measures: OUT1.3, OUT1.5, OUT2.1 – OUT2.8, OUT3.6, OUT3.7 and OUT4.1 – OUT4.5.

Wider Impact Metrics: IMP1.1 – increase in stakeholder engagement and participation; IMP1.2 – increase in innovation adoption due to informed decision-making; IMP1.3 – improvement in societal well-being and community resilience; IMP1.4 – increased inclusiveness of diverse voices in decision-making; IMP1.5 – expansion of the innovation ecosystem within the agricultural sector; IMP1.6 – improved resilience demonstrated in the face of external challenges or disruptions.

IMP2 – sustainability performance and competitiveness in the areas covered by Cluster 6 are improved through further deployment of digital and data technologies as key enablers;

AGROBOOST Pathway to Impacts: **AGROBOOST** will revolutionize traditional agricultural processes through the implementation of robotic solutions and augmented reality applications for hazardous or unattractive tasks such as harvesting, weeding, crop monitoring and animal husbandry, enhancing precision and efficiency while minimizing resource use [WP3]. The proposed Decision Support System [T4.5] will promote (a) circular agriculture through the integration of waste reduction, recycling and resource reuse principles and (b) climate-resilient agricultural practices that can mitigate and adapt to changing climate conditions. A unique monitoring system that will analyse the consequences of the adoption of digital technologies will be developed [T5.11] that will provide evidence and advocate for informed decision-making at multiple levels. It will offer a deeper understanding of the impact of digital technologies on various facets of agriculture, including working conditions, environmental sustainability and economic competitiveness. This evidence-driven advocacy will ensure the continuous improvement of agricultural practices, aligning with the overarching goals of enhancing sustainability and competitiveness in the sector.

Related Outcomes' measures: OUT1.5, OUT2.1 – OUT2.8, OUT3.3 – OUT3.6 and OUT4.1 – OUT4.5,

Wider Impact Metrics: IMP2.1 – reduction of resource consumption, emissions and waste generation in additional agricultural fields; IMP2.2 – reduction in production costs; IMP2.3 – improved productivity in the agricultural sector; IMP2.4 – increased employment rate and income level of workers in agriculture; IMP2.5 – improved physical and mental well-being of workers and farmers; IMP2.6 – expansion of market access and diversification of agricultural products.

IMP3 – stakeholders and end users including primary producers and consumers are better informed and engaged thanks to effective platforms such as AKIS;

AGROBOOST Pathway to Impacts: **AGROBOOST** will connect with established robust AKIS platforms to disseminate the project's findings, technological advancements, insights, lessons learnt and best practices to various stakeholders' groups such as primary producers, food processors, retailers and consumers. These platforms will serve as centralized hubs for sharing information, providing real-time updates on project developments and offering accessible resources and training courses for farmers and consumers to stay informed about the latest innovations in agricultural robotics and augmented reality. Their interactive features, such as webinars, forums and user-friendly interfaces, will foster a collaborative environment, encouraging active participation and feedback from end users. In addition, **AGROBOOST** will implement targeted awareness campaigns [T6.2] to promote these AKIS platforms through strategic partnerships with agricultural extension services, farmer associations and consumer advocacy groups, ensuring their widespread adoption. Training programs will be conducted [T6.4] to empower farmers with the necessary digital skills to navigate and benefit from AKIS tools, while consumer-focused content will be tailored to enhance understanding and appreciation for sustainable and technologically advanced agricultural practices. Last, the project will establish multi-stakeholder fora and advisory groups within the AKIS platforms, encouraging dialogue and knowledge exchange and facilitating direct engagement among actors in the agricultural ecosystem.

Related Outcomes' measures: OUT1.3, OUT1.5, OUT2.1 – OUT2.8, OUT3.1 – OUT3.7 and OUT4.1 – OUT4.5,

Wider Impact Metrics: IMP3.1 – increased access of active users on the AKIS platform; IMP3.2 – increased volume (from additional resources) and type of content users are interacting with; IMP3.3 – increased user engagement and interaction; IMP3.4 – increased growth in the AKIS user community; IMP3.5 – improved consumer awareness and understanding of sustainable agricultural practices; IMP3.6 – increased number of cross-sector collaborative projects initiated through connections made on AKIS.

IMP4 – strengthened EU and international science-policy interfaces to achieve the Sustainable Development Goals;

AGROBOOST Pathway to Impacts: **AGROBOOST** will actively engage with relevant science-policy platforms, institutions and policymakers at both EU and international level, establishing strong collaborations, participating in policy dialogues and contributing to expert working groups in order to facilitate the rapid integration of the project's research findings and recommendations into the policy discourse. To ensure a robust science-policy interface, **AGROBOOST** will prioritize the development of evidence-based policy recommendations aligned with specific SDGs, such as SDG 1: No poverty, SDG 2: Zero Hunger, SDG 3: Good health and well-being, SDG 8: Decent Work and Economic Growth, SDG10: Reduced Inequalities, SDG 12: Responsible Consumption and Production and SDG 13: Climate Action. These recommendations will be communicated to policymakers through targeted reports and presentations, emphasizing on the tangible benefits of **AGROBOOST**'s technological innovations for achieving

sustainable development. Moreover, **AGROBOOST** will contribute to the SDGs by actively participating in international forums and initiatives related to agricultural sustainability in order to strengthen its influence on shaping policies that promote SDG-aligned practices and address pressing challenges in the agricultural sector. Last, the project will contribute to capacity building by organizing training programs and workshops for policymakers, ensuring they are well-equipped to understand and integrate its findings into policy frameworks.

Related Outcomes' measures: **OUT2.1 – OUT2.8, OUT3.6, OUT4.2 – OUT4.3**

Wider Impact Metrics: **IMP4.1** – increased number of actionable policy recommendations from AGROBOOST incorporated into EU and international agricultural policies; **IMP4.2** – positive impact on shaping and influencing agricultural policies at the EU and international levels; **IMP4.3** – increased number of international collaborations established to advance science-policy interfaces; **IMP4.4** – increase in public awareness and advocacy efforts related to AGROBOOST's policy recommendations; **IMP4.5** – increased use of AGROBOOST's scientific publications and reports in policymaking documents; **IMP4.6** – improved integration of policy recommendations into national and international policies addressing SDGs.

2.1.2 *Barriers, obstacles and conditions that determine impact's success*

In addition to the strategic impacts outlined in the call, **AGROBOOST** aims to generate substantial effects achievable either during the project lifecycle or following the comprehensive adoption of project outcomes by a diverse group of stakeholders. The consortium is fully dedicated to mobilizing all necessary resources to realize these envisioned impacts. Dissemination, stakeholder engagement, and communication activities (refer to section 2.2) are anticipated to play a crucial role in amplifying the anticipated **AGROBOOST** impacts. However, despite the tangible nature of the project impacts, certain external factors and barriers, beyond the scientific, business, and technical risks identified in section 3, exist over which the project consortium has limited control. These factors may potentially impede the achievement of said impacts. Given their broader scope, these barriers will be meticulously considered and appropriately addressed, as they have the potential to significantly influence a critical success factor such as user acceptance. The subsequent table outlines these external assumptions and barriers.

	Barriers	AGROBOOST Counter Actions
Political	- Changes in agricultural policies or regulations at the national or European levels may pose challenges to the adoption of AGROBOOST's technologies. - Evolving regulatory landscapes could introduce uncertainties or barriers to the approval and deployment of advanced agricultural solutions. - Shifting political priorities and budgetary constraints within governments may influence the level of support and funding allocated to agricultural research and innovation projects.	AGROBOOST will establish proactive engagement with policymakers in order to stay informed about potential policy changes but also participate in advocacy groups to contribute to policy discussions. Evidence-based research will be provided to policymakers to showcase the positive impact of the project's results on agricultural practices [T2.3]. AGROBOOST consortium will work closely with regulatory authorities to understand their requirements and expectations. It will engage in regular dialogue with regulatory agencies to address concerns and provide clarifications [T6.5]. AGROBOOST will conduct cost-benefit analyses highlighting the potential return on investment and economic advantages [T6.3]. It will also showcase how it aligns with broader national goals, such as job creation, economic growth and sustainable agriculture.
Ethical	- Ethical concerns related to the unauthorized use of personal or sensitive information. - Automation technologies may lead to concerns about job displacement within the agricultural workforce. - Certain communities or farmers are unable to afford or access the innovations, exacerbating existing social and economic disparities.	AGROBOOST will prioritize data privacy and security by implementing stringent protocols aligned with relevant EU regulations [T1.4]. Transparent communication about data handling practices and obtaining informed consent will be central to fostering trust among stakeholders. AGROBOOST will invest in training curricula to equip workers with the skills needed in the age of automation [T6.4], facilitating a smooth shift for workers toward higher-skilled roles and alternative employment opportunities. AGROBOOST will actively engage with local communities, incorporating their perspectives to tailor solutions that meet specific needs [T6.2]. Collaborations with CSOs and local governments will assist towards ensuring that the benefits of these innovations reach all communities, particularly those facing financial constraints.
Social	Societal norms and cultural practices within certain agricultural communities may impede the acceptance of new technologies.	AGROBOOST will implement community engagement programs that involve local leaders and community members involving them in the co-design [T2.1] and evaluation processes [T5.10], thus building trust and acceptance and addressing any potential misconceptions. AGROBOOST will collaborate closely with FAD, the Spanish associations of women in agriculture, to develop targeted empowerment programs and ensure equal access to and benefits from

	- Gender inequalities in agriculture could pose challenges for AGROBOOST's success. - Language barriers can impede effective communication, especially in multicultural or multilingual communities. - Negative societal perceptions of agriculture as a low-status occupation may deter the younger generation from pursuing careers in farming.	project technologies [T2.3]. The project will also prioritize inclusive decision-making processes to incorporate gender perspectives at all levels. AGROBOOST will employ multilingual communication strategies, providing translation services and tools within AKIS platforms [T6.2]. Training sessions and workshops [T6.4] will be conducted in local languages, fostering better understanding and adoption of the proposed technologies. AGROBOOST will collaborate with AJP, the Portuguese youth association in agriculture to design activities showcasing the modern and technology-driven aspects of farming [T2.3]. These engagement activities will highlight innovative careers and entrepreneurship opportunities within the agricultural sector. Targeted education and awareness campaigns will challenge stereotypes, emphasizing the sector's importance in sustainable development while showcasing success stories of young individuals thriving in agriculture [T6.4].
Technological	- Absence of industry standards for certain AGROBOOST technologies, may lead to variations in equipment and compatibility challenges. - Incompatibility and integration (with other farm management systems) issues may hinder the overall effectiveness of the technological solutions. - Unreliable connectivity in rural areas may limit the real-time functionality of AGROBOOST's technologies, (impacting data transmission and communication between robotics, AR, etc.)	AGROBOOST will actively engage with standardization bodies and industry associations, contributing to the development of clear standards for its technologies [T6.5]. This collaboration will ensure that the proposed innovations are designed in a flexible and modular manner to accommodate evolving norms and practices. AGROBOOST will design its technologies with open standards and APIs, facilitating seamless integration with existing farm management systems [T5.2]. Collaborations with other technology providers will be pursued to ensure interoperability. Documentation (manuals and reports) and support mechanisms will be provided to guide farmers through the integration process. AGROBOOST will liaise with telecommunication providers and focus on expanding their existing network infrastructure in rural areas. The proposed technologies will be designed to operate effectively in offline or low-connectivity modes, ensuring functionality even in areas with intermittent network access.

2.2 Measures to maximise impact #@COM-DIS-VIS-CDV@#

2.2.1 Communication, Dissemination and Exploitation Strategy

To maximize the impact of **AGROBOOST**'s results, the consortium will develop from the very beginning a comprehensive and structured strategy (under WP6) towards efficient communication, dissemination and exploitation, with concrete objectives to (i) ensure broad visibility and raise awareness about **AGROBOOST**, spreading knowledge about the project and its results, establishing a distinctive and recognizable identity that will support promotional and marketing efforts; (ii) reach, stimulate and engage a critical mass of stakeholders from the various domains of intervention to ensure that **AGROBOOST**'s results are effectively showcased, leading to validation, improvement and possibly further adoption of the developed technologies and concepts and ensure the maximum possible participation of innovators, SMEs, start-ups and industry players in the planned Open Calls; (iii) facilitate the exploitation of the project's outcomes and promote innovative solutions based on the **AGROBOOST** technologies and concepts; (iv) foster impactful contribution to relevant scientific domains and standardization bodies, as appropriate; (v) ensure close coordination with relevant research, standardisation bodies and working groups, and establish liaisons with initiatives in research and innovation; (vi) ensure OA to all project results.

Stage ① - Awareness creation and communication foundation [M01-M012]: To design the dissemination, exploitation and communication strategy and plan, including refinement of target groups and selection of dedicated tools and community-building activities, and inform all relevant stakeholders about the **AGROBOOST** scope and objectives. This phase will also define the liaisons and interaction mechanisms with the rest of the domain entities and players, including relevant research and innovation projects to identify common practices, opportunities of collaboration and jointly organised activities. The key stakeholders will be refined and a first pool of key players will be created. A plan for the project targeted events will be drafted as well.

Stage ② - Community outreach, initial results dissemination [M13-M40]: To actively reach out to the main target stakeholders and generate interest in **AGROBOOST** activities, Open Calls (OCs) and outputs, set a solid foundation for the planned dissemination, showcasing and policy making / intervention activities, organization of workshops, webinars and training activities, targeted promotion of the Open Calls, participation to events, publications and press releases, open access publication of the project results.

Stage ③ - Global outreach and sustainable impact [M41-M60]: To actively engage and support the adoption and deployment of the concepts, technologies and tools offered by **AGROBOOST** through dedicated promotional activities, promotion of the selected 3rd party projects and their use cases and outputs for both the OCs, publish further scientific publications, development and distribution of promo materials, open accessible project results from the

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

project website and open platforms, participation in selected events, organization of planned workshops and exploitation activities, participation in exhibitions, organization of the project promotional event and extend liaisons with relevant initiatives, promotion of the key project results with special campaigns, the impact evaluation activities (mainly surveys and workshops). It is expected that standardization efforts will intensify as the technologies are developed, and the trials will mature along the project's lifetime.

2.2.2 Key Target Group and Audiences

Farmers, farm workers and farm advisors (*with special focus on the young people and people from rural areas*)

Benefits / messages: Take advantage from the offered solutions, liaise with the experts and technology providers providing direct feedback, being informed about the advancements in the domain, participating in discussions about new business models and opportunities. **Interaction channels/measures:** *participation and presentation in targeted project events, newsletter, project website, social media, targeted communications, showcase events and videos.*

Providers of technical solutions (*including industry, SMEs, startups, innovators, hardware providers, etc.*):

Benefits/messages: Benefit from AGROBOOST related solutions and combined technologies and resources to minimize up-time-to-market for their applications and services, increased market visibility, participation in the two project Open Calls, benefit from the demonstrated project's and 3rd party projects' use cases. **Interaction channels/measures:** *participation and presentation in targeted events, newsletter, project website, project conference, social media, targeted communications, Open Calls webinars, showcase events and videos.*

Research institutes and individual Researchers (*academic and corporate researchers and scientists in the relevant fields*) **Benefits/messages:** engaged in the definition and implementation of the research challenges and spread of the relevant results. AGROBOOST will foster active participation in the discussion and knowledge exchange around the processes and tools of the project, involvement in the project activities and surveys, and create synergies and interactions with the industry and the market through the project ecosystem. **Interaction channels/measures:** *participation and presentation at scientific events, publications in international conferences, book chapters and magazines, targeted communications, webinars, online and offline presence and materials, project website, project workshops, presentations, newsletters.*

Experts in sustainability, health and ethics aspects related to the discussed solutions **Benefits/messages:** Being informed about the innovations and contributions of the project results towards sustainable, healthy and legally compliant solutions, liaison with the technology providers and the end users towards more efficient planning and interaction. **Interaction channels/measures:** *invitation for participation to project events, special publications and targeted campaigns, involvement in project online discussions and forums, online communication.*

Policy makers, policy analysts, legal experts (*public national and EU organisations developing relevant regulatory frameworks, governmental initiatives, public authorities, etc.*) **Benefits/messages:** Make informed strategic decisions and plan targeted activities, investments and calls on AGROBOOST for the good of our economies and societies, liaise with the industry, market and research worlds. **Interaction channels/measures:** *presentations, participation at local / national events and exhibitions, but also to key policy making events at European level, project website, project conference, newsletters.*

Research and innovation initiatives (*projects funded under the H2020 or HE Programmes, liaised with the planned activities*) **Benefits/messages:** AGROBOOST will liaise with these projects that have been or are implemented in the relevant domains with the main goal to: engage them in the research and in the analysis of the project's results, to retrieve valuable and validated best practices and lessons learnt, to jointly promote the results achieved, to engage their existing and live communities, to ensure the most possible visibility and promotion of the project OCs. **Interaction channels/measures:** *targeted communications, social media, project website, project conference, participation in events, promotional material, joint workshops / events / publications / press releases.*

Standardisation bodies, pre-standardisation and open-source initiatives (*Standardisation bodies and entities focused on the development of interoperable solutions and services in agriculture focusing on the project emerging technologies*) **Benefits/messages:** Support technology transfer, liaising with the private sector, innovators, researchers, policy makers, share / promote standards and relevant strategies and success stories, active contributions of AGROBOOST. **Interaction channels/measures:** *publications, participation and presentation in targeted events, newsletter, project website, social media, showcase events.*

Civil society and community at large (*social civic organisations, NGOs, the public, media, special focus on vulnerable groups towards social inclusion*) **Benefits/messages:** Inform for project advancements, best practices, outcomes, liaise with AGROBOOST stakeholders, awareness on social aspects around project suggested activities and solutions, including the results and use cases of the 3rd party projects. **Interaction channels/measures:** *press and media communications, publications in dedicated press, organisation of and participation at domain-focused events, newsletter campaigns, social media, project website, activities towards inclusion, healthy and remote working.*

2.2.3 Communication channels and measures

A comprehensive set of communication and promotional mechanisms and tools will be established early on to properly support all the impact creation activities and especially the promotion of the 2 Open Calls of the project. At the beginning of the project, a core task will be to identify and refine the most effective communication tools, channels, and messages best tailored to the specific project objectives and targeted stakeholders, rooting into the communities gathered around the Consortium and their broad networks and the selected pilot sites/use cases. The main tools AGROBOOST plans to start with are:

(i) **AGROBOOST visual identity and promo kit**, comprising brand guidelines, colour and font codes, and the AGROBOOST logo variants needed for all applicable online or offline channels and collaterals. (ii) **AGROBOOST website**, a fully functional website that will contain comprehensive information on AGROBOOST aims and objectives with easy access and a very friendly interface to retrieve information and any public material generated within the project activities and fully support the OCs entire process. (iii) **Social media**: The project will establish its presence via social media by creating Twitter and LinkedIn accounts, and others as relevant and creating a dedicated YouTube channel, which will be exploited to build a community around the project's work and activities. These channels will be animated by echoing and cross-posting on the partners and relevant liaised initiatives and associations channels for broader reach, taking advantage of the partners' activities, participation and a great network. (iv) **Promo materials**, both online and offline. AGROBOOST will produce a variety of dedicated materials presenting the project and its achievements, the organization of the two OCs and also the results of the 3rd party projects. (v) **Newsletter/newsflashes**: regular updates about the project's activities and work will be pushed out via a dedicated news channel hosted on the AGROBOOST website, and a dedicated newsletter will be issued every four months and shorter newsflashes every month to promote highlights and push out announcements of interest in an ecosystem-wide perspective. (vi) **Dedicated events (workshops, exhibitions, webinars)** to help increase the visibility of the developed technologies and trials, showcasing achievements and promoting the increased deployment of AGROBOOST results and creating sustainable ecosystems around the AGROBOOST and its 3rd party projects ecosystems and use cases; (vii) **Videos**: AGROBOOST will produce a series of videos in order to engage the general public and to build awareness about AGROBOOST activities, exposure among a wide audience will establish strong connections with local press, online blogs and social press to improve the media interest and coverage. The majority of the generated content will be in digital format, with a deliberate avoidance of printed materials, emphasizing the project's commitment to environmental sustainability and its "green" initiative.

2.2.4 Dissemination channels and measures

Events organisation: Throughout the project's duration, **participation and organization of events** in the form of webinars, workshops, demos, trials, exhibitions, expert panels' discussions, training, etc., will play a crucial role. The consortium envisages organizing at least 13 workshops/sessions/webinars pursuing co-location with flagship domain related events or events that are planned from the liaised projects, coordinating on this with related national and international initiatives when possible, including: 2 OCs webinars (WP5), at least 2 co-design workshops (T2.3), at least 3 informative webinars (T6.2), one final conference (T6.2), 5 pilot cases workshops (T6.2).

Scientific/technology/engagement events: Towards efficient dissemination and promotion of the project results that are to be used by external stakeholders and the broader ecosystem, the consortium will have **at least 12 participations/presentations** in relevant events.

Journals/magazines/press: The consortium will organize at least 1 special issues in NATURE, Springer (Discover Food Journal) and Elsevier (Sustainable Agriculture Journal, Smart Agricultural Technology Journal, Sustainable Horizons Journal), IEEE Transactions on Industrial Informatics, or other high-impact journals.

To widen the outreach of the AGROBOOST and maximise its impact the Consortium will pursue and ensure close coordination with the EC, the various ongoing relevant H2020 and HE projects/initiatives, as well as any other relevant national and international ones and it will build **liaisons and synergies** with the EU and National Initiatives and will take advantage from the already extended network of the partners participating actively.

2.2.5 Monitoring the Communication, Dissemination and Exploitation activities

Table 1: Key Performance Indicators for dissemination and communication (KPIs)

Measure	Indicators	Target
e-brochures	Number of e-brochures distributed	≥ 200 per year
Posters	Number of e-posters produced	≥ 3 in total
High-level materials for policy makers	Number of sets (mission statement, slide-deck, e-brochure)	≥ 1 per year
AGROBOOST website	Number of unique visitors	≥ 1000 visitors/year
Social networks	Number of followers in: Twitter, LinkedIn	≥ 500 / ≥ 250
e-Newsletter	Number of newsletters (published every 6 months)	≥ 10
Workshops, events, webinars	Number of workshops and number of participants	≥ 10 workshops (100 participants/event)
Videos	Number of videos published on the project's YouTube channel and average number of views	≥ 5 videos and ≥ 1000 views per video
Scientific publications	Number of peer-reviewed papers/articles	≥ 10 by the end of the project
Participation to events and presentations	Number of external events partners attended to promote the project, including scientific conferences, and number of demos and or presentations	≥ 12

2.2.6 Engagement of Citizens, Civil Society and End-users

AGROBOOST will adopt a participatory and co-creation approach to engage diverse stakeholder groups at various levels and ensure that the developed technologies are addressing the daily needs while enhancing their working

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

conditions. The co-creation of workshops at the early stages of the project will facilitate open dialogue among various actors such as farmers, agricultural workers, competent authorities and civil society organizations, leading to an ideation process which will allow them to share insights and challenges related to working conditions and technological advancements in agriculture. Particular attention will be paid on the element of inclusiveness in agriculture assisting vulnerable people to actively participate in and benefit from the proposed technological advancements, thereby fostering empowerment, equal opportunities and sustainable development within agricultural communities. In addition, the project has designed specific participatory design sessions, where several stakeholders will actively contribute to the design and development process of robotic technologies, AI-enabled services and augmented reality tools described in **WP3** and **WP4**. This user-centric approach will ensure that the technologies resonate with the daily operations and preferences of those who will be using them in real-world agricultural settings. Last, community empowerment and training programs will be organised, focusing on the capacity building of local communities, providing them with the knowledge and skills needed to effectively use the developed technologies. **AGROBOOST** will foster a collaborative and inclusive environment, ensuring that the developed technologies address the social, cultural and economic needs of the communities involved.

2.2.7 Exploitation Plan

AGROBOOST will ensure the exploitation and sustainability of the project results after its end by adopting a business-facing approach to the project visibility through integrated actions interacting between exploitation, dissemination, standardization and communication. The blend of the consortium which consists of industrial partners, academia, think tanks and farmers associations that have the expertise to deliver innovative solutions to enhance the working conditions and strengthening the work force in agriculture through digital technologies and have the experience and capacity to bring them to the market while also to use the project outputs internally, ensuring the exploitation success. To this end, the project will employ a 7-phase structured exploitation strategy for the exploitation of the produced framework, technology tools and other exploitable results such as data, algorithms and policy recommendations. The underlying approach adheres to the recommended best practices for the “Plan for Exploitation of Results (PEDR)” for research project with joint ownership schemes.

Phase ① – Asset Taxonomy: During this phase, the innovative Exploitable Assets/Results and their taxonomy will be identified, as well as their potential use as stand-alone future market offerings or as aggregations into value-added services and product offerings. Results will be mapped to assets and to potential products, addressing technological components, patents and methodologies stemming from the project activities.

Phase ② – Marketing Strategy Aspects: This phase aims at addressing notions of marketing strategy towards the formulation of the **AGROBOOST** solutions for the market. In principle, several online surveys will be conducted, including Likert scaling and qualitative analysis, that will assist in the extraction of several marketing plan key ideas, such as business initiators and market catalysts, optimal channels for exploitation of results, positioning, promotional strategies that will be followed, scalability, maintenance fees after the project’s end, and optimal pricing schemes.

Phase ③ – Emerging Trends, Competition and Positioning: This phase is dedicated to the preliminary market analysis and relevant positioning of **AGROBOOST**. During this phase, the exploitable assets per potential stakeholders, per maturity level and per route to market’s potential will be mapped, in parallel with potential competition and other alternatives analysis. The consortium will perform a thorough research which will aim at the identification of the market, its segmentation, its maturity and scarcity of alternatives, as well as the positioning of the current competitors to define potential commercial and non-commercial exploitation models.

Phase ④ – Exploitation Workshops and Business Models:

Dedicated online exploitation workshops will be held during this phase via collaborative tools with the participation of all partners. Persona Canvases for ideal adopter, business initiator and potential customer will be derived. Furthermore, optimal business models, one for each of the project key assets will be showcased, by employing the novel methodology of the Platform Innovation Kit which provides a toolset to design, launch and scale impactful services and platforms. Specifically, the designs will be based on the Platform Business Model Canvas, which allows mapping the whole platform business operation, including, among others, multiple customer segments, network effects, multiple revenue streams. In parallel, a stakeholder persona canvas will also be created to perform a deep dive into the pains and gains of our platform’s stakeholders, from a business perspective, also validating the project’s views and assumptions.

Phase ⑤ – Individual Exploitation Pathways: The exploitation leader (**MAR**) will assist partners both in their individual as well as in their joint plans, with a special focus



Figure 9: **AGROBOOST**'s seven phase exploitation strategy structure.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

on key sub-value chains and collaborations among select partners with external stakeholders, with established companies and with other projects of similar focus. An applicable derived business model and a master “Grounds Identification” mapping will complement each exploitable asset. Namely, to categorize partner motivations in target exploitation action categories include promote to market, produce, redistribute, exploit internally, assign or license to third-parties, research further, integrate within line of business or pre-existing networks, intention to offer complementary services such as training and consulting, each with tangible actions and plans. **AGROBOOST** partners will be assisted to exemplify their Route to Market Roadmap for each of the exploitable results.

Phase (6) – Joint Exploitation and Sustainability of Results: During this phase, two joint exploitation channels will be set in parallel: **(a)** The predominant exploitation, promotion and distribution channel of the **AGROBOOST** results will be an online platform, showcasing distinct services in freemium versions at first, for critical mass creation, first adopter/customer attraction and a schema for subscription-based services at the next steps. **(b)** The proposed schema for commercial and research penetration of the **AGROBOOST** platform shall be a jointly owned and governed by the partners developing it, based on a separate agreement during the IPR management activities, either via exploiting it through internal business units or through a spin-off. On top of that, the exploitation workshops shall also exemplify sub-value chains of few partners combining exploitable results towards common future market targets into augmented novel products. Conveniently, the pre-existing market access of one partner can be used for the promotion of the key foreground produced by another. Exploitation and business plans for each **exploitable result presented in Section 1.2.4**, will be drafted, complementing the overall business plan developed for the project, with key result the **AGROBOOST** innovations and strategies. All key exploitable results will be also submitted to the Innovation Radar, while we will also seek collaboration with other EC schemas, like the Horizon Results Platform, Horizon Results Booster, IPR Booster and Horizon Standardisation Booster.

Phase (7) – Direct Links to Industry: During this phase, direct industry links with national and European agencies, standardisation bodies, technology providers (large corporates and SMEs in agritech), as well as potential adopters and envisioned first customers will be targeted, in which each partner is expected to contribute with at least 4 key synergies that will lead to 4 complementary products/synergies from the real market based on their pre-existing networks of clients, vendors, collaborators etc. Even if early adopters and future customers are not captured, this will ensure valuable direct feedback from the real market.

2.2.8 Individual dissemination and exploitation plans

All **AGROBOOST** partners will participate in common dissemination and communication events and activities, as well as in the exploitation and business sustainability tasks of the project. Below however, individual dissemination and exploitation plans per partner category are briefly illustrated for each group of partners.

Exploitation plan
Technology Providers (Large Enterprises and SMEs) Technology providers will strategically position themselves to capitalize on the innovations generated by the project, including the development of various business models based on different commercialization schemes, such as licensing agreements, partnerships and direct sales to end-users in the agricultural sector. They will actively engage in the creation of scalable solutions with potential applications beyond the project's scope, exploring opportunities for integration into broader agricultural markets through the identification of key stakeholders and the establishment of collaborative networks to facilitate technology transfer and adoption. Last, they will leverage the outcomes of the project to strengthen their market presence, showcasing the technological advancements through participation in industry events, conferences, and targeted marketing campaigns thus creating additional sustainable revenue streams.
Academic Partners (Universities and Research Centres) and Associations The research institutes will establish partnerships to leverage research outcomes and integrate the proposed innovative technologies into academic curricula. The foreseen joint research initiatives, data sharing and collaborative publications, will enhance their academic reputation. Additionally, these institutes will provide expert guidance, validating the deployed technologies and conducting field trials to ensure that the project's outcomes align with real-world agricultural needs. For associations of young farmers and women in agriculture, the exploitation plan focuses on creating tailored training programs and capacity-building initiatives, to empower these groups with the necessary skills to effectively deploy and benefit from the developed technologies.
Dissemination plan
Technology Providers (Large Enterprises and SMEs) Technology providers will ensure widespread adoption and impactful integration of the project's innovations through their participation in key industry conferences, workshops, fairs, expos and trade shows that will give them the opportunity to demonstrate the practical applications and advantages of the developed robotic and AR technologies. They will also leverage their digital channels, including their accounts in the social media, to reach a broader audience and facilitate continuous communication with potential users and partners, e.g., agribusinesses, agricultural cooperatives, and machinery manufacturers.
Academic Partners (Universities and Research Centres) and Associations The academic partners will be heavily engaged through multilateral collaborations to ensure that the research findings and technological advancements from the project are shared with the research community and the wide audience through academic publications, conferences, symposia and workshops. Moreover, they will integrate the project's outcomes into academic curricula, fostering ongoing academic discourse on the application of robotic technologies and augmented

reality in agriculture. Associations of young farmers and women in agriculture will use their vast network of members to disseminate the project's findings and create tailored outreach activities such as dedicated workshops, training sessions and webinars in order to provide them with insights into the project's innovations.

2.2.9 Knowledge management, IPR & data management

The proper handling of the Intellectual Property (either pre-existing or developed within the project) is essential to stimulate an effective cooperation among the partners, encourage an innovative contribution from all participants independently from their work share, and facilitate further research developments and commercial exploitation activities while keeping the commercial interests of all parties protected. The initial IPR strategy of the project, which will adhere to the guidelines outlined for the HE Programme, will be drafted in the Consortium Agreement, as appropriate. More specifically, an IPR log will be set up to record the background and foreground IP, ownership status, access rights to the project results, as well as the overall protection rules established for the project duration and beyond. In brief, the rule to follow is that the consortium members agree to grant each other royalty-free access to the background and project results as needed for the smooth implementation of the project. Special attention will be given on guaranteeing the confidentiality of the partners' background assets and knowledge. Any disclosure of confidential information to third parties will be on a need-to-know basis for project implementation and will be managed by **WIS**. The IPR log will be revisited periodically and if needed, it will be updated in exploitation deliverables accordingly.

2.2.10 Policy feedback measures

AGROBOOST is committed to contributing to policy shaping and supporting the implementation of new policy initiatives and decisions following the EU's digital strategy. As such, the consortium plans to contribute to the following initiatives through (a) the planned stakeholders engagement activities including policy makers and experts of the domain, (b) dedicated networking and community building activities with key initiatives and actors, (c) participation in policy discussions and relevant events, (d) implement outreach and awareness activities, (e) produce publications and recommendations as output mainly of the impact maximisation and assessment, (f) getting feedback from real world use case scenarios and their ecosystems, (g) creating direct links between the targeted groups and involve them in relevant discussions and (h) providing Policy Recommendations for the Adoption of Digital Technologies for Enhanced Working Conditions in Agriculture as direct input to the existing strategies, policies and recommendations. Targeted policies at initial phase include the following, since this list will be revisited and updated as needed during the implementation of the project: The **Farm to Fork Strategy** that is at the heart of the **European Green Deal** and aims at making food systems fair, healthy and environmentally-friendly; The **common agricultural policy (CAP) 2023-27** that is essential to secure the future of agriculture and forestry, as well the achievement of the objectives of the European Green Deal; The **Digital Decade** policy programme, in relation to the targets and objectives for 2030 that guide Europe's digital transformation; The **Digital Compass** that aims towards a secure human-centred digital ecosystem, where citizens are empowered, and businesses prosper from the digital potential; The **EU Industrial Strategy** that considering the new circumstances following the COVID-19 crisis targets to drive the transformation to a more sustainable, digital, resilient and globally competitive economy; The **new circular economy action plan (CEAP)** (as one of the key parts of European Green Deal) with focus on reducing pressure on natural resources and creating sustainable growth and jobs; The **European data strategy** towards a single market for data that allows it to flow freely within the EU and across sectors for the benefit of businesses, researchers and public administrations; and the **Sustainable Development Goals (SDGs)** related policies and guidelines.

2.3 Impact summary #§COM-DIS-VIS-CDV§#

Specific Needs	Expected Results	D & E & C Measures
- Need for transition to a fair, healthy and resilient agriculture. - Need for enhancement of the sustainability performance of the sector, including social sustainability, and competitiveness in agriculture. - Need for support the further deployment of digital and data technologies as key enablers. - Need for enhanced working conditions - Need for lowered environmental impacts and productions costs and increased product quality. - Need for reduced share of risky or unattractive actions/tasks performed by workers. - Need for mitigation of shortages of work force in agriculture - Need for cost-effective innovative digital and data technologies.	- Autonomous Robotic Systems for Pruning & Thinning Tasks. - AR-based System for Automated Mechanical Weeding. - AR-enabled Control Algorithms for Selective Harvesting. - Gamified VR Environment for Robotic Systems. - AR-powered Operational & Training for Sustainable Agriculture. - Deep Learning for Complex Scene Recognition. - Animal-Robot Interaction for Poultry Management. - Human-Robot Collaboration for Safe Working Conditions. - Automation and AI Innovations in Pathway Planning and Navigation Control.	Dissemination [D]
		- Peer-review scientific publications in journals and conferences ≥ 10. - Participation to events and presentations ≥ 12. - Workshops, events, webinars ≥ 10 workshops (≥ 100 participants/event). - Videos ≥ 5 and ≥ 1000 views per video. - High-level materials for policy makers ≥ 3. - Industrial booths, fairs or expos ≥ 5.
		Communication [C]
		Website Online by M06; ≥ 1.000 unique visitors by M36.

▪ Need for improved working conditions. ▪ Need for business models for the use of innovative technologies under consideration of various farm structures and inter-farm linkages. ▪ Need for system analysis of the consequences for farmers and rural communities. ▪ Need for inclusivity and affordability of digital solutions. ▪ Need for compliance of digital solutions with relevant (and forthcoming) EU regulations.	▪ XAI-enabled Decision Support System for Socially Responsible Agricultural Practices. ▪ Monitoring System for Analyses of the Consequences for Farmers and Rural Communities. ▪ Policy Recommendations for the Adoption of Digital Technologies for Enhanced Working Conditions in Agriculture.	▪ e-Newsletters ≥ 10. ▪ Size of online community (in social media) ≥ 750. ▪ Impressions ≥ 500. ▪ Press releases ≥ 6.
		Exploitation [E]
		Open version of the project's methods, models, tools and data repository and practical guidelines/ services under a fee (post-project) to ensure sustainability.

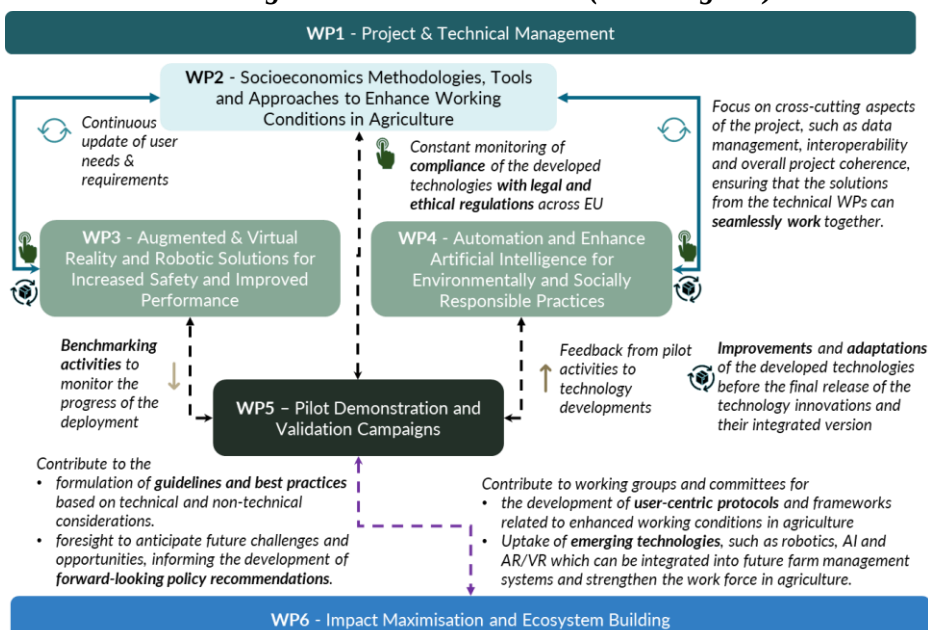
Target Groups	Outcomes	Impacts
▪ Farmers ▪ Farm workers ▪ Young professionals ▪ Women in agriculture ▪ Farmers with disabilities ▪ Local communities ▪ Agricultural technology providers ▪ Local Governments ▪ Policymakers ▪ Research Institutes and Universities ▪ Robotics suppliers and manufacturers ▪ Health and safety organisations	OUT1.1 $\geq 20\%$ accidents reduction; OUT1.2 $\geq 20\%$ reduction of the workers absenteeism related to farmers' injuries or health issues caused by unsafe working conditions; OUT1.3 $\geq 60\%$ improvement of the workplace ergonomics; OUT1.4 $\geq 25\%$ reduction of the costs associated with safety incidents, healthcare expenses and operational disruptions; OUT1.5 $\geq 95\%$ of the participants in the pilot activities provide positive feedback; OUT2.1 $\geq 20\%$ reduction of the water resources used in agricultural activities; OUT2.2 $\geq 20\%$ reduction in overall carbon footprint associated with farming operations; OUT2.3 $\geq 40\%$ in the use of fertilizers and pesticides per unit of production; OUT2.4 $\geq 30\%$ reduction in the time needed for specific agricultural tasks (e.g., harvesting, weeding, etc.) through robotics and automation; OUT2.5 $\geq 15\%$ in the overall farm profitability, considering the impact of cost reductions and improved product quality but also the affordability of the proposed solutions; OUT2.6 $\geq 25\%$ reduction of system failure rates and downtime; OUT2.6 $\geq 20\%$ reduction of the energy consumed per unit of work performed; OUT2.7 $\geq 25\%$ reduction of the costs associated with operational disruptions; OUT2.8 $\geq 15\%$ reduction in agricultural waste (in particular food loss reduction through precise and automated harvesting that assists in minimizing damage to crops and reduce losses during harvest); OUT3.1 $\geq 30\%$ increase of overall tasks automation; OUT3.2 $\geq 85\%$ increase of the user acceptance for the AI-generated recommendations; OUT3.3 $\geq 35\%$ decrease in tasks that require manual handling by workers; OUT3.4 $\geq 80\%$ reduction in both mental and physical effort for harvesting tasks; OUT3.5 $\geq 50\%$ decrease in the average manual labour required for weeding 0.1 hectares, compared to traditional manual mechanical weeding practices; OUT3.6 ≥ 10 new roles or jobs related to robot supervision, maintenance and technical support; OUT3.7 $\geq 30\%$ reduction in harvesting time compared to traditional methods; OUT4.1 $\geq 30\%$ increase in overall labour productivity; OUT4.2 $\geq 30\%$ decrease in the number of unfilled job positions; OUT4.3 $\geq 35\%$ increase of the inclusivity across different demographic groups, with a specific focus on youth, women and persons with disabilities; OUT4.4 $\geq 75\%$ increase in the	IMP1.1: increase in stakeholder engagement and participation; IMP1.2: increase in innovation adoption due to informed decision-making; IMP1.3: improvement in societal well-being and community resilience; IMP1.4: increased inclusiveness of diverse voices in decision-making; IMP1.5: expansion of the innovation ecosystem within the agricultural sector; IMP1.6: improved resilience demonstrated in the face of external challenges or disruptions; IMP2.1: reduction of resource consumption, emissions and waste generation in additional agricultural fields; IMP2.2: reduction in production costs; IMP2.3: improved productivity in the agricultural sector; IMP2.4: increased employment rate and income level of workers in agriculture; IMP2.5: improved physical and mental well-being of workers and farmers; IMP2.6: expansion of market access and diversification of agricultural products; IMP3.1: increased access of active users on the AKIS platform; IMP3.2: Increased volume (from additional resources) and type of content users are interacting with; IMP3.3: increased user engagement and interaction; IMP3.4: increased growth in the AKIS user community; IMP3.5: improved consumer awareness and understanding of sustainable agricultural practices; IMP3.6: increased number of cross-sector collaborative projects initiated through connections made on AKIS; IMP4.1: increased number of actionable policy recommendations from AGROBOOST incorporated into EU and international agricultural policies; IMP4.2: positive impact on shaping and influencing agricultural policies at the EU and international levels; IMP4.3: increased number of international collaborations established to advance science-policy interfaces; IMP4.4: increase in public awareness and advocacy efforts related to AGROBOOST's policy recommendations; IMP4.5: increased use of AGROBOOST's scientific publications and reports in policymaking documents; IMP4.6: improved integration of policy

retention rate of skilled workers within targeted sub-sectors; OUT4.5 $\geq$ 35% increase in the allocation of the workforce towards skilled roles;	recommendations into national and international policies addressing SDGs. #IMP-ACT-IA\$#
--	--

3 Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

3.1.1 Work Packages and their interrelation (Pert Diagram)



AGROBOOST is a multidisciplinary, pilot-driven project that builds its value proposition upon six (6) WPs as presented in the Pert diagram of the Figure on the left which illustrates the clear delineation of project activities and their critical paths providing a structured framework for the project's sequence and dependencies. **WP1** is responsible for the day-to-day project and financial management, the communication with the EC, the quality of the work performed and delivered and the safeguard of the regulatory framework. **WP2** is focusing on understanding the existing challenges of the working conditions and labour shortages in

agriculture, eliciting the user requirements for the development of the innovative solutions through co-creation activities and examining strategies for inclusive technologies and incentives for their rapid adoption. WP3 and WP4 include the core activities of the project with the development and deployment of the AR and robotic solutions for improved performance [WP3] and the automation-based solution enhanced through AI techniques for recommendations on environmentally and socially responsible practices [WP4]. The validation campaigns are designed, demonstrated and validated in WP5 incorporating additional solutions through the process of Open Calls. Last, WP6 coordinates the foreseen communication and dissemination activities along with the exploitation aspects of the proposed innovations and the policy recommendations towards enhanced working conditions in agriculture. Concrete milestones have been set towards the achievement of the objectives defined for the project.

3.1.2 List of work packages (Table 3.1a)

WP	WP Title	Leader	PMs	Start	End
WP1	Project and Technical Management	UCD	64	M01	M60
WP2	Socioeconomics Methodologies, Tools and Approaches to Enhance Working Conditions in Agriculture	FIB	133	M01	M60
WP3	Augmented & Virtual Reality and Robotic Solutions for Increased Safety and Improved Performance	INE	129	M07	M48
WP4	Automation and Enhance Artificial Intelligence for Environmentally and Socially Responsible Practices	DTU	122	M07	M48
WP5	Pilot Demonstration and Validation Campaigns	NTU	261	M07	M60
WP6	Impact Maximisation and Ecosystem Building	MAR	135.5	M01	M60
Total			844.5		

3.1.3 Summary of consortium effort (Table 3.1.f)

	UCD	NTU	KUL	AUA	INE	LIN	NVR	NCI	SSI	WIS	MAR	AJP	DTU	HWU	FAD	G&D	FIB	EPF	Total PMs	%
WP1	24	2	1	2	2	3	1	4	3	6	4	2	2	1	1	1	1	4	64	8%
WP2	4	10	1	4	0	2	4	0	19	18	0	10	1	2	14	21	22	1	133	16%
WP3	25	4	5	20	28	24	5	0	0	0	0	0	0	18	0	0	0	0	129	15%
WP4	17	3	17	5	0	23	0	15	0	0	0	0	26	16	0	0	0	0	122	14%
WP5	16	25	17	14	18	23	21	23	2	12	19	15	4	6	3	0	18	25	261	31%
WP6	4	11.5	2.5	6.5	2	5	9	3	8	7	44	6	1	3	6	12	4	1	135.5	16%
Total	90	55.5	44	52	50	80	40	45	32	43	67	33	34	46	24	34	45	31	844.5	

3.1.4 Timing of the work packages and tasks (Gantt chart)

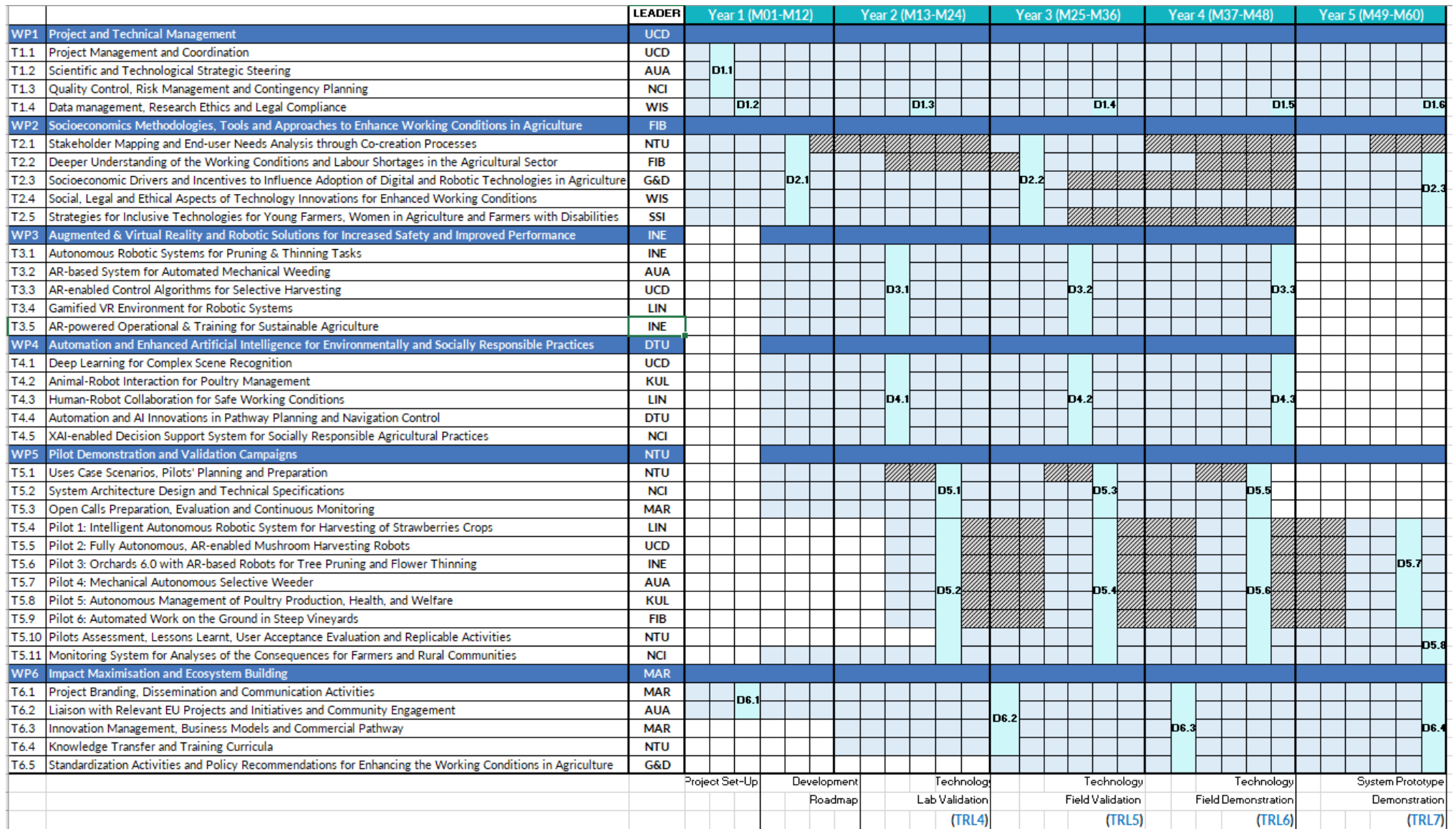


Figure 10: The Gantt chart of **AGROBOOST** is shown in the Figure below, illustrating the timing of WPs and Tasks over a total project duration of 60 months. As the project follows an agile approach a number of iterations per WP and Task have been foreseen. Careful planning has been made in order not to have many deliverables submitted in a single month. The advances of the project's developments (in terms of TRL) are also shown at the end of the Gantt linked with specific months guiding the development and deployment effort of the consortium.

3.1.5 Work package description (Table 3.1b)

WP Number	WP1	WP Title	Project and Technical Management
Objectives: (1) To define and issue AGROBOOST management plan, including all the subsidiary plans; (2) To effectively monitor the project, in administrative, technical and financial terms; (3) To ensure the project's strategic steering and daily operations in accordance with the overall project plan, available resources and timing; (4) To offer the necessary interface to the EU services and external actors; (5) To assure the high quality of the project outcomes and to ensure that the project development will adhere to existing and emerging standards and application guidelines; (6) To identify and manage project risks by performing an effective risk management;			
T1.1 - Project Management & Coordination [M01-M60] (UCD, All Partners) This task will provide the overall project coordination. It will perform self-assessment activities (resources planning distribution, project monitoring, corrective actions if needed) lead by UCD and assisted by all partners, including quality assurance for the deliverables, risk management, and consideration of external factors orchestrated by NCI. Work of the project management structures and procedures described in Section 3.2, will be facilitated within this task, including maintenance of the decision-making process and conflict management for technical and other related aspects. This task will also ensure proper communication with the EC and perform related administrative and support activities, including: (i) project administration; (ii) distribution of relevant information within the project; (iii) preparation of management reports and submission of project deliverables; (iv) preparation of project reviews and administration meetings at all levels; (v) keeping project accounts and (vi) interaction with external stakeholders.			
T1.2 – Scientific and Technological Strategic Steering [M01-M60], (AUA, UCD, INE, DTU) This task covers all technical and research activities of the project. Within this task, initiatives will be undertaken by AUA for proposing technical solutions and fine-tuning technical and scientific orientations whenever necessary. Moreover, control of the scientific and technical work will be carried out in the related tasks and propose technical modifications and reallocation of resources in collaboration with the project coordinator, UCD, as necessary for achieving the project objectives. Finally, reporting about technical and scientific deviations from the work plan will be performed along with the constant co-operation with the project's WP leaders, INE and DTU.			
T1.3 – Quality Control, Risk Management and Contingency Planning [M01-M60], (NCI, UCD, NTU, INE, MAR) This task aims to ensure the proper implementation of the project plan and the satisfaction of its objectives. It will include the establishment of Quality procedures and Risk Management assessment, that will lead to the generation of the relevant Quality Management Plan (QMP) assuring the conformity and the quality of all project deliverables with the respective requirements, thus all WP leaders (FIB, INE, DTU, NTU, MAR) will also report any potential risks that may rise during the project's lifespan. The QMP will be overviewed by NCI and allow the realisation of the internal assessment of project's quality on a 6-month basis. The Risk Management will cover the risks that will be constantly assessed and evaluated along with their mitigation measures described in detail in the project's Periodic Progress Reports.			
T1.4 – Data Management, Research Ethics and Legal Compliance [M01-M60], (WIS, UCD) Building on the draft considerations outlined in Section 1.2.11, this task will define the data management plans (DMPs), describing the overall data management lifecycle process for the data anticipated to be collected, processed, generated and further exchanged throughout the project duration. In alignment with the principle of making a project's data findable, accessible, interoperable and reusable (FAIR), the defined DMP will include details on (a) how the data will be handled during and after the project's completion, (b) what data will be collected and/or processed, generated and further exchanged, (c) any standards that will be applied, (d) identifying whether the data will be made open access and (e) how the data will be preserved, also considering the post project period. WIS will also investigate the ethical and legal aspects of the project by considering both the technical work within the project, as well as the regulatory and policy developments at EU level. Starting from the applicable regulations within EU, as well as other EU policy instruments, WIS will first produce, early in the project, guidelines to support project partners, thus facilitating compliance with the most relevant regulations.			
WP Number	WP2	WP Title	Socioeconomics Methodologies, Tools & Approaches to Enhance Working Conditions in Agriculture
Objectives: (1) To map the current status of the working conditions in agriculture; (2) To involve stakeholders in the design and testing of the systems; (3) To identify the labour trends and implications on European farms and farm enterprises; (4) To develop a set of recommendations for policies and initiatives that can incentivize farmers to integrate sustainable and technology-driven agricultural practices; (5) To provide a deeper understanding of the potential broader implications of the proposed technologies; (6) To derive the Societal Readiness Level SRL for each proposed technology; (7) To develop novel strategies that will ensure the inclusive adoption of digital and robotic technologies in agriculture.			
T2.1 – Stakeholder Mapping and End-user Needs Analysis through Co-creation Processes [M01-M10, M25-M36, M49-M53], (NTU, UCD, KUL, AUA, LIN, NVR, SSI, WIS, AJP, DTU, HWU, FAD, G&D, FIB, EPF) This task aims to map the current status of the working conditions in agriculture in terms of core concepts, processes, responsibilities, procedures, challenges and supporting technologies in order to enhance the sustainability performance of the sector, including social sustainability, and competitiveness. Most importantly, NTU will identify the operational challenges that farmers and on-farm workers face with an emphasis on reducing risky or unattractive tasks. SSI will perform a systematic literature review strengthened by interviews with all the stakeholders of the consortium. Focus groups will be organised by D&G and FIB (supported by the pilot leaders) with different stakeholder groups (including young farmers, women in agriculture, or farmers with disabilities), to better understand their perspectives, needs, and concerns related to the			

current working conditions in agriculture. The insights gained from these focus groups will be analysed by **SSI, D&G** and **FIB** and used to inform the technical partners, i.e., **UCD, AUA, INE, LIN, KUL, DTU, NCI** regarding the development of the overall systems and the associated policies and practices. Last, **D&G** will include in the analysis, the connection points of sustainable and inclusive practices and processes with specific SDGs such as sustainable food systems from farm to fork, good health and well-being, decent work and economic growth. Co-creation processes will be employed by **NTU** to involve stakeholders in the design and testing of the systems and ensure that it meets their needs.

T2.2 – Deeper Understanding of the Working Conditions and Labour Shortages in the Agricultural Sector [M01-M16, M27-M40, M49-M60], (FIB, UCD, AUA, SSI, WIS, AJP, FAD, G&D) This Task will use qualitative and quantitative research methods to delve into the working conditions and labour availability, needs and productivity in the agricultural sector. Using FADN data at national and regional NUTS2, NUTS3 level, **FIB, SSI, AJP, FAD** and **WIS** will identify the labour trends and implications on European farms and farm enterprises in the last 20 years. This analysis will include the exploration of trends on labour quantity and cost including the use of external labour and an econometric investigation of the determinants of labour dependence. The microeconomics analysis will be complemented and put into perspective through exploring the past and current macro-economic trends through a light-touch review with experts, tackling the question of labour conditions, needs and availability in main economic sectors conducted by **FIB, SSI, G&D**. In parallel, **FIB** and the pilot leaders (**UCD, AUA, INE, LIN, KUL**) will conduct qualitative interviewing of farmers, agricultural workers and advisors to gather in-depth insights into their work experiences and perceptions. The results will be communicated to WP5. During each pilot demonstration campaigns, participant observation including narrative interviews with farmers, agricultural workers and technology developers will be conducted to assess the changes introduced by the **AGROBOOST** robotic systems in terms of labour and working conditions. Results will be analysed in terms of the trajectory of change in narrative content and structure to highlight the areas where digital and robotic technologies can make a significant impact, informing the development of solutions.

T2.3 – Socioeconomic Drivers and Incentives to Influence Adoption of Digital and Robotic Technologies in Agriculture [M01-M30, M49-M60], (G&D, UCD, SSI, WIS, AJP, FAD, NVR, FIB) During this Task, **SSI** will conduct a literature review to identify existing studies on the socioeconomic drivers and incentives that influence the adoption of emerging technologies in agriculture. Relevant modelling exercises will be integrated by **FIB** considering economic studies, environmental footprint and the inputs from the stakeholders (**NVR, AJP, FAD**) to integrate knowledge and identify areas of improvements and of further developments. The results will be then analysed by **G&D** and **SSI** in order to identify key socioeconomic factors that have the greatest influence on farmers' behaviour and the types of incentives that have been shown to be effective in encouraging the adoption of sustainable farming practices, thereby informing targeted strategies for promoting environmentally conscious and economically viable agricultural approaches. Interviews and surveys with farmers and relevant stakeholders will be organised by **G&D, FIB** and **SSI** to better understand the specific socioeconomic drivers and incentives that are most relevant to the local context. The outcomes of these activities will be used to develop a set of recommendations for policies and initiatives that can incentivize farmers to integrate sustainable and technology-driven agricultural practices. Engagement activities with local farmers and farmers associations, governmental agencies, civil society organisations, etc., will be performed in close collaboration with WP6.

T2.4 – Social, Legal and Ethical Aspects of Technology Innovations for Enhanced Working Conditions [M01-M60], (WIS, DTU, UCD, NTU, AUA, LIN, SSI, FAD) The goal of this task is to provide a deeper understanding of the potential broader implications of the proposed ground-breaking technologies and develop recommendations and mitigation measures for their responsible and ethical implementation. A framework for ethical considerations will be established by **WIS** to facilitate collaboration between various stakeholders such as the technology partners (**UCD, AUA, NTU, LIN, KUL, DTU, NCI**) and actors within the agricultural domain. This framework will incorporate practical strategies and guidelines based on established international standards and values. Connections will be made with Responsible Innovation principles through **SSI**, to guide the development of socio-technical solutions and innovations for global and regional challenges. Furthermore, the Societal Readiness Level SRL for each proposed technology will be derived by **SSI** before their deployment in the pilot campaigns, ensuring the acceptance of the technologies, methods and processes by stakeholders in agriculture. This Task will also assist towards the production of appropriate strategies and incentivization mechanisms for relevant stakeholders to adopt the proposed innovations in T2.5.

T2.5 – Strategies for Inclusive Technologies for Young Farmers, Women in Agriculture and Farmers with Disabilities [M01-M30, M49, M60], (SSI, NTU, WIS, AJP, FAD, G&D, FIB) This task focuses on the development and implementation of strategies that will ensure the inclusive adoption of digital and robotic technologies in agriculture, specifically targeting young farmers, women in agriculture and farmers with disabilities. Capitalizing on the outcomes of T2.1, **SSI** will conduct a thorough user needs assessment for the abovementioned stakeholder groups in order to understand their specific challenges, preferences and requirements. **NTU** will assist in addressing the usability concerns of the stakeholders and propose suitable technology adaptations to meet their diverse needs. **FIB** and **G&D** will translate these strategies into recommendations for policies that promote the inclusive adoption of technologies in agriculture and advocate for supportive regulations and incentives that meet the specific needs of young farmers, women and farmers with disabilities. Moreover, **NTU** will also incorporate best practices into the training curricula that will be tailored to the identified groups, focusing on building their capacity to effectively use and benefit from the adopted technologies.

WP Number	WP3	WP Title	Augmented/Virtual Reality and Robotic Solutions for Increased Safety and Improved Performance
Objectives: (1) To develop autonomous robotic platforms for performing agricultural tasks; (2) To develop the components for the remote monitoring and control of the robotic platforms; (3) To develop AR solutions for enhanced			

user interaction; (4) To develop advanced control algorithms that interpret AR data to informed decisions; (5) To deliver the necessary software components for a VR environment that will enable users to train and control a robotic system remotely; (6) To develop an AR-powered platform targeting operational support and stakeholders' training.

T3.1 – Autonomous Robotic Systems for Pruning & Thinning Tasks [M07-M48], (INE, NVR) During this task, **INE** will develop an intelligent autonomous robotic system able to learn the required pruning and thinning tasks from representative datasets and human input. The development will be done in two stages: first, using already existent (related) data offered by **NVR**; second, using data acquired through the decision support system, which will make use of video overlayed with relevant information. The system will include algorithms and models for processing of the acquired visual data (sensing of the operation environment and target for the robotic platform) trained on representative datasets. The initial dataset will be augmented by new demonstrations (visual data and corresponding actions) from skilled labour. The autonomous platform will be built upon **INE**'s robots Modular-E solution which will be extended with attached robotic module (Prune-I) with dexterous manipulators and with advanced high-resolution cameras. Information will be made available to the user, i.e., a skilled labourer, by means of video overlayed with relevant information in a mobile device, and its inputs captured to provide feedback to the autonomous platform. This data will also be automatically captured through the AR environment for the incremental improvement of the algorithms and models.

T3.2 – AR-based System for Automated Mechanical Weeding [M07-M48], (AUA, UCD) This task consists of four distinct but highly interdependent subtasks: (a) an autonomous navigation (SLAM) system will be developed by **AUA** that enables the HUSKY AGV to navigate organic fields autonomously; (b) a computer vision solution relying on colour camera streams will be developed by **AUA** to enable weed detection; (c) the weed detection information will be translated to mechanical actuation, moving the mechanical weeder implement blades in the correct position to destroy the weeds without damaging the crop. To attain optimum real-time speeds, all three subtasks will be deployed on an onboard processing unit (e.g., GPU, VPU); (d) development of AR solutions by **UCD** to be coupled with the automated weeding system. The AR solutions will allow the end-user to see the field as the robot sees it, using the same computer vision algorithm. At the same time, a second computer vision detection algorithm will be developed capable of recognizing the different system components, thus allowing non-experts to easily operate, troubleshoot and perform minor repairs.

T3.3 – AR-enabled Control Algorithms for Selective Harvesting [M07-M48], (UCD, NTU, HWU) This task focuses on the development of an AR-based recognition system by **UCD** which will start with the data collection of diverse crop images, followed by the development and integration of novel machine learning models into an AR platform for real-time data overlay. The next task involves integrating this AR system with selected harvesting machinery, such as robotic arms and drones, which includes **NTU** equipping these machines with necessary cameras and sensors and developing interfaces for seamless software-hardware communication. Advanced control algorithms that interpret AR data to informed real-time decisions will be employed by **HWU**, requiring algorithm design, integration with AR data and extensive testing through simulations and field implementation.

T3.4 – Gamified VR Environment for Robotic Systems [M07-M48], (LIN, HWU) This task will deliver the necessary software components that will enable users to train and control a robotic system remotely. The VR solution will be developed by **LIN** using professional level equipment, such as the Varjo XR4, to ensure high quality by exploiting the increased graphical processing capabilities of the equipment in combination with high hand and gesture tracking accuracy. The virtual content will be developed using the Unity engine due to its ability to generate high fidelity graphics. The VR robot training environment will display various fruit cluster configurations and will prompt the user to perform harvesting of specific fruit from the cluster. To increase the user's engagement, and maximise the generation of training data, **LIN** will additionally introduce gamification features inspired by elements of movement-based VR games. In addition to the training environment, **HWU** and **LIN** will develop the telepresence environment that will allow the user to monitor remotely the operation of the robotic system in real time. This will be achieved by appropriately processing and projecting the feed captured from the robot's RGBD sensors to the user's headset. The user will be able to virtually move around within the robot's remote environment for obtaining an optimal perspective of the robot's operation.

T3.5 – AR-powered Operational & Training for Sustainable Agriculture [M07-M48], (INE, UCD, KUL, LIN, HWU) During this Task, **INE** will develop an AR-powered platform targeting operational support and stakeholders' training for sustainable agricultural practices. Real-time data on crop health, soil conditions and weather patterns will be collected from all the pilot leaders, i.e., **UCD, AUA, INE, LIN, KUL, FIB**, in order to develop a visualization feature that will allow farmers and farm workers to see this information overlaid on their fields through AR glasses or mobile devices. **LIN** will also develop interactive AR training modules to offer immersive learning experiences, assisting relevant stakeholders in practicing sustainable farming techniques in simulated (realistic) environments. Last, the AR-based platform will incorporate decision support tools developed in **T4.5**, providing farmers with personalized recommendations on sustainable agricultural practices such as crop management, irrigation and pest control.

WP Number	WP4	WP Title	Automation and Enhance Artificial Intelligence for Environmentally and Socially Responsible Practices
Objectives: (1) To develop deep learning algorithms for complex scene recognition; (2) To develop a computer vision solution able to detect birds at flock level; (3) To develop a collaborative human-robot framework; (4) To develop a pathway planning and navigation control software stack; (5) To develop a Decision Support System for socially responsible agricultural practices.			
T4.1 – Deep Learning for Complex Scene Recognition [M07-M48], (UCD, AUA, LIN, DTU) This Task will focus on the implementation of deep learning algorithms by UCD for complex scene recognition used by a mushroom harvesting robot. Initially, a diverse dataset of images or videos depicting various scenes encountered during mushroom			

harvesting in the greenhouse will be collected and labelled to signify different objects, structures and relevant environmental conditions. Subsequently, the collected data will be pre-processed to enhance its quality and suitability for deep learning algorithms, incorporating tasks such as resizing, normalization, and augmentation. Various deep learning algorithms will be tested by **AUA** and **LIN**, and an appropriate architecture for scene recognition will be selected. The most suitable deep learning model will then be tested using the labelled dataset, with parameters adjusted through backpropagation, and its performance will be optimized based on a defined loss function. Evaluation metrics, including accuracy, precision, recall and F1-score, will be employed to quantify the model's effectiveness. Hyperparameter tuning will be conducted by **UCD** to further optimize the deep learning model's performance, involving adjustments to parameters like learning rate, batch size and network architecture. The trained model will be integrated into the mushroom harvesting robot's control system using ROS, facilitating real-time communication with other components such as sensors and actuators. Finally, the deployed DL model will undergo extensive testing in real-world environments, evaluating its performance under various conditions encountered during mushroom harvesting operations.

T4.2 – Animal-Robot Interaction for Poultry Management [M07-M48], (KUL, DTU, HWU) During this task, **KUL** will develop a computer vision solution able to detect birds at flock level and combine these data to infer the risk of emergence of abnormal behaviour. A suitable controller will be designed by **HWU**, that will position the remote stimulation tool in an optimum position to trigger as a function of the emerging group behaviour. For the mobile platform, the already developed quadrupled legged robot will be used further integrating a robotic arm equipped with RGB-D and thermal sensors for poultry evaluation and eggs manipulation and collection. The robot will employ visual serving control strategies combined with high level predictive control schemes developed by **KUL**. Locomotion adaption of the platform to the poultry behaviour will be achieved using reinforcement learning techniques that guarantee safe interaction with the animals through the ability of learning the way they move, while also tracking assets using the sensor information mounted on the robot. Robot locomotion control assisted by **DTU**, sensor information processing and decision-making management will be monitored remotely, via a cloud system. This integration of technology will lead to better decisions on the different learning architectures to be implement in the pilot demonstration campaigns. Last, during this task, **KUL** will explore the potential of overlaying animal models with information about their condition.

T4.3 – Human-Robot Collaboration for Safe Working Conditions [M07-M48], (LIN, DTU, HWU) In this specific task, robot control algorithms will be developed by **LIN** and **DTU** that not only enhance the efficiency of agricultural operations but also prioritize the safety of human workers within the collaborative human-robot framework. The machine learning models used for robot control by **DTU** will be designed to optimize the performance of the robotic solutions and ensure that the interactions between humans and robots are inherently safe. The recorded user demonstrations in the gamified virtual reality environment (T3.4) will serve as a valuable training ground, allowing the machine learning models to understand and adapt to various scenarios, thereby contributing to a safer working environment. Furthermore, the incorporation of self-tuned machine learning methods through **HWU**, will add an extra layer of safety by enabling the system to dynamically adjust and adapt to different operational conditions. The formulation of machine learning controllers as time-invariant dynamical systems, coupled with analytical proofs of stability using Lyapunov theory, will provide a rigorous foundation for ensuring the stability and reliability of the robotic system under various circumstances.

T4.4 – Automation and AI Innovations in Pathway Planning and Navigation Control [M07-M48], (DTU, UCD, AUA, LIN) In this task **DTU** will develop a pathway planning and navigation control software stack for **AGROBOOST**'s agricultural robots. First, the focus will be on developing modular pipelines that will serve as baseline. These modular pipelines will consist of individual modules for perception, localization, planning and control. Later and as project-specific data are gathered, end-to-end approaches that optimize the combined task will be applied by **UCD**, from raw perception to control signals, as a single Deep Learning task. Well-established planning approaches for global planning with Deep Reinforcement Learning will be followed, specifically Proximal Policy Optimization, for local planning. Adaptability to novel situations, such as changing weather conditions or unexpected obstacles, will also be considered by both **LIN** and **AUA** optimizing the developed approaches for continuous performance improvement. The proposed implementations will focus on real-time performance and their potential for implementation on edge computing devices.

T4.5 – XAI-enabled Decision Support System for Socially Responsible Agricultural Practices [M07-M48], (NCI, UCD, NTU) During this Task, **NCI** will design and develop a decision support system that will collect, fuse and analyse a wide range of data coming from the farm management systems and the deployed robotic technologies deployed to provide insightful recommendations pertinent to socially responsible agricultural practices such as conservation practices to minimize the use of water, crop rotation and other agroecological practices that prioritize biodiversity and ecosystem resilience. The fusion will be performed based on robust, unbiased, fair, Deep Learning-based techniques such as Recurrent Neural Networks able to capture the dependencies and patterns in datasets. Ensemble methods like Extreme Gradient Boosting (XGBoost) and Random Forest will also be deployed by **NTU** and **UCD** to enhance recommendation accuracy and reduce inherent biases in individual models. Moreover, XAI techniques described in Section 1.2.4 will be integrated into the framework to interpret the rationale behind AI model recommendations, ensuring transparency and user comprehensibility, promoting trust and thus, acceptance among solution providers and agricultural practitioners.

WP Number	WP5	WP Title	Pilot Demonstration and Validation Campaigns
Objectives: (1) To define the system architecture and its technical specifications; (2) To draft the use case scenarios that will be followed in the pilot demonstration campaigns; (3) To implement the projects cascade funding mechanisms that will integrate third party solutions into the project's system; (4) To prepare and execute the pilot demonstrations based on real-world scenarios with the active participation of various stakeholder groups; (5) To assess the TRL, SRL and LRL of the develop technologies; (6) To consolidate lesson learnt and best practices from the pilot validation campaigns; (7)			

To design and develop a monitoring system that will assess the environmental, economic and social impacts resulting from the integration of the proposed technologies.

T5.1 – Uses Case Scenarios, Pilots' Planning and Preparation [M07-M16, M21-28, M33-M40], (NTU, UCD, KUL, AUA, INE, LIN, NVR, AJP, FAD, FIB, EPF) This task will develop the detailed plans for the AGROBOOST demonstrations and validation in the context of the six pilot use cases [T5.4 - T5.9] against the operational user needs and requirements extracted in T2.1. The primary activities in the task will be NTU's responsibility and include (a) elaboration of the use case scenarios to align with the technological maturity and integration plans, (b) definition of the target number of participants, (c) the design of a testing protocol including a sound timeline of activities and definition of roles and responsibilities, (d) identification of required datasets, preparation and calibration of robotics, AR and the automation-based solutions and the (e) alignment of the ethical aspects. To lay the groundwork for the evaluation, the current performance of the technologies will be catalogued by the use case leader, i.e., LIN, UCD, INE, AUA, KUL and FIB and updated KPIs for each of the pilots will be defined to provide a baseline measure for their assessment in T5.10. AJP and FAD will provide the social element of the pilots incorporating at least one scenario dedicated to the use of robotic technologies and AR by young farmers and women in agriculture respectively.

T5.2 – System Architecture Design and Technical Specifications [M07-M40], (NCI, UCD, NTU, AUA, INE, DTU, LIN, HWU, WIS, EPF) Based on the results of T5.1, NCI with the active involvement of all the technology providers, i.e., LIN, UCD, INE, AUA, KUL, DTU, HWU, EPF and NTU, will define the overall system architecture in terms of interfaces, information flows, components interactions, deployment views, etc. Initially, the user requirements will be translated into functional requirements and technical specifications that will guide the development of the proposed robotics and AR solutions. A security-and-privacy by design approach will be followed focusing on a more proactive and preventive security-and-privacy rather than on reactive and remedial approach. End-to-end security-and-privacy in the design, including data protection, database security, identity and access management will be considered. Moreover, Open APIs and Connectors will be developed among system components and external systems (e.g., farm management platforms) and devices, ensuring interoperability in the respective interactions. This task will include iterative reports that detail how the methodologies have been operationalised during the project, feeding T2.4.

T5.3 – Open Calls Preparation, Evaluation and Continuous Monitoring [M05-M43], (MAR, UCD, NTU, AUA, WIS) This task will define and implements the AGROBOOST's 3rd party projects cascade funding and implementation strategy. AGROBOOST will consider a number of eight 3rd party projects in total through two open calls that will act as a mechanism to attract EU-based (and in associated countries) actors, organizations, SMEs, startups, research groups and industrial players, under a selection process based on specific criteria defined by the Open Call Committee (OCC, See Annex). The open calls packages will be created based on the project's planning and priorities and will include a full set of reference material for applicants and a "Guide for Applicants – Information document" briefing pack, along with the required application form(s). The open call will be widely promoted through a number of channels in collaboration with WP6 including AGROBOOST's network, social media and other online marketing and other HE and EC communication means (including the Official Journal of the EC) and utilizing the existing ecosystems of the partners and especially the piloting sites. This task will manage the launch and implementation of the open calls, taking care of the evaluation of proposals against a given set of criteria through internal evaluation, selection of 3rd parties, initiation, contracting and management of the projects through to completion, along with the implementation, support, mentoring, monitoring and evaluation of the piloting activities. Full details on the strategy and implementation processes can be found in Annex. The OCC will be led by the MAR and will include the following partners involved in technical and financial aspects of the open call: the Project Coordinator (UCD), Technical Coordinator (AUA), key technical partners (NTU), the legal and ethical expert (WIS) and the pilots' leaders (LIN, UCD, INE, AUA, KUL, FIB) that will provide material and guidance during the implementation of the Information document, provide recommendations in terms of engaged external experts, provide support during the evaluation process and mentoring the 3rd party projects during their implementation. UCD will be responsible for the contractual and financial aspects of the FSTP mechanism in collaboration with MAR.

T5.4 – Pilot 1: Intelligent Autonomous Robotic System for Harvesting of Strawberries Crops [M17-M60], (LIN, WIS, AJP, DTU, HWU)

T5.5 – Pilot 2: Fully Autonomous, AR-enabled Mushroom Harvesting Robots [M17- M60], (UCD, WIS, AJP)

T5.6 – Pilot 3: Orchards 6.0 with AI-based Robots for Tree Pruning and Flower Thinning [M17- M60], (INE, NVR, WIS, AJP)

T5.7 – Pilot 4: Mechanical Autonomous Selective Weeder [M17- M60], (AUA, WIS, AJP, DTU)

T5.8 – Pilot 5: Autonomous Management of Poultry Production, Health and Welfare [M17- M60], (KUL, HWU, WIS, AJP)

T5.9 – Pilot 6: Automated Work on the Ground in Steep Vineyards [M17- M60], (FIB, EPF)

These Tasks [T5.4 - T5.9] are dedicated to the validation of the AGROBOOST solutions developed in WP3 and WP4 for each one of the respective use cases. The responsible partners (LIN, UCD, INE, AUA, KUL, FIB) will be internally trained for the technical and operational functionalities of the system before deploying the system in the pilot sites. The pilots will have a duration of 6 months and will be divided into four phases (M17-M22, M29-M34, M41-M46, M53-M58). The kick-off of the demonstrators will firstly test the early-stage prototypes to refine the overall solutions towards their full functionality. The initial evaluation results from the first phase will provide feedback to the next versions of the prototypes for the next pilot iterations and ensure alignment of technology offerings with end user needs. Next steps will deploy the additional features of the framework and evaluate the complete set of functionalities. Special attention will be

given to covering all evaluation KPIs, including both those identified in the project and any additional that may have been identified whilst developing the evaluation methodology.

T5.10 – Pilots Assessment, Lessons Learnt, User Acceptance Evaluation and Replicable Activities [M23-M60], (NTU, UCD, LIN, SSI, WIS, AJP, FAD, FIB, EPF) This task will validate that both the developed solutions within the timeframe of the project as well as the whole **AGROBOOST** framework have reached the desired technology, **NTU**, societal (how ready the society is to adopt the proposed solutions), **SSI**, and legal (if the proposed framework is legally compliant), **WIS**, readiness level. During this task the results of the pilots will be evaluated by **NTU**, analysing the feedback received from the involved stakeholders and the pilot leaders after the end of each pilot cycle. Usability assessment and user acceptance evaluation of the overall solution will also be performed by **NTU**, verifying that the targeted KPIs set in **T5.1** have been successfully met. Objective and subjective feedback collected from internal and external participants attending the pilot validation campaign events will be analysed to improve the functionality offered and enhance the user acceptance of the proposed solutions. Additionally, during this task, **FIB** will identify replicable activities that can be scaled and implemented for different EU countries with varying socioeconomic criteria. This feasibility study will contribute to the refinement of the proposed technologies to fine-tune them for broader applicability.

T5.11 – Monitoring System for Analyses of the Consequences for Farmers and Rural Communities [M07-M60], (NCI, NTU, NVR, WIS, MAR, AJP, FIB) During this Task, **NCI** will design and develop a monitoring system that will assess the environmental, economic and social impacts resulting from the integration of the proposed technologies into existing farm management practices. It will collect data from various sources, such as IoT-enabled sensors and agricultural machinery, on-site surveys, etc. facilitated by **WIS**, to analyse the effects of the proposed solutions on the overall farm sustainability. The monitoring system will leverage advanced data analytics techniques developed by **NCI**, to continuously monitor and evaluate the consequences of the technology adoption by the farmers emphasizing on the reduction of the environmental footprint, the improvements in resources and the fluctuations in the socio-economic dynamics within farming communities. Feedback mechanisms from the farmers will be developed by **NTU** coupled with in-depth assessments performed by **FIB** to capture and prioritize barriers and insights from the farmers. Several indicators will be monitored based on a co-creation process performed by **AJP** with the agricultural community.

WP Number	WP6	WP Title	Impact Maximisation and Ecosystem Building
-----------	-----	----------	--

Objectives: (1) To ensure widespread dissemination of **AGROBOOST** results to diverse stakeholders maximizing impact (2) To engage farmers, farm workers and relevant stakeholders and liaise with EU and international initiatives and clusters (3) To strategically plan the exploitation of **AGROBOOST** results by identifying opportunities for technology transfer, commercialization and collaboration with industry partners (4) To evaluate the feasibility of **AGROBOOST** 's technologies within the market context and developing a robust business model (5) To generate policy recommendations, highlighting the potential operational implications and suggesting measures for effective integration into existing frameworks.

T6.1 – Project Branding, Dissemination and Communication Activities [M01-M60], (MAR, All Partners) This task will be responsible for the dedicated **AGROBOOST** communication and dissemination activities. The planned work will be led by **MAR** and **all partners** will contribute towards the achievement of the respective KPIs. The first months will be dedicated to the definition of a well-articulated communication and outreach strategy to engage target groups as well as media and groups of stakeholders. The strategic approach will set the overall framework and will give directions for all publications, liaisons, events and general communication activities. This includes all means and measures defined in Section 2.2. This task is also responsible for creating, implementing and monitoring the project's brand identity as the basis for all communication tools: the project website, social media channels, graphic material for online and offline promotion, market insights (e.g., white papers), practical guides, policy briefs, PR outputs (press releases). This task will also coordinate with all WPs in the design and presentation of proper knowledge sharing material and will support open access to all public deliverables and outputs of the project and plan campaigns for the promotion of the Open Calls.

T6.2 – Liaison with Relevant EU Projects and Initiatives and Community Engagement [M01-M60], (AUA, UCD, AUA, LIN, NCI, SSI, WIS, MAR, AJP, G&D, FIB) To ensure awareness about the project's achievements and results, foster engagement of target stakeholders and grow the overall **AGROBOOST** community, the consortium will organise dedicated events disseminating the outcomes and the activities of the project. These events will be overviewed by **AUA** and include (a) A dedicated workshop for each one of the pilot cases organized by **LIN, UCD, INE, AUA, KUL** and **FIB** during the third and fifth year of the project, presenting the pilot developments and lessons learnt to each relevant community; (b) A final conference to be held after M53 to present the project's outcomes to the wide public and the media. Different relevant stakeholders and think tanks will be invited by **FIB, WIS** and **SSI** to participate and debate around **AGROBOOST**; (c) Informative tutorials and online webinars on the proposed solutions for enhanced working conditions will be organised by **NTU**. Last, **G&D** will also examine the coordination with other relevant EU initiatives of the Green Deal sector to explore complementarities with other EC projects seeking cross-fertilization of ideas, exchange of good practices and networking with relevant actors.

T6.3 – Innovation Management, Market Analysis and Commercial Pathway [M13-M60], (MAR, NTU, KUL, AUA, INE, LIN, NCI, WIS, HWU) The purpose of this Task is twofold. First, to ensure the coordination of all activities connected to technology alignment to user needs, technology transfer, management of innovation, including IPR management and identification of market opportunities for the rapid exploitation of the **AGROBOOST** innovations. The innovation manager, **MAR** will foster the exploitation of innovation and connect with other stakeholders to promote the openness of the approach. This will be translated into the following activities: a) **NTU** will link and refine user needs to monitor and ensure alignment of technology innovation to user needs while feeding the exploitation plan with the inputs

from the stakeholders and the advisory board; b) **MAR** will manage the execution of the overall exploitation plan of the project and support the partners in setting up their individual business plans, in order to exploit the results; c) **AUA** will manage the knowledge produced during the project lifecycle and assess the opportunity for applying for patents or declaring copyrights, through maintaining all innovations descriptions, screening and ownership. To ensure the sustainability of **AGROBOOST** well after the project end, **WIS** will deliver a complete IPR Plan which will include results from the IPR management strategies, ownership and governance. Moreover, **MAR** will also perform a market analysis and create a business plan that will ensure the sustainability of **AGROBOOST**'s innovations by (a) analysing the best practice cases in relevant industries in the market in close collaboration with large industrial partners (**NCI**), (b) elaborating and refining scenarios and revenue models for sustainability, including funding requirements, potential growth and long-term profitability and ROI to identify those most appropriate and (c) developing business models and commercial roadmaps for the most relevant scenarios, as well as collaborate with **INE** to recommend business models and strategies for the use of the developed innovative technologies under consideration of various farm structures and inter-farm linkages as well as of various biogeographic and socio-economic framing conditions.

T6.4 – Replicable Activities, Knowledge Transfer and Training Curricula [M13-M60], (**NTU**, **UCD**, **AUA**, **LIN**, **NVR**, **SSI**, **WIS**, **FAD**, **FIB**) During this task, **NTU** will interact with the technology providers and **SSI**, **FAD**, **NVR** and **WIS** to create training curricula for the adoption of robotic, AR and AI technologies that enhance the working conditions of workers in agriculture. For this purpose, a comprehensive set of training material for competent authorities (local, national and EU level) and fishermen/fisherwomen will be developed by the technology providers (**UCD**, **AUA**, **LIN**) under the guidance of **NTU**, based on the following pillars: (a) training curricula emphasizing on the understanding of additional capabilities of modern technologies used to enhance the working conditions of farm workers, (b) dedicated online and offline e-learning training courses and seminars based on the project's innovative solutions. Feedback will be obtained at the end of each training session to improve the developed technologies, to assess their inclusivity and enhance their user interfaces before the different phases of the validation campaigns. Training material will be made publicly available assisting project members to communicate the results to a wider. Last, during this Task, **FIB**, **WIS** and **SSI** will examine several EU countries with varying socio-economic criteria where replicable activities can be performed, deploying the proposed innovations and inclusive technologies in real-world agricultural settings.

T6.5 – Standardization Activities and Policy Recommendations [M25-M60], (**G&D**, **NVR**, **SSI**, **WIS**, **MAR**, **AJP**, **FAD**) The main objective of the task is twofold. On the one hand to engage with the community to co-create a roadmap with targeted policy recommendations that will support a systematic approach towards the implementation of Farm to Fork priorities and the Green Deal initiative. This target will be achieved through a concrete definition of the vision of the roadmap on the implementation of the Farm to Fork and Green Deal priorities based on a literature analysis of reports, deliverables from other EU funded projects and other publications from existing national, EU and international initiatives performed by **G&D** and in close collaboration with **FAD**, **AJP** and **SSI**. Subsequently, this Task will align the produced roadmap to existing policies, strategies and agendas on a regional, national and European level by organising ad hoc bilateral meetings with main actors through **NVR**, and co-design events for the finalisation of the roadmap. On the other hand, this Task will also contribute to standardization working groups, mapping the key exploitable innovations to the standardization objectives and preparing a concrete plan to submit contributions to relevant standardization bodies.

3.1.6 List of deliverables (Table 3.1c)

Del. No	Deliverable Name	Deliverable Description	WP No.	Lead	Type	Diss lev.	Deliv date
D1.1	Project management handbook	Description of the project's management plan, setting the guidelines for quality assurance of the project's activities, administration and deliverables preparation.	WP1	UCD	R	SEN	M04
D1.2 – D1.6	Data management, legal and ethics analysis	Reports for the data management policy of the project along with the protocol that AGROBOOST will adhere to. Details on privacy, ethical and legal compliance of the AGROBOOST solutions will be provided.	WP1	WIS	R	SEN	M06, M20, M34, M48, M60
D2.1 – D2.3	Socioeconomic methodologies, tools and approaches towards enhanced working conditions in agriculture	Reports describing the developed methodologies, tools and approaches to improve the working conditions in agriculture. The initial version will detail the overall framework, outlining proposed strategies for enhancing safety. The interim version will provide progress updates, incorporating feedback based on initial findings. In its final iteration, it will offer a detailed analysis of their effectiveness in promoting sustainable and socially responsible practices.	WP2	FIB	R	PU	M10, M28, M60
D3.1 – D3.3	Augmented & Virtual Reality and robotic	Reports describing the developed AR/VR and robotic applications aimed at enhancing safety and performance in agricultural operations. The	WP3	INE	R, DEM	PU	M18, M32, M48

	solutions for increased safety and improved performance	initial version will outline the initial technological developments of each solution and its maturity level, while the interim version will provide updates on the ongoing progress, incorporating insights gained from their application in the pilot validation campaigns. In its conclusive form, the TRL7 technologies will be described along with any potential modification performed.					
D4.1 – D4.3	Automation and enhanced Artificial Intelligence for environmentally and socially responsible practices	Reports describing the developed AI-based and automation-driven applications for environmentally and socially responsible agricultural practices. The initial version will outline the initial technological developments of each solution while the interim version will provide updates on the ongoing progress, incorporating insights gained from the pilot validation campaigns. In its conclusive form, the TRL7 technologies will be describe along with any potential modification performed.	WP4	DTU	R, DEM	PU	M18, M32, M48
D5.1, D5.3, D5.5	Use case scenarios, system architecture and progress of open calls.	Report elaborating on the use case scenarios that will be followed during the pilot validation campaigns. In addition, the overall system architecture and the technical specifications will be described. The progress of the Open Calls processes will be included along with the thorough description of the chosen third party solutions.	WP5	NCI	R, DEM	PU	M22, M34, M46
D5.2, D5.4, D5.6, D5.7	Pilot preparation, execution, lesson learnt and replicable activities,	Report describing the preparatory measures for the implementation of the project validation campaigns. It will summarise the lesson learnt based on stakeholders' feedback and detail the replicable activities that can be performed in other EU countries with different socioeconomic criteria.	WP5	NTU	R, DEM	PU	M22, M34, M46, M58
D5.8	Pilot overall assessment, user acceptance and analysis of consequences for rural communities.	Report describing the final evaluation of the project's pilots' results, lessons learnt, through assessment of the pilots, demonstrating detailed users' feedback and evaluation. It will also present the achieved outcomes in terms of TRL, SRL and LRL metrics.	WP5	NTU	R	PU	M60
D6.1 – D6.4	Outreach, engagement and exploitation strategy	Reports on the strategy and plan towards the outreach, engagement and exploitation activities of the project. The material used for the training curricula and the feedback received by the stakeholders will also be described in every version. The produced policy recommendations will be presented.	WP6	MAR	R	PU	M06, M26, M40, M60

3.1.7 Project milestones (Table 3.1.d)

Milestone number	Milestone name	Related WPs	Month	Means of verification
MS1	Project Set up	WP1, WP6	M06	Successful submission of D1.1, D1.2, D6.1
MS2	First working prototypes	WP2-WP4	M18	Successful submission of D2.1, D3.1, D4.1
MS3	First Pilot Evaluation	WP5	M22	Successful submission of D5.1, D5.2
MS4	Final working prototypes	WP2-WP4	M48	Successful submission of D3.3, D4.3
MS5	Final Pilot Evaluation	WP5	M59	Successful submission of D5.7, D5.8
MS6	Project Consolidation	WP1, WP2, WP6	M60	Successful submission of D1.6, D2.3, D6.4

3.1.8 Critical risks for implementation (Table 3.1.e) #@RSK-MGT-RM@#

The management of risks will be part of the regular meetings of the Executive Board in which the project coordinator, quality assurance manager and WP leaders participate. The Table below summarises the critical risks, their

Probability [**P**] and Impact [**I**] (**L**: Low, **M**: Medium, **H**: High) on the successful implementation of the project and the proposed mitigation actions.

Description of risk [WP(s)], [Prob, Impact]	Proposed risk-mitigation measures
Some modules exhibit lower accuracy or performance than expected. [WP3, WP4], [Prob. L, Impact M]	Each of the modules provided through AGROBOOST are offered by at least one technical partner and are currently at least at TRL3/4, having proven their readiness level. In case that technology still underperforms alternative methods will be explored through state-of-the-art to achieve the foreseen result.
Although the system requirements are covered there is not high acceptability from end-users. [WP2, WP5], [Prob. M, Impact M]	Timely involvement of stakeholders (project partners and external end-users participating from the very beginning of the project) for providing the requirements. Users will participate during the whole project and perform the system tests and validation.
Errors in robotic planning due to insufficient background knowledge or imprecise environment interpretation. [WP3-WP4], [Prob. M, Impact M]	The components developed for handling robotic planning activities have dependencies on multiple other system modules and procedures, while also being prone to facing runtime difficulties in the real experimental setting. To this end, module integration, deployment and evaluation will be of high priority.
The AR/VR interface proves too cumbersome for the relevant stakeholders. [WP3-WP4], [Prob. M, Impact M]	Care will be taken to make all AR/VR functionalities also available through a mouse-operated web interface, even at the cost of reduced immersion. Continuous feedback loops and usability testing will be performed to refine the interfaces.
Data collected to formulate the necessary AR/VR and robotic frameworks lacks quality or is not in the desired standards. [WP3-WP4], [Prob. M, Impact M]	The identification and collection of real-world data will be performed with thorough state-of-the-art analyses carried out by multiple partners that have significant experience in that field. Historical data from the fields will be leveraged while common formats will be used suitable for AR and robotics applications.
Proposed solutions are not aligned with the European core values and citizen expectations, [WP1-WP6], [Prob. L, Impact H]	Societal concerns are taken into account at an early stage and communicated to all partners, while the dedicated T2.4 will continuously monitor any relevant new laws and rules by the European Nations. #§RSK-MGT-RM§#

3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.2.1 AGROBOOST consortium complementarity

The **AGROBOOST** consortium members have been carefully selected to cover all the required research and development areas of the project in a well-balanced way, leveraging their expertise and prior collaborations to match the project's objectives successfully. The partnership structure is based on excellence, complementarity, transnationality and multidisciplinary. The rationale for selecting organisations was to achieve • a balanced consortium with an adequate inclusion of experts in all the related domains, • scientific excellence, • high exploitation and dissemination potential, • top quality level of management. Overall, **AGROBOOST** asserts that (i) a critical mass of skilled resources will be committed to the project, (ii) it has close collaboration and complementary strengths, and (iii) the scientific and development team has a long track record and scientific credentials. The consortium includes all the essential actors of the value network including researchers in the agricultural domain, associations of stakeholders (young farmers and women in agriculture), communication experts, technology providers, impact multipliers, civil society organizations, policy makers, etc. across Europe. The scientific and technical expertise needed can be found within the current members of the consortium ensuring that the project will start quickly and provide ground-breaking results. The way the main roles have been assigned among **AGROBOOST** partners guarantees a balanced work contribution, as each partner has clearly defined its role in each of its tasks, as well as the lead he has taken in the WPs and tasks. The consortium brings together the resources of **18** participating organisations from **11** European countries each excelling in their respective field and with significant R&D experience. The **AGROBOOST** interdisciplinary consortium consists of:

UCD, the leading centre for education and research in the application of engineering principles to agriculture, food and robotics in Ireland that will lead the project's activities, develop novel AR-enabled control algorithms for selective harvesting and deploy their solution in a dedicated pilot targeting fully autonomous, AR-enabled mushroom harvesting robots.

NTU, serving as the link between academic knowledge, taught at the National Technical University of Athens, and their practical application in leading industrial enterprises. NTU will lead the user requirement elicitation that will guide the development of the robotic and AR-based applications and will also coordinate the implementation of the six pilot validation campaigns.

KUL, the largest university in Belgium in terms of research funding and expenditure, conducting fundamental and applied research in a wide range of academic disciplines with a clear international orientation. KUL will develop novel AR and robotic technologies for anima-robot-interaction and demonstrate their functionality in a pilot case targeting autonomous management of poultry production, health and welfare.

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

EPF is a top ranked Swiss university specializing in engineering and natural sciences. EPF will provide valuable expertise and support to the project, leveraging its strengths in cutting-edge research and innovation on morph robots and heavily assisting in the respective pilot demonstration campaign.

AUA and its Smart Farming Technology Group which belongs to the Department of Natural Resources Development and Agricultural Engineering, will oversee the technical activities of the project while leading the development of AR-based systems for mechanical weeding and deploying them in a dedicated pilot in Greece.

INE, an associate laboratory with 37 years of experience in R&D and technology transfer, will focus on the development of autonomous systems in agriculture coupled with computer vision and AI technologies and work in close collaboration with NVR to deploy these innovative solutions in a pilot.

LIN is rapidly rising university that has established its international reputation on both research and teaching while demonstrating an exceptional track record on agro-robotics research. LIN will overtake the overview of the AR-related technologies leading WP3 and demonstrate its developed technologies in a dedicated pilot.

NVR is the State Laboratory of the Ministry of Agriculture and Food, that develops research activities in the agricultural and veterinary fields, providing laboratory services and scientific and technical support to both farmers and industry. NVR will validate the proposed AR and robotics solutions for tree pruning and flower thinning.

DTU is an international elite technical university where education, scientific advice, and innovation rest on a solid foundation of world-class research. DTU will focus on autonomous robotic systems bringing its expertise in robotic perception using machine learning competencies in real-life application environments.

HWU a renowned university recognized for its excellence in research, innovation and commitment to providing a global learning environment. HWU will co-develop AR technologies for operational and training platforms towards sustainable agriculture demonstrating this solution in the pilot addressing autonomous robotic system for livestock farm management.

FIB is one of the world's leading research and information centres for organic agriculture with close links between different fields of research and the rapid transfer of knowledge from research to advisory work and agricultural practice. FIB will coordinate the activities in WP2, laying the foundations for a deeper understanding of the working condition and labour shortages in the agricultural domain.

NCI, one of the biggest ICT service providers in Europe with a vast expertise in complex integration of software systems and in developing IT systems for the EC. NCI will lead the development of the XAI-enabled Decision Support System for socially responsible agricultural practices and will also lead the activities of the monitoring system analyses of the consequences for farmers and rural communities.

SSI, a Romanian SME that promotes collaborative efforts while embracing a strong research orientation in social innovation and field research. SSI will lead the strategies targeting inclusiveness for young farmers, women in agriculture and farmers with disabilities.

WIS, a customer-focused SME in Cyprus, offering legal and ethical advice services and monitoring the compliance of the project's activities in respect to ethics, legal, privacy and data protection norms (including GDPR).

MAR is dynamic Swiss-based SME with many years' experiences managing innovation projects across Europe and worldwide. MAR will coordinate the Open call process and lead the outreach activities of the project, drawing on its proven track record of effective communication and dissemination management and its extensive network to ensure the successful engagement of stakeholders from the agricultural sector.

G&D is a Greek SME committed to driving progress towards the green and digital transition that is currently reshaping our society. G&D will leverage its broad range of stakeholders from academia, government, business and civil society and provide tailor-made services in the areas of policy analysis, policy advice and strategic analysis, with a focus on sustainable agriculture and food systems.

FAD is a progressive organization that fights to achieve equality and progress for women who live and work in rural areas, improving their daily life. FAD will assist towards the adoption of the proposed solutions by women working in the agriculture shaping the project's offerings to be more inclusive.

AJP, an association of young Portuguese farmers offering technical assistance and professional training to several countries while having a permanent representation delegation in Brussels to monitor and raise awareness of the community instances on the Common Agricultural Policy. AJP will enhance the technology adoption by young farmers strengthening the inclusivity of the proposed solutions and replicating the training curricula in Portugal.

3.2.2 AGROBOOST breakdown of roles and capacity.

The Project Coordination Team (PCT) shall be responsible for the planning, execution and controlling of **AGROBOOST**; the PCT will operate under the supervision of the coordinator and includes the following activities: a) Administration and scientific coordination activities, b) Implementation of all action plans, c) Establishment of a budget and schedule-controlling system, d) Implementation of a 360o quality assurance system, e) Provision of clear guidance on Intellectual Property issues, f) Developing and maintaining a communication and reporting attitude. The following key project persons constitute the PCT: • Project Coordinator (**UCD**), being responsible for the overall management, communication, and coordination of **AGROBOOST** and chairing the two main project bodies, the PCT and the Plenary Board. The PC will be the only official channel that interacts with the EC, especially with regards to the submission of deliverables and aspects related to the **AGROBOOST** consortium. • Scientific and Technical Manager (**AUA**), ensuring that the scientific and technological objectives of the project will be met. • Data Manager

HORIZON-CL6-2024-GOVERNANCE-01-7 - Enhancing working conditions and strengthening the work force through digital and data technologies – the potential of robotics and augmented reality in agriculture.

(**WIS**), defining the data management policy of **AGROBOOST** and monitor the data management processes (i.e., data collection, access, sharing, processing, storage, etc.) throughout the project life cycle. • Quality Manager (**NCI**), being responsible for the implementation of the quality procedures and the verification of the project results. • Dissemination Manager (**MAR**), being responsible for all dissemination and communication actions of **AGROBOOST**, including contact with stakeholders and synergies with other projects and initiatives. • Ethics and Legal Advisor (**WIS**), monitoring the compliance of project's activities in respect to ethics, legal, privacy and data protection norms (including GDPR).

3.2.3 Industrial/Commercial involvement to ensure exploitation.

AGROBOOST enjoys a strong industrial involvement, both from Large Enterprises (LEs) and SMEs which combined with the prestigious academic and research organisations will ensure the exploitation of the project's ground-breaking results, providing the necessary resources, expertise and infrastructure to bring the project outcomes to market. The industrial partners are specialised in a wide spectrum of activities, necessary to carry out project tasks, such as technology providers in AI, AR, Big Data and Data Analytics, Cloud Computing, IoT, etc., bringing in their commercialisation experience, which runs throughout the respective value chain, i.e., from market analysis and business planning to standardisation and setting up a fair, focused, and viable IPR strategy. The involvement of end-users (association of young farmers and the association of women in agriculture) will ensure that the project is aligned with the stakeholders' needs and priorities and thus increase the likelihood of successful adoption. Industrial partners will collaborate both with each other and with academic partners to lay down a concrete exploitation strategy and an analytical exploitation plan [T6.3]. Targeting new business development upon patented inventions will significantly raise the exploitation potential of the project. For this purpose, a suitable exploitation instrument will be explored evaluating a) which technologies from the proposed ones (if not all), b) under which conditions and c) based on which criteria can be brought to market.

3.2.4 Access to critical infrastructures

UCD has several agricultural robots such as ROSbot 2 PRO and Husarion Panther Mobile platform, a UAV (DJI Phantom 4 Multispectral with RTK module), IoT enabled μ METOS weather stations, a high-precision GPS system (GPS - U-BLOX EVK-7P), an inertial navigation system and several digital testbeds which are flexible for crop growth monitoring and harvesting tasks with easy integration of additional sensors and cameras (e.g., Lidar LDR20 - Velodyne VLP-16 PUCK CAM04, interZED 2i Stereo Camera, Intel® RealSense Depth cameras, Specim IQ hyperspectral camera, time-lapse Cam Pro) and Oculus Quest VR Headsets to support VR/AR guided robotic applications. **AUA** has a high-precision GNSS system with less than 2 cm error, a vegetation index data sensor for NVI, LAI (crop Circle and MicaSense Rededge2), several agricultural robots such as Zues customised ground vehicle and HUSKY commercial platform, the UAV (DJI S-800 EVO) II and the Spraying UAV platform which is a flexible aerial platform for crop monitoring tasks with easy integration of additional sensors and cameras. **INE** will build upon the Modular-E solution that will be extended with attached robotic module (Prune-I) with dexterous manipulators and with advanced high-resolution cameras. The Prune-I incorporates the most advanced hardware for Deep Learning models deployment (GPU, TPU and FPGA) to interpret the environment and the plant tree structure. These robots are developed considering ROS2 ecosystem, where several advanced navigation technologies are deployed, such as VineSLAM a semantic SLAM approach that considers multiple GNSS receivers, LiDAR, RADAR and camera-based technologies. **NVR** has a vast number of sensors that will be used in the demonstration phase of the project in the a 7-year-old apple orchard (*Malus domestica*), with 3175 trees/ha (3.5m x 0.9 m). **LIN** has a strawberry research farm at the Risholm campus, multiple RGBD sensors such as the Intel RealSense, robotic mobile platforms including the Husky UGV and Boston Dynamic's Spot, soft grippers developed for fruit harvesting, the industrial grade Varjo XR4 VR headset and Franka Emica robotic manipulator. **DTU** maintains multiple robot lab facilities equipped with numerous mobile robots, an OptiTrack motion capture system to observe motions with a high degree of accuracy, and a plethora of 2D and 3D visual sensing devices. More precisely, DTU's Autonomous Systems Test Arena (ASTA) facility will be used for the needs of this project. DTU ASTA is a modern covered facility, housing a floor area larger than that of 2 professional basketball courts and with a height up to 14 m. Within the facility, a volume of 16 x 12 x 8m equipped with a 16 cameras motion capture system (OptiTrack) where to experiment with multiple robotic systems. **NCI**, as a large ICT provide in Europe has robust data storage and management systems, necessary to ensure that the data that will be used for the project's activities are secure, accessible by the partners and well-organized.

MAR will contribute with the following resources: Personnel costs (a1) 737'000.00 €; Travel & subsistence (c1) 25'000.00 €; Other goods, works and services (c3) 35'000.00 €; Indirect costs (25%) 184'250.00 €; Total Financial Contributions 996'250.00 €; **FIB** will contribute with the following resources: Personnel costs (a1) 508'950.00 €; Travel & subsistence (c1) 17'000.00 €; Other goods, works and services (c3) 9'000.00 €; Indirect costs (25%) 133'737.50 €; Total Financial Contributions 668'687.50 €; **EPF** will contribute with the following resources: Personnel costs (a1) 279'000.00 €; Travel & subsistence (c1) 20'000.00 €; Other goods, works and services (c3) 51'000.00 €; Indirect costs (25%) 90'500.00 €; Total Financial Contributions 452'500.00 €; **MAR, FIB, EPF** are non-associated third country partner with headquarters in Switzerland and their costs will be funded via the Swiss State Secretariat for Education, Research, and Innovation (SERI).

#§CON-SOR-CS§# #§PRJ-MGT-PM§#



Digitally sealed by the European Commission

Date: 2024.02.28 15:20:25 CET

Reason: Acknowledgement of Receipt

This electronic receipt is a digitally signed version of the document submitted by your organisation. Both the content of the document and a set of metadata have been digitally sealed.

This digital signature mechanism, using a public-private key pair mechanism, uniquely binds this eReceipt to the modules of the Funding & Tenders Portal of the European Commission, to the transaction for which it was generated and ensures its full integrity. Therefore a complete digitally signed trail of the transaction is available both for your organisation and for the issuer of the eReceipt.

Any attempt to modify the content will lead to a break of the integrity of the electronic signature, which can be verified at any time by clicking on the eReceipt validation symbol.

More info about eReceipts can be found in the FAQ page of the Funding & Tenders Portal.

(<https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/support/faq>)